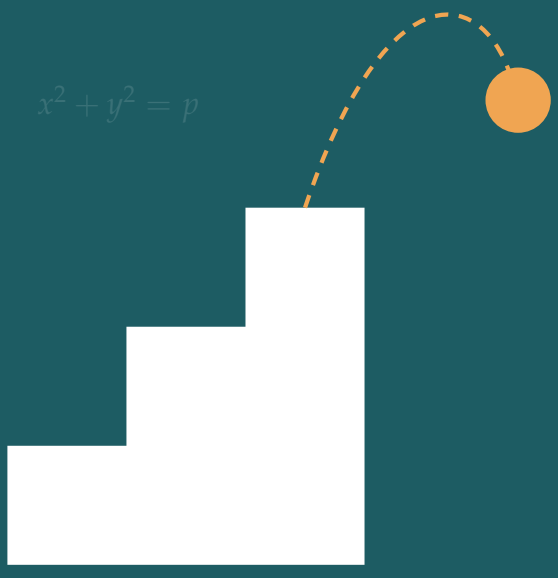
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2009 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2009 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2009 Worked Solutions

Worked solutions to all 39 questions of the 2009 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2009, Question 1

Pure • Counting Divisors

STEP I 2009, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q1

The key idea

Every factor of $p^a q^b$ (with p, q distinct primes) is $p^i q^j$ with $0 \leq i \leq a$ and $0 \leq j \leq b$, and different choices of (i, j) give different factors by unique factorisation. So the number of factors is a *product* $(a+1)(b+1)$, and the same holds with any number of primes. Counting proper factors just means subtracting 2, and both parts become questions about writing a number as a product.

Solution

Write $d(N)$ for the number of positive factors of N, including 1 and N. If

$$N = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

with distinct primes p_i and $a_i \geq 1$, then a positive factor of N is $p_1^{e_1} \cdots p_k^{e_k}$ with $0 \leq e_i \leq a_i$; there are $a_i + 1$ choices for each e_i, and by unique factorisation distinct choices give distinct factors. Hence

$$d(N) = (a_1 + 1)(a_2 + 1) \cdots (a_k + 1).$$

For $N > 1$, the factors 1 and N are different, so the number of proper factors is $d(N) - 2$.

Part (i)

For $3^2 \times 5^3$ we get $d = (2+1)(3+1) = 12$, so there are $12 - 2 = 10$ proper factors, as required.

Now take $3^m \times 5^n$. For this to be an integer with a proper factor at all we need $m, n \geq 0$ (a negative power of 3 or 5 cannot be cancelled by the other prime). The factors are $3^i 5^j$ with $0 \leq i \leq m$, $0 \leq j \leq n$, so $d = (m+1)(n+1)$ (this also covers $m = 0$ or $n = 0$). We need

$$(m+1)(n+1) = 12,$$

that is, an ordered factorisation of 12 into two positive integers. These are

$$1 \times 12, \quad 2 \times 6, \quad 3 \times 4, \quad 4 \times 3, \quad 6 \times 2, \quad 12 \times 1,$$

giving

$$(m, n) = (0, 11),\ (1, 5),\ (2, 3),\ (3, 2),\ (5, 1),\ (11, 0).$$

Each gives a different integer, again by unique factorisation. One of them, $(2, 3)$, is the given number, so there are exactly **5 other** integers of this form with 10 proper factors: 5^{11}, 3×5^5, $3^3 \times 5^2$, $3^5 \times 5$ and 3^{11}.

Part (ii)

We need $d(N) = 426 + 2 = 428$, and

$$428 = 2^2 \times 107,$$

where 107 is prime (it is not divisible by 2, 3, 5 or 7, and $11^2 > 107$).

? **Ask yourself** Which shapes of prime factorisation can N possibly have?

The numbers $a_i + 1$ are each at least 2 and multiply to 428. Since 107 is prime, exactly one of them is a multiple of 107, and it is 107, 214 or 428; the remaining ones multiply to 4, 2 or 1 respectively. Ignoring order, the possibilities for the list of $a_i + 1$ are

$$\{428\}, \qquad \{214, 2\}, \qquad \{107, 4\}, \qquad \{107, 2, 2\},$$

so the multiset of exponents is

$$\{427\}, \qquad \{213, 1\}, \qquad \{106, 3\}, \qquad \{106, 1, 1\}.$$

? **Ask yourself** Given the exponents, which primes should carry them?

To make N as small as possible with a given list of exponents, we should use the smallest primes, and put the largest exponent on the smallest prime. Using the smallest available primes is clearly best: replacing a prime by a smaller unused one makes N smaller. And if $p < q$ but p carries exponent a and q carries exponent b with $a < b$, then swapping them multiplies N by

$$\frac{p^b q^a}{p^a q^b} = \left(\frac{p}{q}\right)^{b-a} < 1,$$

so the swap makes N smaller. So the best candidate of each shape is

$$2^{427}, \qquad 2^{213} \times 3, \qquad 2^{106} \times 3^3, \qquad 2^{106} \times 3 \times 5.$$

Comparing them:

- $2^{106} \times 3 \times 5 = 2^{106} \times 15 < 2^{106} \times 27 = 2^{106} \times 3^3$;

- $2^{106} \times 27 < 2^{106} \times 2^5 = 2^{111}$, which is far less than $2^{213} \times 3$ and 2^{427}.

Hence

$$N = 2^{106} \times 3 \times 5.$$

As a check, $d(N) = 107 \times 2 \times 2 = 428$, so N has 426 proper factors.

Checks

- For a small case, list the factors of $3 \times 5^5 = 9375$ directly: $3^i 5^j$ with $i \in \{0,1\}$, $j \in \{0,\ldots,5\}$ gives 12 factors, so 10 proper ones.
- A short computer search over all exponent patterns $2^a 3^b 5^c 7^d$ with $(a+1)(b+1)(c+1)(d+1) = 428$ confirms that the smallest is $2^{106} \times 15 \approx 1.22 \times 10^{33}$, with $2^{106} \times 27$ next.

Common slips

- Answering 6 in part (i): the question asks for the *other* integers, so the given one is excluded.
- Forgetting the cases $m = 0$ or $n = 0$ (that is, 5^{11} and 3^{11}), which are of the required form.
- Assuming the answer to part (ii) must use as many primes as possible: the primality of 107 forces one huge exponent, and only the split of the factor 4 is a real choice.

Beyond STEP

The function $d(N)$ is a *multiplicative* function: $d(mn) = d(m)d(n)$ whenever m and n are coprime, and that is exactly the product formula used here. Numbers that have more factors than every smaller number are Ramanujan's *highly composite numbers*, and the exponent-swapping argument in part (ii) is the first step in his description of them. Davenport's *The Higher Arithmetic* is a good place to read about unique factorisation and arithmetic functions.

WORKED SOLUTION

STEP I 2009, Question 2

Pure • Tangents to Cubic Curves

STEP I 2009, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q2

This question has a full guided treatment in *Beyond STEP*, Section 1.9, *Tangents to a Cubic*: https://beyondstep.keithmcnulty.org/ch01/tangents-cubic.html

The key idea

A tangent meets a curve at its point of contact *twice*: when we substitute the tangent line into the curve's equation, the point of contact gives a repeated root. Once we know a repeated root of a cubic, the remaining root drops out by factorising, and that third intersection point turns out to have rational coordinates. Clearing denominators then turns the curve's equation into a sum of cubes.

Solution

The tangent at $(-a, b)$

First, $(-a, b)$ really is on the curve, since $x^3 + a^3 + b^3 = -a^3 + a^3 + b^3 = b^3 = y^3$.

Differentiating $y^3 = x^3 + a^3 + b^3$ implicitly with respect to x (where a and b are constants) gives

$$3y^2 \frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2, \qquad \text{so} \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^2}{y^2}$$

wherever $y \neq 0$. At $(-a, b)$, where $b > 0$, the gradient is a^2/b^2. So the tangent is

$$\begin{aligned} y - b &= \frac{a^2}{b^2}(x + a) \\ b^2 y - b^3 &= a^2 x + a^3 \\ b^2 y - a^2 x &= a^3 + b^3, \end{aligned}$$

as required.

The intersections when $a = 1, b = 2$

Now the curve is $y^3 = x^3 + 9$ and the tangent at $(-1, 2)$ is $4y - x = 9$, that is, $y = \frac{1}{4}(x + 9)$. Substituting into the curve's equation:

$$\begin{aligned} \frac{(x+9)^3}{64} &= x^3 + 9 \\ x^3 + 27x^2 + 243x + 729 &= 64x^3 + 576 \\ 0 &= 63x^3 - 27x^2 - 243x - 153. \end{aligned}$$

Every coefficient is divisible by 9, and dividing through gives

$$7x^3 - 3x^2 - 27x - 17 = 0,$$

which is the cubic we were asked for.

Finding p, q, r, s

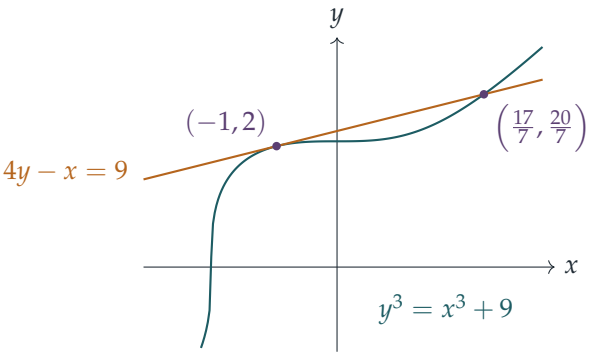
Ask yourself We already know one root of this cubic. Why should it be a *double* root?

The line touches the curve at $x = -1$, so $x = -1$ is a repeated root: near the point of contact the line and the curve agree to first order, so their difference vanishes to second order there. Hence $(x + 1)^2 = x^2 + 2x + 1$ is a factor. Comparing the x^3 and constant terms, the other factor must be $7x - 17$, and we can check this:

$$(x + 1)^2(7x - 17) = 7x^3 + 14x^2 + 7x - 17x^2 - 34x - 17 = 7x^3 - 3x^2 - 27x - 17.$$

So the tangent meets the curve again where $x = \frac{17}{7}$, and there

$$y = \tfrac{1}{4}\left(\tfrac{17}{7} + 9\right) = \tfrac{1}{4} \cdot \tfrac{80}{7} = \tfrac{20}{7}.$$



Since this point lies on $y^3 = x^3 + 9$,

$$\left(\frac{20}{7}\right)^3 = \left(\frac{17}{7}\right)^3 + 9.$$

Multiplying by 7^3 gives $20^3 = 17^3 + 9 \cdot 7^3$. The last term is not a cube, but $9 = 1 + 8 = 1^3 + 2^3$, so

$$9 \cdot 7^3 = 7^3 + 2^3 \cdot 7^3 = 7^3 + 14^3.$$

(This is where the original form $a^3 + b^3$ with $a = 1$, $b = 2$ comes back.) Therefore

$$20^3 = 17^3 + 14^3 + 7^3,$$

and we can take $p = 20$, $q = 17$, $r = 14$, $s = 7$ (the last three in any order).

Checks

Directly: $17^3 + 14^3 + 7^3 = 4913 + 2744 + 343 = 8000 = 20^3$. Also, the three roots of the cubic are $-1, -1, \frac{17}{7}$, whose sum is $\frac{3}{7}$ and product is $\frac{17}{7}$, which matches $3/7$ and $17/7$ read off from the coefficients. A numerical derivative of $y = \sqrt[3]{x^3 + a^3 + b^3}$ at $x = -a$ agrees with a^2/b^2.

Common slips

- Not using the double root: trying to solve the cubic from scratch instead of dividing out $(x + 1)^2$.
- Stopping at $20^3 = 17^3 + 3087$. We need positive integers whose cubes add up, and $3087 = 9 \cdot 7^3$ has to be split as $7^3 + 14^3$.

Beyond STEP

This is the classical *tangent method* for curves of degree three. A line meets a cubic curve in three points, counted with multiplicity, so two rational points (or one counted twice, as here) produce a third. On elliptic curves this chord-and-tangent construction becomes a group law, the starting point of a large area of modern number theory. Davenport's *The Higher Arithmetic* discusses Diophantine equations of this kind, and Silverman and Tate's *Rational Points on Elliptic Curves* develops the geometry in full.

WORKED SOLUTION

STEP I 2009, Question 3

Pure • Counting Real Roots

STEP I 2009, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q3

The key idea

Get rid of the root by raising both sides to the right power, but keep track of what that loses. A square root is never negative, so squaring is only reversible if we also insist that $x \geq 0$; a real cube root has no such restriction, because every real number has exactly one real cube root. After that, each count comes down to the roots of a quadratic, and the signs of those roots can be read off from their sum and product.

Solution

Part (i): the square root equation

Here $(a-x)^{\frac{1}{2}}$ means the non-negative square root, so any solution has $x \geq 0$, and squaring gives $x^2 = a - x$, that is $x^2 + x - a = 0$. Conversely, if $x \geq 0$ and $x^2 + x - a = 0$, then $a - x = x^2 \geq 0$ and $(a-x)^{\frac{1}{2}} = \sqrt{x^2} = |x| = x$. So

$$x = (a-x)^{\frac{1}{2}} \iff x^2 + x - a = 0 \text{ and } x \geq 0,$$

and we have to count the *non-negative* roots of the quadratic $x^2 + x - a = 0$. If its roots are α, β (when real), then

$$\alpha + \beta = -1, \qquad \alpha\beta = -a.$$

Case $a > 0$. The discriminant $1 + 4a$ is positive, so there are two real roots, and their product $-a$ is negative. So one root is positive and the other negative: exactly one solution.

Case $a = 0$. The quadratic is $x(x+1) = 0$, with roots 0 and -1. Only $x = 0$ is non-negative: exactly one solution.

Case $a < 0$. If $a < -\frac{1}{4}$ the discriminant is negative and there are no real roots at all. If $-\frac{1}{4} \leq a < 0$ the roots are real, their product $-a$ is positive, so they have the same sign, and their sum is -1, so both are negative. Either way there is no non-negative root, hence no solution.

So the equation has one real solution when $a \geq 0$ and none when $a < 0$, as required.

Part (i): the cube root equation

Every real number t has exactly one real cube root, and $x = t^{\frac{1}{3}} \iff x^3 = t$, with no sign condition. So the equation is equivalent to

$$x^3 - (1+a)x + a = 0.$$

The coefficients sum to $1 - (1+a) + a = 0$, so $x = 1$ is always a root, and dividing out,

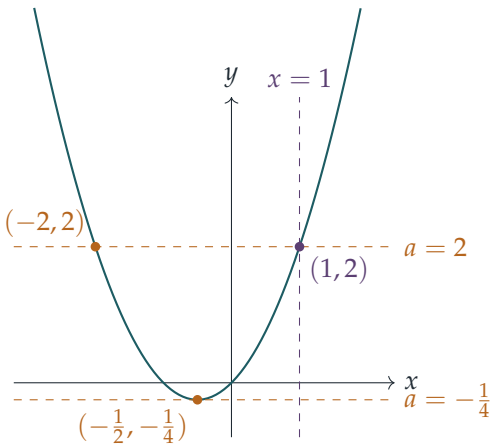
$$x^3 - (1+a)x + a = (x-1)(x^2 + x - a).$$

The same quadratic has appeared, and the solutions are $x = 1$ together with the real roots of $x^2 + x - a = 0$, now with no sign restriction.

> **Ask yourself** Could a root of the quadratic coincide with $x = 1$, or with each other?

Both happen, and they are exactly the cases that reduce the count. The quadratic has a repeated root when $1 + 4a = 0$, that is $a = -\frac{1}{4}$, the root being $-\frac{1}{2}$. And $x = 1$ is a root of the quadratic when $1 + 1 - a = 0$, that is $a = 2$; then $x^2 + x - 2 = (x-1)(x+2)$, so the roots are 1 and -2.

Writing the quadratic as $a = x^2 + x$, its real roots are where the horizontal line at height a meets the parabola $y = x^2 + x$, whose lowest point is $(-\frac{1}{2}, -\frac{1}{4})$. The extra solution $x = 1$ corresponds to the point $(1, 2)$ on the parabola.



Putting the cases together, the number of distinct real solutions is:

- $a < -\frac{1}{4}$: the quadratic has no real roots, so **one** solution, $x = 1$;
- $a = -\frac{1}{4}$: the solutions are 1 and $-\frac{1}{2}$, so **two**;
- $-\frac{1}{4} < a < 2$ or $a > 2$: two distinct roots of the quadratic, neither equal to 1, plus $x = 1$, so **three**;

- $a = 2$: the solutions are 1 and -2, so **two**.

Part (ii)

Exactly as in part (i), the square root is non-negative, so

$$x = (b+x)^{\frac{1}{2}} \iff x^2 - x - b = 0 \text{ and } x \geq 0,$$

(the converse holds because $x \geq 0$ and $b + x = x^2$ give $(b+x)^{\frac{1}{2}} = |x| = x$). Now the roots of $x^2 - x - b = 0$ have sum 1 and product $-b$, and the discriminant is $1 + 4b$.

Case $b < -\frac{1}{4}$. No real roots, so **no** solutions.

Case $b = -\frac{1}{4}$. The quadratic is $(x - \frac{1}{2})^2 = 0$, and $x = \frac{1}{2} \geq 0$, so **one** solution.

Case $-\frac{1}{4} < b < 0$. Two distinct real roots with positive product $-b$, so they have the same sign, and positive sum 1, so both are positive: **two** solutions.

Case $b = 0$. The quadratic is $x(x-1) = 0$, and both 0 and 1 are non-negative: **two** solutions. (Indeed $0 = \sqrt{0}$ and $1 = \sqrt{1}$.)

Case $b > 0$. Two real roots with negative product, so one is positive and one negative: **one** solution.

In summary: no solutions for $b < -\frac{1}{4}$; one for $b = -\frac{1}{4}$; two for $-\frac{1}{4} < b \leq 0$; one for $b > 0$.

Checks

- Graphically, part (ii) asks where the line $y = x$ meets the half-parabola $y = \sqrt{b+x}$, which starts at $(-b, 0)$. For $b = -\frac{1}{5}$ the solutions are $\frac{1}{2} \pm \frac{1}{2\sqrt{5}}$, about 0.276 and 0.724, and both satisfy $x = \sqrt{x - 0.2}$ to many decimal places.
- In the cube root equation with $a = 2$, both $x = 1$ and $x = -2$ work: $(3 \cdot (-2) - 2)^{\frac{1}{3}} = (-8)^{\frac{1}{3}} = -2$.
- A computer counted the roots of each equation on a fine grid for a range of values of a and b either side of every boundary ($-\frac{1}{4}$, 0, 2), and the counts agreed with the case analysis above.

Common slips

- Squaring and forgetting that $x \geq 0$: this would give two solutions in part (i) for every $a > -\frac{1}{4}$, which is wrong.
- Imposing $x \geq 0$ on the cube root equation as well. Cube roots of negative numbers are perfectly good real numbers.
- Missing the case $a = 2$, where the root $x = 1$ of the linear factor is also a root of the quadratic, so two of the three solutions merge.

Beyond STEP

The key point is that squaring is not a reversible operation on $\mathbb{R}$, while cubing is: $x \mapsto x^3$ is a bijection of $\mathbb{R}$, but $x \mapsto x^2$ is not, so "the" square root is only defined once we choose the non-negative branch. Thinking of equations like $x = f(x)$ as fixed points, and counting them as a parameter varies, is the start of bifurcation theory, where solutions are born in pairs (as at $a = -\frac{1}{4}$) or exchange places (as at $a = 2$). For a careful account of functions, inverses and continuity, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 2009, Question 4

Pure • Cosine Rule

STEP I 2009, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q4

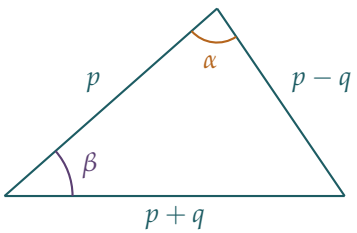
The key idea

In a triangle the largest angle is opposite the longest side and the smallest angle is opposite the shortest side. So the cosine rule gives $\cos\alpha$ and $\cos\beta$ directly in terms of p and q, and the identity is just algebra. For the second part, $\alpha = 2\beta$ lets us write $\cos\alpha$ in terms of $\cos\beta$ with a double-angle formula. The identity then becomes an equation in $\cos\beta$ alone.

Solution

The identity

Since $p - q < p < p + q$, the largest angle α is opposite the side $p + q$ and the smallest angle β is opposite the side $p - q$.



The cosine rule for α gives

$$\cos\alpha = \frac{p^2 + (p-q)^2 - (p+q)^2}{2p(p-q)} = \frac{p^2 - 4pq}{2p(p-q)} = \frac{p-4q}{2(p-q)},$$

and for β it gives

$$\cos\beta = \frac{p^2 + (p+q)^2 - (p-q)^2}{2p(p+q)} = \frac{p^2 + 4pq}{2p(p+q)} = \frac{p+4q}{2(p+q)}.$$

We cancelled p, which is allowed because $p > 0$. Then

$$1 - \cos\alpha = \frac{2(p-q) - (p-4q)}{2(p-q)} = \frac{p+2q}{2(p-q)},$$
$$1 - \cos\beta = \frac{2(p+q) - (p+4q)}{2(p+q)} = \frac{p-2q}{2(p+q)},$$

so

$$4(1-\cos\alpha)(1-\cos\beta) = \frac{(p+2q)(p-2q)}{(p-q)(p+q)} = \frac{p^2-4q^2}{p^2-q^2}.$$

On the other side,

$$\begin{aligned}\cos\alpha + \cos\beta &= \frac{(p-4q)(p+q) + (p+4q)(p-q)}{2(p^2-q^2)} \\ &= \frac{(p^2-3pq-4q^2) + (p^2+3pq-4q^2)}{2(p^2-q^2)} = \frac{p^2-4q^2}{p^2-q^2}.\end{aligned}$$

The two expressions agree, so $4(1-\cos\alpha)(1-\cos\beta) = \cos\alpha + \cos\beta$, as required.

(For the triangle to exist we need $(p-q) + p > p + q$, that is, $p > 2q$. Then both sides of the identity are positive, but the algebra above doesn't depend on this.)

The case $\alpha = 2\beta$

Write $c = \cos\beta$. Then $\cos\alpha = \cos 2\beta = 2c^2 - 1$, so $1 - \cos\alpha = 2(1-c^2) = 2(1-c)(1+c)$. Substituting into the identity gives

$$8(1-c)^2(1+c) = 2c^2 + c - 1 = (2c-1)(c+1).$$

Ask yourself Can we divide both sides by $1 + c$?

Yes. β is an angle of a triangle, so $0 < \beta < \pi$ and $c = \cos\beta > -1$, which means $1 + c \neq 0$. Dividing by it gives

$$8(1-c)^2 = 2c - 1 \quad\Longrightarrow\quad 8c^2 - 18c + 9 = 0 \quad\Longrightarrow\quad (4c-3)(2c-3) = 0.$$

So $c = \frac{3}{4}$ or $c = \frac{3}{2}$. A cosine can't exceed 1, so we reject $\frac{3}{2}$. Therefore $\cos\beta = \frac{3}{4}$, as required.

The ratio of the sides

Using the formula for $\cos\beta$ from the first part,

$$\frac{p+4q}{2(p+q)} = \frac{3}{4} \quad\Longrightarrow\quad 4p + 16q = 6p + 6q \quad\Longrightarrow\quad p = 5q.$$

This satisfies $p > 2q$, so the triangle exists. Its sides are $p - q = 4q$, $p = 5q$ and $p + q = 6q$, so the ratio of the sides is

$$4 : 5 : 6.$$

Checks

In the $4,5,6$ triangle, the cosine rule gives $\cos\beta = \frac{25+36-16}{60} = \frac{3}{4}$ and $\cos\alpha = \frac{16+25-36}{40} = \frac{1}{8}$. Also $2\cdot\left(\frac{3}{4}\right)^2 - 1 = \frac{1}{8}$, which confirms that $\alpha = 2\beta$ (both angles lie in $(0,\pi)$, where cosine is one-to-one). Numerically, $\beta \approx 0.7227$ and $\alpha \approx 1.4455$ radians. In this triangle the identity reads $4\cdot\frac{7}{8}\cdot\frac{1}{4} = \frac{7}{8} = \frac{1}{8}+\frac{3}{4}$. We also tested the identity on 10^5 random triangles with $p > 2q$, and the two sides agreed to within 3×10^{-15}.

Common slips

- Matching the angles to the wrong sides. The largest angle is opposite the *longest* side, $p+q$.
- Cancelling the factor $1+c$ without saying why it is non-zero, or keeping the root $c = \frac{3}{2}$.
- Giving the answer as $p = 5q$ instead of the ratio of the three sides, $4:5:6$.

Beyond STEP

Up to scaling, the $4,5,6$ triangle is the only triangle whose sides are in arithmetic progression and whose largest angle is twice its smallest. Questions like "which triangles have integer sides and angles in a given ratio?" lead quickly to Diophantine equations and to parametrising rational points on curves, much as the Pythagorean triples do. Davenport's *The Higher Arithmetic* is a good place to see these ideas develop, and Siklos's *Advanced Problems in Mathematics* has more STEP-style geometry and trigonometry.

WORKED SOLUTION

STEP I 2009, Question 5

Pure • Constrained Optimisation

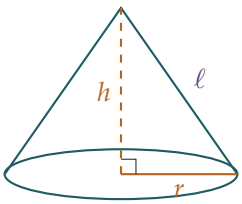
STEP I 2009, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q5

The key idea

There are three lengths but only two are free, since r, h and ℓ are tied together by Pythagoras: $\ell^2 = r^2 + h^2$. Holding one quantity fixed takes away one more freedom, so in each part we write the quantity to be made stationary as a function of r alone and differentiate. Square roots make the algebra messy, so it pays to work with V^2 or A^2 instead: these are stationary exactly where V or A is.



Solution

Throughout, r, h and ℓ are positive and, from the right-angled triangle in the cross-section,

$$\ell^2 = r^2 + h^2.$$

Since $V > 0$ and $A > 0$, the derivative of V^2 is $2V\dfrac{\mathrm{d}V}{\mathrm{d}r}$, which vanishes exactly when $\dfrac{\mathrm{d}V}{\mathrm{d}r}$ does; similarly for A. So we may find stationary points of the squares instead.

Part (i)

With A fixed, $\ell = \dfrac{A}{\pi r}$, so

$$h^2 = \ell^2 - r^2 = \frac{A^2}{\pi^2 r^2} - r^2.$$

(For a genuine cone we need $h > 0$, that is $A > \pi r^2$.) Then

$$V^2 = \tfrac{1}{9}\pi^2 r^4 h^2 = \tfrac{1}{9}\pi^2 r^4 \left(\frac{A^2}{\pi^2 r^2} - r^2\right)$$
$$= \tfrac{1}{9}\left(A^2 r^2 - \pi^2 r^6\right).$$

Differentiating with respect to r, keeping A constant,

$$\frac{\mathrm{d}(V^2)}{\mathrm{d}r} = \tfrac{1}{9}\left(2A^2 r - 6\pi^2 r^5\right) = \tfrac{2}{9} r\left(A^2 - 3\pi^2 r^4\right).$$

Since $r > 0$, this is zero exactly when

$$A^2 = 3\pi^2 r^4,$$

which is the first result. Taking the positive square root, $A = \sqrt{3}\,\pi r^2$, and so

$$\ell = \frac{A}{\pi r} = \frac{\sqrt{3}\,\pi r^2}{\pi r} = \sqrt{3}\, r,$$

as required. (Note that $A = \sqrt{3}\,\pi r^2 > \pi r^2$, so this really is a cone.)

The derivative of V^2 changes from positive to negative as r increases through this value, so the stationary value is a maximum: the cone of greatest volume for a given curved surface area. From $\ell^2 = r^2 + h^2$ we also get $h^2 = 3r^2 - r^2 = 2r^2$, so $h = \sqrt{2}\, r$.

Part (ii)

Now V is fixed, so $h = \dfrac{3V}{\pi r^2}$, and

$$A^2 = \pi^2 r^2 \ell^2 = \pi^2 r^2 (r^2 + h^2) = \pi^2 r^4 + \pi^2 r^2 \cdot \frac{9V^2}{\pi^2 r^4}$$
$$= \pi^2 r^4 + \frac{9V^2}{r^2}.$$

Differentiating with respect to r, keeping V constant,

$$\frac{\mathrm{d}(A^2)}{\mathrm{d}r} = 4\pi^2 r^3 - \frac{18V^2}{r^3},$$

which is zero exactly when

$$V^2 = \tfrac{2}{9}\pi^2 r^6, \qquad \text{so} \qquad V = \tfrac{\sqrt{2}}{3}\,\pi r^3$$

(taking the positive root, as $V > 0$).

Ask yourself We have found V in terms of r, but the question asks for h. How do we get there?

Use the volume formula once more: $V = \frac{1}{3}\pi r^2 h$. Setting this equal to $\frac{\sqrt{2}}{3}\pi r^3$ and cancelling $\frac{1}{3}\pi r^2$ gives

$$h = \sqrt{2}\,r.$$

This time the derivative goes from negative to positive (for small r the term $9V^2/r^2$ is huge, for large r the term $\pi^2 r^4$ is), so this is the cone of least curved surface area for a given volume.

? **Ask yourself** Is it a coincidence that both parts give the same shape, $h = \sqrt{2}\,r$?

No. If a cone maximises volume for its area, then it also minimises area for its volume: any cone with the same volume and smaller area could be scaled up to the original area, and it would then have a larger volume, which is impossible. Since the problem only fixes the shape up to scale, both parts must pick out the same ratio $h : r$.

Checks

- Take $r = 1$ in part (i): then $A = \sqrt{3}\,\pi$, $\ell = \sqrt{3}$, $h = \sqrt{2}$, and $\pi r \ell = \sqrt{3}\,\pi$ as it should be. Nudging r to 0.99 or 1.01 with A held at $\sqrt{3}\,\pi$ gives a slightly smaller volume.
- Numerically maximising V for random fixed A, and minimising A for random fixed V, always gave $h/r \approx 1.41421$, which is $\sqrt{2}$.
- Dimensions: A^2 and $\pi^2 r^4$ are both (length)^4; V^2 and $\pi^2 r^6$ are both (length)^6.

Common slips

- Forgetting that ℓ depends on r and h: treating ℓ as a constant in part (i), or h as a constant in part (ii), when in fact only A (or V) is fixed.
- Differentiating V directly with the square root still inside; it works, but it is easy to make an error. Squaring first avoids the chain rule entirely.

Beyond STEP

Finding the extreme value of one quantity while another is held fixed is a *constrained optimisation* problem. At university you meet the method of Lagrange multipliers, which handles the constraint without eliminating a variable, and which explains the "duality" between the two parts here. For a clear first treatment of multivariable calculus and Lagrange multipliers, try Stewart, *Calculus*; Siklos, *Advanced Problems in Mathematics*, has more STEP-style optimisation practice.

WORKED SOLUTION

STEP I 2009, Question 6

Pure • Integration by Substitution

STEP I 2009, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q6

The key idea

The limits $1/m$ and m are swapped by $x \mapsto 1/x$, so that substitution is the natural one to try: it maps the interval to itself and trades a power x^{-n} for a power u^{n-1}. Part (ii) then lets us turn the awkward pieces of part (iii) into copies of the friendly integral from part (i). In (iii)(b) the interval is no longer symmetric, but the same substitution sends $[1,2]$ to $[\frac{1}{2},1]$, and the two halves glue back together.

Solution

Part (i)

Dividing x^2 by $x+1$,

$$\frac{x^2}{x+1} = x - 1 + \frac{1}{x+1},$$

so, since $x+1 > 0$ on the interval (as $m > 0$),

$$\int_{1/m}^{m} \frac{x^2}{x+1}\,\mathrm{d}x = \left[\tfrac{1}{2}x^2 - x + \ln(x+1)\right]_{1/m}^{m}.$$

The logarithmic part gives

$$\ln(m+1) - \ln\left(\frac{1}{m}+1\right) = \ln\frac{m+1}{(m+1)/m} = \ln m.$$

The polynomial part gives

$$\begin{aligned}\tfrac{1}{2}\left(m^2 - \frac{1}{m^2}\right) - \left(m - \frac{1}{m}\right) &= \left(m - \frac{1}{m}\right)\left[\tfrac{1}{2}\left(m + \frac{1}{m}\right) - 1\right] \\ &= \frac{m^2-1}{m}\cdot\frac{m^2-2m+1}{2m} \\ &= \frac{(m-1)(m+1)(m-1)^2}{2m^2} = \frac{(m-1)^3(m+1)}{2m^2}.\end{aligned}$$

Adding the two parts gives the required result.

Part (ii)

Put $x = 1/u$, so $\mathrm{d}x = -\mathrm{d}u/u^2$. When $x = 1/m$, $u = m$; when $x = m$, $u = 1/m$. Also

$$x^n(x+1) = \frac{1}{u^n} \cdot \frac{1+u}{u} = \frac{u+1}{u^{n+1}}.$$

Hence

$$\begin{aligned}\int_{1/m}^{m} \frac{1}{x^n(x+1)}\,\mathrm{d}x &= \int_{m}^{1/m} \frac{u^{n+1}}{u+1} \cdot \left(-\frac{1}{u^2}\right)\mathrm{d}u \\ &= \int_{1/m}^{m} \frac{u^{n-1}}{u+1}\,\mathrm{d}u,\end{aligned}$$

where the minus sign was used to swap the limits back. This is the required result.

Part (iii)

(a) Split the integrand:

$$\frac{x^5+3}{x^3(x+1)} = \frac{x^2}{x+1} + \frac{3}{x^3(x+1)}.$$

The interval is $[\frac{1}{2}, 2]$, which is part (ii)'s interval with $m = 2$. Using part (ii) with $n = 3$,

$$\int_{1/2}^{2} \frac{3}{x^3(x+1)}\,\mathrm{d}x = 3\int_{1/2}^{2} \frac{u^2}{u+1}\,\mathrm{d}u,$$

which is three times the first piece. By part (i) with $m = 2$,

$$\int_{1/2}^{2} \frac{x^2}{x+1}\,\mathrm{d}x = \frac{1^3 \cdot 3}{2 \cdot 4} + \ln 2 = \frac{3}{8} + \ln 2.$$

So the integral is $4\left(\frac{3}{8} + \ln 2\right)$, that is,

$$\int_{1/2}^{2} \frac{x^5+3}{x^3(x+1)}\,\mathrm{d}x = \frac{3}{2} + 4\ln 2.$$

(b) Now

$$\frac{x^5+x^3+1}{x^3(x+1)} = \frac{x^2}{x+1} + \frac{1}{x+1} + \frac{1}{x^3(x+1)}.$$

Ask yourself The interval is now $[1,2]$, not of the form $[1/m, m]$. Can part (ii) still help?

Yes: the substitution $x = 1/u$ works on any interval of positive numbers; it just moves the limits. Exactly the same computation as in part (ii), with $x = 1$ giving $u = 1$ and $x = 2$ giving $u = \frac{1}{2}$, shows

$$\int_{1}^{2} \frac{1}{x^3(x+1)}\,\mathrm{d}x = \int_{1/2}^{1} \frac{u^2}{u+1}\,\mathrm{d}u.$$

This has the same integrand as the first piece, on the neighbouring interval, so the two together make the integral of part (i) with $m = 2$:

$$\int_1^2 \frac{x^2}{x+1}\,\mathrm{d}x + \int_{1/2}^1 \frac{x^2}{x+1}\,\mathrm{d}x = \int_{1/2}^2 \frac{x^2}{x+1}\,\mathrm{d}x = \frac{3}{8} + \ln 2.$$

The remaining piece is

$$\int_1^2 \frac{1}{x+1}\,\mathrm{d}x = \ln 3 - \ln 2.$$

Adding,

$$\int_1^2 \frac{x^5+x^3+1}{x^3(x+1)}\,\mathrm{d}x = \frac{3}{8} + \ln 2 + \ln 3 - \ln 2 = \frac{3}{8} + \ln 3.$$

Checks

- In part (i), $m = 1$ gives an integral over a single point, and indeed the right-hand side is $0 + \ln 1 = 0$. Replacing m by $1/m$ should negate the integral (the limits swap), and the formula does change sign.
- Simpson's rule gives 4.272589 for (iii)(a), matching $\frac{3}{2} + 4\ln 2 \approx 4.272589$, and 1.473612 for (iii)(b), matching $\frac{3}{8} + \ln 3 \approx 1.473612$.
- Sanity check for (b): on $[1,2]$ the integrand dips from 1.5 to about 1.37 and then rises to about 1.71, so an answer near 1.47 is reasonable.

Common slips

- Forgetting the minus sign from $\mathrm{d}x = -\mathrm{d}u/u^2$, or forgetting to swap the limits; you must do both, and they cancel.
- In (iii)(b), applying part (ii) as if the interval were $[\frac{1}{2}, 2]$. The substitution moves $[1,2]$ to $[\frac{1}{2}, 1]$, and that is exactly what makes the pieces fit together.

Beyond STEP

The map $x \mapsto 1/x$ is a symmetry of the interval $[1/m, m]$, and using symmetries of the domain to relate integrals is a recurring trick in analysis. For example, it is how one shows $\int_0^\infty \frac{\ln x}{1+x^2}\,\mathrm{d}x = 0$. In complex analysis the same inversion $z \mapsto 1/z$ swaps the inside and outside of the unit circle and becomes a basic tool. Needham's *Visual Complex Analysis* has a beautiful geometric treatment of inversion.

WORKED SOLUTION

STEP I 2009, Question 7

Pure • Integration by Parts

STEP I 2009, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q7

The key idea

The opening integral is the classic "integrate by parts twice and the integral comes back" trick: after two steps the original integral reappears on the right-hand side, and we solve for it like an unknown in an equation. Once we have a formula for $\int_0^{2\pi} \mathrm{e}^x \cos mx \,\mathrm{d}x$ for *every* integer m, the later parts become bookkeeping: products of sines and cosines are turned into sums of single cosines, each of which we can integrate at once.

Solution

The general integral

Write $I_m = \int_0^{2\pi} \mathrm{e}^x \cos mx \,\mathrm{d}x$, and let $E = \mathrm{e}^{2\pi} - 1$ for short. Integrating by parts with $u = \cos mx$ and $\mathrm{d}v = \mathrm{e}^x \,\mathrm{d}x$,

$$I_m = \Big[\mathrm{e}^x \cos mx\Big]_0^{2\pi} + m \int_0^{2\pi} \mathrm{e}^x \sin mx \,\mathrm{d}x.$$

Since m is an integer, $\cos 2m\pi = 1$, so the bracket is $\mathrm{e}^{2\pi} - 1 = E$. Integrating by parts again, now with $u = \sin mx$,

$$\int_0^{2\pi} \mathrm{e}^x \sin mx \,\mathrm{d}x = \Big[\mathrm{e}^x \sin mx\Big]_0^{2\pi} - m \int_0^{2\pi} \mathrm{e}^x \cos mx \,\mathrm{d}x = 0 - mI_m,$$

because $\sin 2m\pi = \sin 0 = 0$, again using that m is an integer. Putting the two together,

$$I_m = E - m^2 I_m, \qquad \text{so} \qquad (m^2+1)I_m = E.$$

Ask yourself We have gone round in a circle and got I_m back. Is that a dead end?

No: that is exactly what we want. The equation $I_m = E - m^2 I_m$ is a linear equation for

the unknown number I_m, and since $m^2+1 \neq 0$ we can divide by it:

$$\int_0^{2\pi} \mathrm{e}^x \cos mx \,\mathrm{d}x = \frac{1}{m^2+1}\left(\mathrm{e}^{2\pi}-1\right),$$

as required. (The case $m=0$ is included: it just says $\int_0^{2\pi} \mathrm{e}^x \,\mathrm{d}x = E$.) Notice that the integer condition was used to make the boundary terms simple; for non-integer m the formula is false.

Part (i)

Expanding with the compound-angle formulae, the $\sin A \sin B$ terms cancel:

$$\cos(A+B)+\cos(A-B) = 2\cos A \cos B.$$

With $A = 6x$ and $B = x$ this gives $\cos x \cos 6x = \frac{1}{2}(\cos 7x + \cos 5x)$. Hence, by the general result with $m=7$ and $m=5$,

$$\begin{aligned}\int_0^{2\pi} \mathrm{e}^x \cos x \cos 6x \,\mathrm{d}x &= \tfrac{1}{2}\left(I_7 + I_5\right) = \tfrac{1}{2}\left(\frac{1}{50}+\frac{1}{26}\right)E \\ &= \tfrac{1}{2}\cdot\frac{26+50}{1300}\,E = \frac{76}{2600}\,E = \frac{19}{650}\left(\mathrm{e}^{2\pi}-1\right),\end{aligned}$$

which is the required result.

Part (ii)

We need two product-to-sum steps. Subtracting rather than adding the compound-angle formulae gives $\cos(A-B)-\cos(A+B) = 2\sin A \sin B$, so with $A = 4x$, $B = 2x$,

$$\sin 2x \sin 4x = \tfrac{1}{2}(\cos 2x - \cos 6x).$$

Multiplying by $\cos x$ and using part (i)'s identity on each product,

$$\begin{aligned}\sin 2x \sin 4x \cos x &= \tfrac{1}{2}\cos 2x \cos x - \tfrac{1}{2}\cos 6x \cos x \\ &= \tfrac{1}{4}(\cos 3x + \cos x) - \tfrac{1}{4}(\cos 7x + \cos 5x).\end{aligned}$$

Now each term is covered by the general result, with $m = 3, 1, 7, 5$:

$$\begin{aligned}\int_0^{2\pi} \mathrm{e}^x \sin 2x \sin 4x \cos x \,\mathrm{d}x &= \tfrac{1}{4}\left(I_3 + I_1 - I_7 - I_5\right) \\ &= \tfrac{1}{4}\left(\frac{1}{10}+\frac{1}{2}-\frac{1}{50}-\frac{1}{26}\right)E \\ &= \tfrac{1}{4}\cdot\frac{65+325-13-25}{650}\,E = \tfrac{1}{4}\cdot\frac{352}{650}\,E.\end{aligned}$$

So

$$\int_0^{2\pi} \mathrm{e}^x \sin 2x \sin 4x \cos x \,\mathrm{d}x = \frac{44}{325}\left(\mathrm{e}^{2\pi}-1\right).$$

Checks

- Differentiating $\dfrac{e^x(\cos mx + m\sin mx)}{m^2+1}$ gives $e^x\cos mx$, so this is an antiderivative; evaluating it at 2π and 0 reproduces the general result.
- Put $x = 0$ in the part (ii) identity: the left side is 0 and the right side is $\frac{1}{4}(1+1-1-1) = 0$. Good.
- Numerically (Simpson's rule), the part (i) integral is $15.6236\ldots$ and the part (ii) integral is $72.3619\ldots$, matching $\frac{19}{650}E$ and $\frac{44}{325}E$ with $E = e^{2\pi} - 1 \approx 534.49$.

Common slips

- Sign errors in the second integration by parts: the derivative of $\sin mx$ is $+m\cos mx$, which with the minus sign from the parts formula gives $-m^2 I_m$.
- Forgetting the factor $\frac{1}{2}$ in each product-to-sum step, so that part (ii) picks up $\frac{1}{2}$ instead of $\frac{1}{4}$.

Beyond STEP

A slicker route is to write $\cos mx = \operatorname{Re} e^{imx}$, so that $I_m = \operatorname{Re}\int_0^{2\pi} e^{(1+im)x}\,dx = \operatorname{Re}\dfrac{E}{1+im} = \dfrac{E}{1+m^2}$, with no integration by parts at all. Breaking a function into its components $\cos mx$, as we did in the later parts, is the start of Fourier series, a central tool in first-year analysis and differential equations. Boyce and DiPrima, *Elementary Differential Equations*, has a readable introduction, and Needham, *Visual Complex Analysis*, shows why the exponential viewpoint works so well.

WORKED SOLUTION

STEP I 2009, Question 8

Pure • Circles and Tangent Lines

STEP I 2009, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q8

The key idea

A circle touches a line exactly when the distance from its centre to the line equals its radius. The circles C all have their centres on the line through the origin that bisects the angle between the x-axis and $3y = 4x$, so each of them touches both lines. That makes C the complete family of circles sitting in that corner, and the incircle in part (ii) must be one of them: we only need to pick the value of t that also makes the circle touch the third side.

Solution

Part (i)

The circle C has centre $P = (2t, t)$ and radius t. Since $t > 0$, the centre lies above the x-axis at a distance t from it, which equals the radius. So C touches $y = 0$, at the point $(2t, 0)$.

The line OP has gradient $\frac{t}{2t} = \frac{1}{2}$, so $\tan\alpha = \frac{1}{2}$ and

$$\tan 2\alpha = \frac{2\tan\alpha}{1 - \tan^2\alpha} = \frac{1}{1 - \frac{1}{4}} = \frac{4}{3},$$

as required.

Now let β be the acute angle between the x-axis and the line $3y = 4x$, whose gradient is $\frac{4}{3}$. Then $\tan\beta = \frac{4}{3} = \tan 2\alpha$. Since $0 < \alpha < \frac{\pi}{4}$ (because $\tan\alpha = \frac{1}{2} < 1$), the angle 2α is also acute, and tan is one-to-one on acute angles, so $\beta = 2\alpha$.

? **Ask yourself** Why does $\beta = 2\alpha$ tell us that C touches $3y = 4x$?

The line OP makes angle α with the x-axis and angle $\beta - \alpha = \alpha$ with the line $3y = 4x$, so OP bisects the angle between the two lines. Dropping perpendiculars from P to each line gives two right-angled triangles with the same hypotenuse OP and the same

angle α at O, so the two perpendicular distances are equal, both being $OP\sin\alpha$. The distance to the x-axis is t, so the distance from P to $3y = 4x$ is also t, the radius. Hence C touches $3y = 4x$.

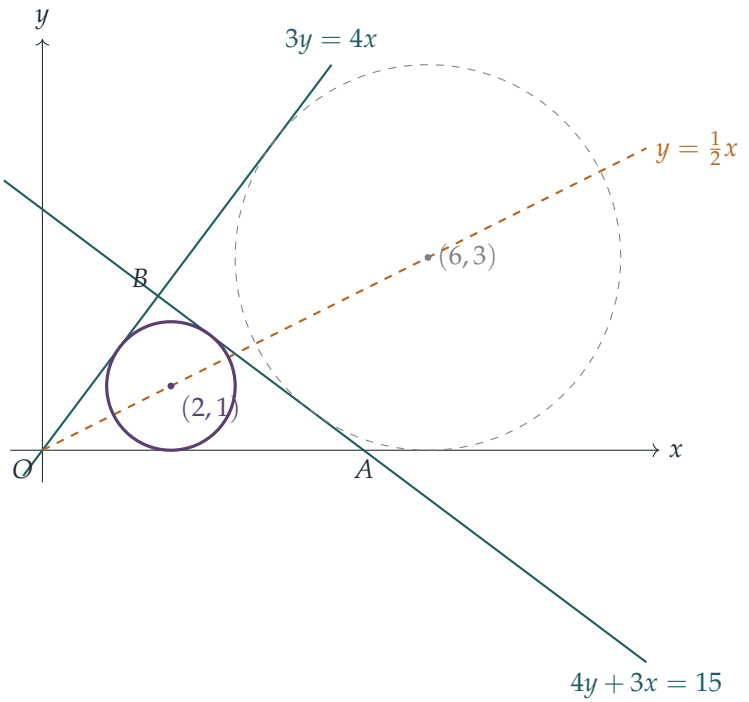
(As a cross-check, the perpendicular distance formula gives the distance from $(2t, t)$ to $4x - 3y = 0$ as $\frac{|8t-3t|}{5} = t$.)

Part (ii)

First we find the triangle. The lines $y = 0$ and $3y = 4x$ meet at $O = (0,0)$; $y = 0$ and $4y + 3x = 15$ meet at $A = (5,0)$. For the third vertex, substitute $y = \frac{4}{3}x$ into $4y + 3x = 15$:

$$\tfrac{16}{3}x + 3x = 15 \quad \Longrightarrow \quad x = \tfrac{9}{5}, \qquad y = \tfrac{12}{5},$$

so $B = \left(\frac{9}{5}, \frac{12}{5}\right)$. The gradients of $3y = 4x$ and $4y + 3x = 15$ are $\frac{4}{3}$ and $-\frac{3}{4}$, whose product is -1, so the triangle has a right angle at B. Its sides are $OA = 5$, $OB = 3$ and $AB = 4$.



The incircle lies inside the triangle, so near O it sits in the acute angle between the positive x-axis and the line $3y = 4x$, touching both lines. Its centre is equidistant from the two lines, so it lies on the bisector of that angle, which by part (i) is the line $y = \frac{1}{2}x$ (the line OP). If the centre is $(2t, t)$ with $t > 0$, its distance from $y = 0$ is t, which must be the radius. So the incircle is one of the circles C from part (i).

It remains to make C touch the third side $3x + 4y - 15 = 0$. The distance from $(2t, t)$ to

this line must equal t:

$$\frac{|6t + 4t - 15|}{5} = t \quad \Longleftrightarrow \quad |10t - 15| = 5t.$$

Either $10t - 15 = 5t$, giving $t = 3$, or $15 - 10t = 5t$, giving $t = 1$.

Both circles touch all three lines, but only one lies inside the triangle. For $t = 3$ the centre $(6, 3)$ has $3x + 4y = 30 > 15$, so it is on the far side of AB from O (which has $3x + 4y = 0 < 15$): this is an *excircle*, the dashed circle in the figure. For $t = 1$ the centre $(2, 1)$ has $3x + 4y = 10 < 15$, and it is above $y = 0$ and below $3y = 4x$, so it is inside the triangle. A circle with its centre inside the triangle that touches all three sides lies inside the triangle, so the incircle has centre $(2, 1)$ and radius 1:

$$(x - 2)^2 + (y - 1)^2 = 1.$$

Checks

For a right-angled triangle with legs a, b and hypotenuse c, the inradius is $\frac{1}{2}(a + b - c)$; here $\frac{1}{2}(3 + 4 - 5) = 1$, as found. The inradius is also area/semi-perimeter $= 6/6 = 1$. Numerically, the incentre computed as the side-weighted average of the vertices, $\frac{4O+3A+5B}{12}$, comes out as $(2, 1)$ exactly, and the distances from $(2, 1)$ to all three lines are each 1.

Common slips

- Solving $|10t - 15| = 5t$ without the modulus and finding only $t = 3$, which is the excircle, not the incircle.
- Finding both roots but not explaining why $t = 1$ is the one inside the triangle.
- In part (i), saying $\tan\beta = \tan 2\alpha$ so $\beta = 2\alpha$ without noting that both angles are acute.

Beyond STEP

Every triangle has one incircle and three excircles, and the four centres are where the internal and external angle bisectors meet; the value $t = 3$ found here is the excircle opposite O. These circles lead into a rich vein of classical geometry, including the nine-point circle and Feuerbach's theorem that it touches all four. Coxeter's *Introduction to Geometry* is a classic place to explore this; Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice.

WORKED SOLUTION

STEP I 2009, Question 9

Mechanics • Colliding Projectiles

STEP I 2009, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

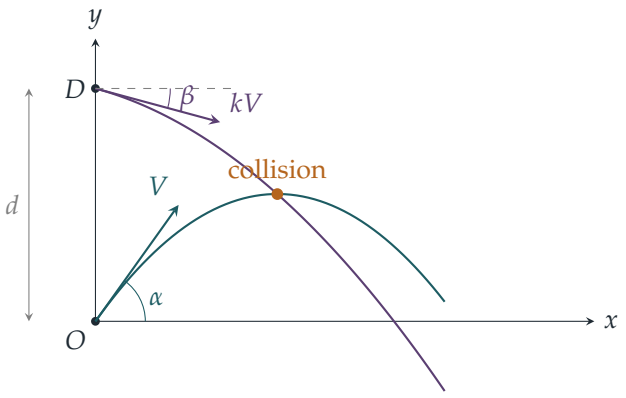
step.maths.org/questions/09-s1-q9

The key idea

Two particles under gravity alone have the *same* acceleration, so the $-\frac{1}{2}gt^2$ terms cancel when we compare their positions: relative to each other, they move in straight lines at constant velocity. Collision means equal horizontal and equal vertical displacements at the same time T. The horizontal equation gives the cosine relation at once. The vertical one involves β, which we remove using $\sin^2\beta = 1 - \cos^2\beta$ and the cosine relation.

Solution

We model P and Q as particles moving freely under gravity, with no air resistance. For them to collide they must be projected in the same vertical plane and in the same horizontal direction. Take O as origin, x horizontal in that direction and y vertically upwards, so D is $(0, d)$.



At time t the positions are

$$\begin{aligned} P: \quad & x = Vt\cos\alpha, && y = Vt\sin\alpha - \tfrac{1}{2}gt^2, \\ Q: \quad & x = kVt\cos\beta, && y = d - kVt\sin\beta - \tfrac{1}{2}gt^2, \end{aligned}$$

where the minus sign in Q's vertical velocity is because β is measured *below* the horizontal.

The cosine relation and the equation for T

At the collision, $t = T > 0$ and the coordinates agree. Horizontally,

$$VT\cos\alpha = kVT\cos\beta,$$

and dividing by $VT \neq 0$ gives $\cos\alpha = k\cos\beta$, as required.

Vertically, the $-\frac{1}{2}gT^2$ terms cancel from both sides, leaving

$$VT\sin\alpha = d - kVT\sin\beta, \qquad \text{so} \qquad kVT\sin\beta = d - VT\sin\alpha.$$

Ask yourself How do we get rid of β?

We know $k\cos\beta$ in terms of α, so we square to bring in $k^2\sin^2\beta$, which is

$$k^2\sin^2\beta = k^2 - k^2\cos^2\beta = k^2 - \cos^2\alpha.$$

Squaring the vertical equation and substituting,

$$\begin{aligned} (k^2 - \cos^2\alpha)V^2T^2 &= d^2 - 2dVT\sin\alpha + V^2T^2\sin^2\alpha, \\ (k^2 - \cos^2\alpha - \sin^2\alpha)V^2T^2 + 2dVT\sin\alpha - d^2 &= 0, \\ (k^2 - 1)V^2T^2 + 2dVT\sin\alpha - d^2 &= 0, \end{aligned}$$

using $\sin^2\alpha + \cos^2\alpha = 1$. This is the required equation.

As a quadratic in T, its roots have product $-d^2/\big((k^2-1)V^2\big) < 0$ because $k > 1$. So exactly one root is positive, and that root is the collision time.

Collision at P's highest point

P's vertical velocity is $V\sin\alpha - gt$, which is zero when $t = V\sin\alpha/g$. So here $T = V\sin\alpha/g$. Substituting into the quadratic,

$$(k^2-1)\frac{V^4\sin^2\alpha}{g^2} + \frac{2dV^2\sin^2\alpha}{g} - d^2 = 0.$$

Multiplying by g^2 and collecting the $\sin^2\alpha$ terms gives $\sin^2\alpha\,\big((k^2-1)V^4 + 2gdV^2\big) = g^2d^2$, so

$$\boxed{\sin^2\alpha = \frac{g^2d^2}{V^2\big((k^2-1)V^2 + 2gd\big)}}$$

The bracket is positive since $k > 1$, so this is well defined.

The inequality

Since $\sin^2\alpha \le 1$ and the denominator is positive,

$$g^2d^2 \le (k^2-1)V^4 + 2gdV^2,$$
$$g^2d^2 - 2gdV^2 + V^4 \le k^2V^4,$$
$$(gd - V^2)^2 \le (kV^2)^2.$$

Both $|gd - V^2|$ and kV^2 are non-negative, so taking square roots preserves the inequality: $|gd - V^2| \le kV^2$. In particular $gd - V^2 \le kV^2$, that is,

$$gd \le (1+k)V^2,$$

as required. Equality would need $\sin^2\alpha = 1$, i.e. P projected vertically.

Checks

- Take $g = 10$, $V = 10$, $k = 2$, $d = 10$. The formula gives $\sin^2\alpha = \frac{100\cdot100}{100(300+200)} = \frac{1}{5}$, so $T = V\sin\alpha/g = 1/\sqrt{5}$. Then the quadratic is $3\cdot100\cdot\frac{1}{5} + 2\cdot10\cdot10\cdot\frac{1}{5} - 100 = 60 + 40 - 100 = 0$. With $\cos\beta = \cos\alpha/2$, both particles are at height 1 at time T. Here $gd = 100 \le 300 = (1+k)V^2$.
- A numerical check stepped both motions forward for thousands of random V, k, α, T, and confirmed the collision, the cosine relation, the quadratic and the formula for $\sin^2\alpha$ to rounding error.
- Limiting case: if $\alpha \to \frac{1}{2}\pi$ then $\cos\beta \to 0$ too, so both move vertically. P rises to height $V^2/(2g)$ in time V/g, while Q falls by $kV^2/g + V^2/(2g)$, so $d = (1+k)V^2/g$: equality, as predicted.

Common slips

- Getting the sign of Q's vertical velocity wrong: projected *below* the horizontal, it is $-kV\sin\beta$.
- Keeping the $\frac{1}{2}gT^2$ terms: they cancel, and the equation for T has no g in it.
- Taking square roots of $(gd - V^2)^2 \le k^2V^4$ as $gd - V^2 \le kV^2$ without noting the modulus. The conclusion is right, but the reason is that $x \le |x|$.

Beyond STEP

The cancellation of gravity is a first taste of changing frames of reference. In a frame falling freely with acceleration g, both particles move in straight lines at constant speed, and collision problems reduce to geometry of relative velocity. This is the idea behind Einstein's equivalence principle, and it makes many projectile-interception problems one-liners. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops relative motion and non-inertial frames carefully.

WORKED SOLUTION

STEP I 2009, Question 10

Mechanics • Connected Particles on Inclines

STEP I 2009, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

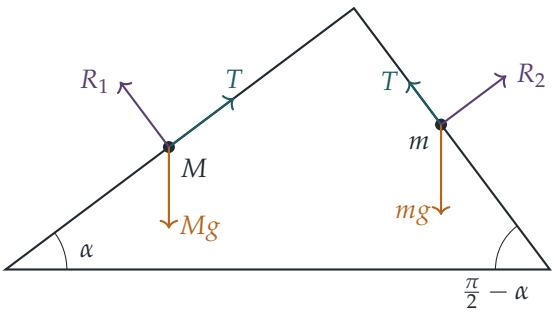
step.maths.org/questions/09-s1-q10

The key idea

Because the string is inextensible and the pulley smooth, the two particles move as one system: when M goes down its face a distance s, m goes up its face the same distance s, and the tension is the same on both sides. So we compare the two components of gravity along the faces, and Newton's second law for each particle gives the common acceleration. The last part is then an ordinary calculus maximisation in the single variable α.

Solution

We model the particles as point masses on smooth faces, with a light inextensible string over a smooth pulley, and assume the system starts from rest. Let T be the tension and R_1, R_2 the normal reactions on M and m.



The angle at the apex is $\pi - \alpha - (\frac{\pi}{2} - \alpha) = \frac{\pi}{2}$, so the faces are perpendicular. Gravity's component down the face of inclination α is $Mg\sin\alpha$, and down the face of inclination $\frac{\pi}{2} - \alpha$ it is $mg\sin(\frac{\pi}{2} - \alpha) = mg\cos\alpha$. The normal reactions are perpendicular to the faces, so they play no part in motion along them.

Part (i)

Take the direction "M down its face, m up its face" as positive, with acceleration a. Resolving along each face:

$$\begin{aligned} M: &\quad Mg\sin\alpha - T = Ma, \\ m: &\quad T - mg\cos\alpha = ma. \end{aligned}$$

Adding eliminates T:

$$a = \frac{g\,(M\sin\alpha - m\cos\alpha)}{M+m}.$$

Since $\cos\alpha > 0$ for $0 < \alpha < \frac{\pi}{2}$,

$$a > 0 \iff M\sin\alpha > m\cos\alpha \iff \tan\alpha > \frac{m}{M}.$$

So if $\tan\alpha > \frac{m}{M}$, the system, starting from rest, accelerates in the positive direction: M slides down its face, as required.

> **Ask yourself** How do we know the string stays taut, so that these equations are the right ones?

From the second equation, $T = m(a + g\cos\alpha)$, which is positive when $a > 0$, so a taut string is consistent. Also, the string couldn't go slack: if it did, each particle would slide *down* its own face, moving both of them away from the pulley, which would need a longer string.

Part (ii)

Now $\tan\alpha = \frac{2m}{M}$, so $m = \frac{1}{2}M\tan\alpha$, and $\tan\alpha > \frac{m}{M}$, so by part (i) the motion is again with M moving down. Substituting $m = \frac{1}{2}M\tan\alpha$ and cancelling M:

$$\begin{aligned} a &= \frac{g\left(M\sin\alpha - \frac{1}{2}M\tan\alpha\cos\alpha\right)}{M + \frac{1}{2}M\tan\alpha} = \frac{g\left(\sin\alpha - \frac{1}{2}\sin\alpha\right)}{1 + \frac{1}{2}\tan\alpha} = \frac{\frac{1}{2}g\sin\alpha}{\frac{1}{2}(2+\tan\alpha)} \\ &= \frac{g\sin\alpha}{\tan\alpha + 2}, \end{aligned}$$

using $\tan\alpha\cos\alpha = \sin\alpha$. That is the required magnitude of the acceleration.

Maximising. The ratio m/M fixes α through $\tan\alpha = 2m/M$, so we maximise

$$f(\alpha) = \frac{\sin\alpha}{\tan\alpha + 2}, \qquad 0 < \alpha < \tfrac{\pi}{2}.$$

By the quotient rule, with $\frac{\mathrm{d}}{\mathrm{d}\alpha}\tan\alpha = \sec^2\alpha$,

$$f'(\alpha) = \frac{\cos\alpha\,(\tan\alpha+2) - \sin\alpha\sec^2\alpha}{(\tan\alpha+2)^2} = \frac{\sin\alpha + 2\cos\alpha - \dfrac{\sin\alpha}{\cos^2\alpha}}{(\tan\alpha+2)^2}.$$

Multiplying the numerator by $\cos^2\alpha > 0$ and using $1 - \cos^2\alpha = \sin^2\alpha$:

$$\cos^2\alpha\,(\sin\alpha + 2\cos\alpha) - \sin\alpha = 2\cos^3\alpha - \sin\alpha\,(1 - \cos^2\alpha) = 2\cos^3\alpha - \sin^3\alpha.$$

So $f'(\alpha)$ has the same sign as $2\cos^3\alpha - \sin^3\alpha = \cos^3\alpha\,(2 - \tan^3\alpha)$. As $\tan\alpha$ increases from 0 to ∞, this is positive while $\tan^3\alpha < 2$ and negative once $\tan^3\alpha > 2$. Hence f increases and then decreases, and its maximum is at the unique stationary point

$$\tan^3\alpha = 2.$$

(You can also see this is a maximum because $f > 0$ on the interval while $f \to 0$ at both ends.) With $\tan\alpha = \frac{2m}{M}$ this says

$$\frac{8m^3}{M^3} = 2, \qquad \text{that is,} \qquad 4m^3 = M^3,$$

as required.

Checks

- Limiting cases of part (i): with $m = 0$ we get $a = g\sin\alpha$, a lone particle on a smooth slope; and when $\tan\alpha = m/M$ we get $a = 0$, the balance point.
- At the optimum, $\tan\alpha = 2^{1/3} \approx 1.26$, so $\alpha \approx 51.6^\circ$ and $m/M = 4^{-1/3} \approx 0.63$. A grid search of f over $(0, \frac{\pi}{2})$ put its maximum at the same α, with value $f \approx 0.240$, so the greatest acceleration is about $0.240g$.
- The general formula in part (i), with random masses and angles, agreed with the formula in part (ii) whenever $\tan\alpha = 2m/M$, and both equations of motion gave the same, positive, tension.

Common slips

- Using $mg\sin\alpha$ for m: its face is inclined at $\frac{\pi}{2} - \alpha$, so the component is $mg\cos\alpha$.
- Maximising with m held fixed and M varying (or the other way round) without noticing that a depends only on α. Writing everything in terms of α makes it a one-variable problem.

Beyond STEP

Systems like this, with bodies linked by constraints, are the starting point for Lagrangian mechanics: you pick one coordinate (here, the distance s moved along the string), and the constraint forces such as the tension and the normal reactions drop out automatically. A first-year course turns this into a systematic method that works for pulleys, pendulums and much more. For a clear treatment of Newtonian mechanics at this level, try Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 2009, Question 11

Mechanics • Collisions and Restitution

STEP I 2009, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q11

The key idea

Every one-dimensional collision is governed by two equations: conservation of momentum and Newton's law of restitution. The trick for the energy is to write each particle's loss as a difference of two squares, $\frac{1}{2}m(u-p)(u+p)$, because momentum conservation ties the factors $u-p$ and $q-v$ together. Once a common factor $u-p$ has been pulled out, the remaining bracket is exactly what the restitution law describes.

Solution

We model the particles as moving along a line on a smooth table, with velocities measured in the direction of $\mathbf{i}$. The particle of mass m is behind and catches up, since $u > v$. With no horizontal external forces, momentum is conserved:

$$mu + Mv = mp + Mq, \qquad \text{so} \qquad M(q-v) = m(u-p). \tag{1}$$

Newton's law of restitution says the speed of separation is e times the speed of approach:

$$q - p = e(u-v). \tag{2}$$



before (velocities along $\mathbf{i}$), $u > v$

Part (i)

The loss of kinetic energy is

$$\Delta = \tfrac{1}{2}m(u^2-p^2) + \tfrac{1}{2}M(v^2-q^2) = \tfrac{1}{2}m(u-p)(u+p) - \tfrac{1}{2}M(q-v)(q+v).$$

By (1) we may replace $M(q-v)$ by $m(u-p)$, so a factor $\frac{1}{2}m(u-p)$ comes out:

$$\begin{aligned}\Delta &= \tfrac{1}{2}m(u-p)\big[(u+p)-(q+v)\big]\\ &= \tfrac{1}{2}m(u-p)\big[(u-v)-(q-p)\big]\\ &= \tfrac{1}{2}m(u-p)\big[(u-v)-e(u-v)\big] \qquad \text{by (2)}\\ &= \tfrac{1}{2}m(u-p)(u-v)(1-e),\end{aligned}$$

as required.

Ask yourself How does a formula for the energy loss tell us anything about p?

A collision can't create kinetic energy, so $\Delta \geq 0$. In the product $\frac{1}{2}m(u-p)(u-v)(1-e)$ we have $m>0$, $u-v>0$ (given) and $1-e>0$ (since $e<1$). So the sign of Δ is the sign of $u-p$, and $\Delta \geq 0$ forces $u \geq p$. The particle of mass m can't speed up in the direction of $\mathbf{i}$, which makes sense since the impulse on it from M acts backwards.

If you'd rather not rely on $\Delta \geq 0$ as a physical fact, you can solve (1) and (2) directly:

$$p = \frac{mu+Mv-Me(u-v)}{m+M}, \qquad \text{so} \qquad u-p = \frac{M(1+e)(u-v)}{m+M} > 0$$

for $e \geq 0$. That confirms $u \geq p$, and hence $\Delta \geq 0$.

Part (ii)

Each particle loses the same non-zero amount of energy, so

$$\tfrac{1}{2}m(u-p)(u+p) = \tfrac{1}{2}M(v-q)(v+q) \neq 0.$$

By (1), $M(v-q) = -m(u-p)$, so the right-hand side is $-\frac{1}{2}m(u-p)(v+q)$ and

$$\tfrac{1}{2}m(u-p)(u+p) = -\tfrac{1}{2}m(u-p)(v+q).$$

The left-hand side is non-zero, so $m(u-p) \neq 0$ and we may divide by $\frac{1}{2}m(u-p)$:

$$u+p = -(v+q), \qquad \text{that is,} \qquad u+v+p+q = 0. \tag{3}$$

Now (2) and (3) give the sum and difference of p and q:

$$p+q = -(u+v), \qquad q-p = e(u-v),$$

so

$$p = -\tfrac{1}{2}\big[(u+v)+e(u-v)\big], \qquad q = -\tfrac{1}{2}\big[(u+v)-e(u-v)\big].$$

We haven't used momentum conservation in its full form yet, and that is what fixes e. Substituting into $mp+Mq = mu+Mv$ and multiplying by 2:

$$\begin{aligned}-(m+M)(u+v)+(M-m)e(u-v) &= 2mu+2Mv,\\ (M-m)e(u-v) &= (m+M)(u+v)+2mu+2Mv\\ &= (M+3m)u+(3M+m)v.\end{aligned}$$

Since $u - v > 0$, and $M - m \neq 0$ when $m \neq M$, we can divide to get

$$e = \frac{(M+3m)u + (3M+m)v}{(M-m)(u-v)},$$

as required.

Checks

- Take $m = 1$, $M = 3$, $u = 1$, $v = -\frac{1}{2}$. The formula gives $e = \frac{6-5}{2\times\frac{3}{2}} = \frac{1}{3}$. Then $p = -\frac{1}{2}$ and $q = 0$, and momentum checks out: $1 - \frac{3}{2} = -\frac{1}{2} + 0$. Each particle loses $\frac{3}{8}$ of energy, and $u + v + p + q = 1 - \frac{1}{2} - \frac{1}{2} + 0 = 0$.
- For $e = 1$ the formula in (i) gives zero loss, as it should for a perfectly elastic collision.
- The computer tested (i) with exact fractions on 2000 random collisions, and (ii) on 455 random cases with $0 \leq e < 1$.

Common slips

- Getting the restitution law back to front: it's $q - p = e(u - v)$, separation over approach, with both measured as positive quantities here.
- Dividing by $u - p$ without saying why it's non-zero. Here that's exactly what "non-zero amount" is there for.

Beyond STEP

The factorisation in (i) is a neat consequence of working in the centre-of-mass frame. In that frame the total momentum is zero, the kinetic energy is $\frac{1}{2}\mu(\text{relative speed})^2$ with reduced mass $\mu = \frac{mM}{m+M}$, and the loss is $\frac{1}{2}\mu(1 - e^2)(u - v)^2$. Reduced mass and the centre-of-mass frame turn up again in two-body orbits and in particle scattering. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops these ideas carefully.

WORKED SOLUTION

STEP I 2009, Question 12

Probability and Statistics • Probability and Inequalities

STEP I 2009, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q12

The key idea

Each probability here is a sum of products of bag counts divided by a power of the total, so every comparison comes down to comparing two polynomials in a, b, c. The opening inequality $x^2 + y^2 \geq 2xy$ is exactly the tool for that. In part (ii), counting "exactly two the same" head-on is fiddly. It is much easier to count the other two cases (all the same, all different) and subtract from 1.

Solution

The inequality

For any real x and y,

$$x^2 + y^2 - 2xy = (x - y)^2 \geq 0,$$

since a real square is never negative. So $x^2 + y^2 \geq 2xy$, with equality exactly when $x = y$.

Part (i)

Each bag holds $a + b$ sweets, and the two picks are independent because they come from different bags. Let S be the event that the two sweets have the same colour and D the event that they differ. Then

$$\mathrm{P}(S) = \mathrm{P}(\text{red, red}) + \mathrm{P}(\text{blue, blue}) = \frac{a}{a+b} \cdot \frac{b}{a+b} + \frac{b}{a+b} \cdot \frac{a}{a+b} = \frac{2ab}{(a+b)^2},$$

$$\mathrm{P}(D) = \mathrm{P}(\text{red, blue}) + \mathrm{P}(\text{blue, red}) = \frac{a}{a+b} \cdot \frac{a}{a+b} + \frac{b}{a+b} \cdot \frac{b}{a+b} = \frac{a^2 + b^2}{(a+b)^2}.$$

The denominators are equal and positive, and $2ab \leq a^2 + b^2$ by the inequality above. Hence $\mathrm{P}(S) \leq \mathrm{P}(D)$, as required, with equality only when $a = b$.

Part (ii)

Write $n = a + b + c$, the number of sweets in each bag. The contents are:

	red	white	yellow
bag 1	a	b	c
bag 2	b	c	a
bag 3	c	a	b

The three picks are independent, so the probability of any particular colour from each bag (in order) is the product of the three counts divided by n^3. Let A be the event that all three sweets are the same colour, T the event that exactly two are, and E the event that all three differ. These three events cover every possibility and don't overlap.

All the same. Reading down each column of the table,

$$\mathrm{P}(A) = \frac{abc + bca + cab}{n^3} = \frac{3abc}{n^3}.$$

All different. The colours from bags 1, 2, 3 form one of the $3! = 6$ orderings of red, white, yellow:

$$\begin{array}{lll} \text{RWY: } a \cdot c \cdot b & \text{RYW: } a \cdot a \cdot a & \text{WRY: } b \cdot b \cdot b \\ \text{WYR: } b \cdot a \cdot c & \text{YRW: } c \cdot b \cdot a & \text{YWR: } c \cdot c \cdot c \end{array}$$

So

$$\mathrm{P}(E) = \frac{a^3 + b^3 + c^3 + 3abc}{n^3}.$$

Ask yourself How do we get from these two answers to $\mathrm{P}(T)$ without listing the 18 remaining outcomes?

Since $\mathrm{P}(T) = 1 - \mathrm{P}(A) - \mathrm{P}(E)$, we need the expansion of $n^3 = (a + b + c)^3$. It has the three cubes, each of the six terms like a^2b with coefficient 3, and abc with coefficient 6:

$$(a + b + c)^3 = a^3 + b^3 + c^3 + 3\left(a^2b + b^2c + c^2a + ab^2 + bc^2 + ca^2\right) + 6abc.$$

Hence

$$\begin{aligned} \mathrm{P}(T) &= \frac{n^3 - 3abc - (a^3 + b^3 + c^3 + 3abc)}{n^3} \\ &= \frac{3\left(a^2b + b^2c + c^2a + ab^2 + bc^2 + ca^2\right)}{(a + b + c)^3}, \end{aligned}$$

which is the given result. The probability that the sweets are all the same colour is

$$\mathrm{P}(A) = \frac{3abc}{(a + b + c)^3}.$$

The deduction. Group the six terms in pairs that share a factor:

$$a^2b + b^2c + c^2a + ab^2 + bc^2 + ca^2 = a\left(b^2 + c^2\right) + b\left(c^2 + a^2\right) + c\left(a^2 + b^2\right).$$

By the opening inequality $b^2 + c^2 \geq 2bc$, and since $a > 0$, multiplying keeps the direction: $a(b^2 + c^2) \geq 2abc$. Similarly $b(c^2 + a^2) \geq 2abc$ and $c(a^2 + b^2) \geq 2abc$. Adding,

$$a^2b + b^2c + c^2a + ab^2 + bc^2 + ca^2 \geq 6abc.$$

Multiplying by $3/n^3 > 0$,

$$\mathrm{P}(T) \geq \frac{18abc}{n^3} = 6 \cdot \frac{3abc}{n^3} = 6\,\mathrm{P}(A),$$

as required. Equality needs $b = c$, $c = a$ and $a = b$ together, so it holds only when $a = b = c$.

Checks

- With $a = b = c$ every bag is one third of each colour, so $\mathrm{P}(A) = 3 \cdot \frac{1}{27} = \frac{1}{9}$ and $\mathrm{P}(E) = \frac{6}{27} = \frac{2}{9}$, leaving $\mathrm{P}(T) = \frac{2}{3} = 6 \cdot \frac{1}{9}$: the equality case.
- With $a, b, c = 1, 2, 3$ ($n = 6$): $\mathrm{P}(A) = \frac{18}{216} = \frac{1}{12}$, $\mathrm{P}(E) = \frac{54}{216} = \frac{1}{4}$ and $\mathrm{P}(T) = \frac{144}{216} = \frac{2}{3}$. These add to 1, and $\frac{2}{3}$ is 8 times $\frac{1}{12}$.
- We checked all three formulas by exact enumeration for every a, b, c from 1 to 8, and by simulation (for $a, b, c = 2, 3, 5$, 200 000 trials gave 0.659 against the exact 0.66 for $\mathrm{P}(T)$).

Common slips

- Writing the bag contents in the wrong order. Bag 2 has *a yellow*, not *a* red; a table like the one above prevents this.
- Assuming every ordering in $\mathrm{P}(E)$ gives abc. Only three do; the orderings RYW, WRY and YWR give a^3, b^3 and c^3.
- Forgetting to say that $a, b, c > 0$ when multiplying an inequality by a, b or c.

Beyond STEP

The final inequality is a case of Muirhead's inequality, a general rule for comparing symmetric sums of monomials; here it says the sum of the terms a^2b dominates the sum of the terms abc. Part (i) is the two-variable case of the same idea, and both rest on the AM–GM inequality. For more on counting arguments and complementary events, see Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP I 2009, Question 13

Probability and Statistics • Arrangements and Runs

STEP I 2009, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q13

The key idea

Only the *positions* of the girls matter, not which girl or boy is which. So rather than juggling $(n+3)!$ orders, we choose the set of three places the girls occupy: every one of the $\binom{n+3}{3}$ sets is equally likely. Counting the favourable sets is then easiest if we line up the n boys first and ask which of the gaps between and around them the girls go into.

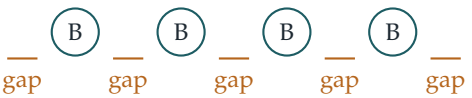
Solution

Since all $(n+3)!$ orders are equally likely, and each set of three positions for the girls arises from the same number, $3!\,n!$, of orders, the set of girls' positions is uniformly distributed over the

$$\binom{n+3}{3} = \frac{(n+3)(n+2)(n+1)}{6}$$

possible sets. Throughout, K can only be 1, 2 or 3.

The n boys in a row create $n+1$ gaps (one before the first boy, one after the last, and $n-1$ between neighbours):



Placing girls into gaps (any number into one gap) gives a line, and every line arises in exactly one way. Girls in the same gap sit next to each other; girls in different gaps are separated by at least one boy.

Part (i)

$K = 3$ means all three girls are in one block, that is, they all go into the same gap. There are $n+1$ choices of gap, so

$$\mathrm{P}(K=3) = \frac{n+1}{\binom{n+3}{3}} = \frac{6(n+1)}{(n+3)(n+2)(n+1)} = \frac{6}{(n+2)(n+3)}.$$

(Equivalently: a block of three consecutive places can start at any of positions $1, 2, \ldots, n+1$.)

Part (ii)

$K = 1$ means no two girls are adjacent, that is, the girls are in three *different* gaps. We choose 3 of the $n+1$ gaps, in $\binom{n+1}{3}$ ways, and put one girl in each. Hence

$$\mathrm{P}(K=1) = \frac{\binom{n+1}{3}}{\binom{n+3}{3}} = \frac{(n+1)n(n-1)}{(n+3)(n+2)(n+1)} = \frac{n(n-1)}{(n+2)(n+3)},$$

as required. (For $n = 0$ or 1 there are too few gaps, and the formula correctly gives 0.)

Ask yourself Why is the count $\binom{n+1}{3}$ and not $\binom{n+1}{3} \times 3!$?

We are counting sets of *positions*, not orders of named children. Choosing which three gaps are used fixes the girls' positions completely; which girl sits where has already been accounted for in the claim that every set of positions is equally likely.

Part (iii)

First we need $\mathrm{P}(K=2)$. Since $K \in \{1,2,3\}$,

$$\begin{aligned}\mathrm{P}(K=2) &= 1 - \frac{6}{(n+2)(n+3)} - \frac{n(n-1)}{(n+2)(n+3)} \\ &= \frac{(n^2+5n+6) - 6 - (n^2-n)}{(n+2)(n+3)} = \frac{6n}{(n+2)(n+3)}.\end{aligned}$$

As a check, count directly: $K = 2$ means a pair of girls in one gap and the third girl in a different gap. Choosing the gap for the pair and then a different gap for the single gives $(n+1)n$ sets, and $(n+1)n/\binom{n+3}{3} = 6n/\big((n+2)(n+3)\big)$, as before.

Now

$$\begin{aligned}\mathrm{E}(K) &= 1 \cdot \frac{n(n-1)}{(n+2)(n+3)} + 2 \cdot \frac{6n}{(n+2)(n+3)} + 3 \cdot \frac{6}{(n+2)(n+3)} \\ &= \frac{n^2+11n+18}{(n+2)(n+3)} = \frac{(n+2)(n+9)}{(n+2)(n+3)} \\ &= \frac{n+9}{n+3}.\end{aligned}$$

Checks

- $n = 0$: there are only girls, so $K = 3$ certainly, and $\frac{n+9}{n+3} = 3$.
- $n = 1$: the lines are GGGB, GGBG, GBGG, BGGG, equally likely, so $K = 3$ or 2 with probability $\frac{1}{2}$ each and $\mathrm{E}(K) = \frac{5}{2} = \frac{10}{4}$.
- As $n \to \infty$, $\mathrm{E}(K) \to 1$: with many boys the girls are almost never neighbours.
- We enumerated all position sets for $n = 0, \ldots, 9$ with exact fractions (for example $n = 3$ gives $\mathrm{P}(K = 1, 2, 3) = \frac{1}{5}, \frac{3}{5}, \frac{1}{5}$ and $\mathrm{E}(K) = 2$), and all $(n+3)!$ orders of labelled children for $n \leq 4$; everything matched.

Common slips

- Mixing counts of ordered arrangements with counts of position sets in the same fraction. Pick one sample space and stick to it.
- Forgetting the two end gaps, and so using $n - 1$ gaps instead of $n + 1$.

Beyond STEP

The "boys make gaps, girls go into gaps" trick is the stars-and-bars idea, and the general question of the longest run in a random arrangement is a classical one: for long random sequences the longest run grows like a logarithm of the length. An undergraduate probability course meets these through indicator variables and generating functions. Grimmett and Welsh, *Probability: An Introduction*, is a good place to see such counting arguments developed.

WORKED SOLUTION

STEP II 2009, Question 1

Pure • Symmetry of Curves

STEP II 2009, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q1

The key idea

Almost nothing here needs computing: symmetry does the work. Both curves are unchanged by reflection in the lines $y = x$ and $y = -x$, so once we have one intersection point, the others come from reflecting it. The rectangle then follows from the shape of the four points. For the area, we never need to find α and β themselves: the answer turns out to need only one combination of them, and we can get that straight from $\alpha^4 + \beta^4 = u$ and $\alpha\beta = v$.

Solution

Lines of symmetry

A curve is symmetric in a line if reflecting any point of the curve in that line gives another point of the curve. The four reflections we need are

$$(x,y) \mapsto (-x,y), \quad (x,y) \mapsto (x,-y), \quad (x,y) \mapsto (y,x), \quad (x,y) \mapsto (-y,-x),$$

which are the reflections in $x = 0$, $y = 0$, $y = x$ and $y = -x$.

The curve $x^4 + y^4 = u$. The expression $x^4 + y^4$ is unchanged by all four maps, because fourth powers ignore signs and the expression is symmetric in x and y. So this curve has four lines of symmetry:

$$x = 0, \qquad y = 0, \qquad y = x, \qquad y = -x.$$

The curve $xy = v$. Swapping x and y leaves xy unchanged, and so does $(x,y) \mapsto (-y,-x)$, since $(-y)(-x) = xy$. Changing the sign of just one coordinate sends xy to $-xy$, and since $v > 0$ the point $(-x,y)$ is then on $xy = -v$, not on our curve. So this rectangular hyperbola has exactly two lines of symmetry:

$$y = x, \qquad y = -x.$$

The four intersection points

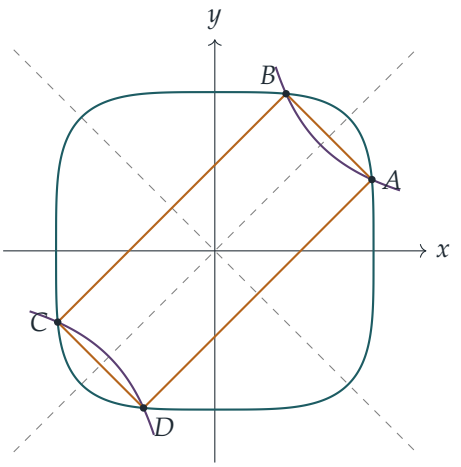
Let S be the reflection in $y = x$ and T the reflection in $y = -x$. Each maps both curves to themselves, so each maps an intersection point to an intersection point. Starting from $A = (\alpha, \beta)$ we get

$$S(A) = (\beta, \alpha), \qquad T(A) = (-\beta, -\alpha), \qquad T(S(A)) = (-\alpha, -\beta).$$

These four points are distinct, because $\alpha > \beta > 0$. There are no other intersections. Substituting $y = v/x$ into the quartic gives $x^8 - ux^4 + v^4 = 0$, a quadratic in x^4 with at most two positive roots, so there are at most four real values of x.

To label them anticlockwise, look at their polar angles. Since $0 < \beta < \alpha$, the point A has angle θ with $0 < \theta < 45°$. Then (β, α) has angle $90° - \theta$, $(-\alpha, -\beta)$ has angle $180° + \theta$ and $(-\beta, -\alpha)$ has angle $270° - \theta$. These angles increase in that order, so

$$B = (\beta, \alpha), \qquad C = (-\alpha, -\beta), \qquad D = (-\beta, -\alpha).$$



(The picture shows the case $u = 81$, $v = 4$: the quartic in teal, the hyperbola in plum.)

$ABCD$ is a rectangle

The diagonals are AC and BD. The midpoint of AC is $\left(\frac{\alpha-\alpha}{2}, \frac{\beta-\beta}{2}\right) = (0, 0)$, and likewise the midpoint of BD is $(0, 0)$. So the diagonals bisect each other, and $ABCD$ is a parallelogram. Their lengths are

$$AC = \sqrt{(2\alpha)^2 + (2\beta)^2} = 2\sqrt{\alpha^2 + \beta^2}, \qquad BD = \sqrt{(2\beta)^2 + (2\alpha)^2} = 2\sqrt{\alpha^2 + \beta^2},$$

which are equal. A parallelogram with equal diagonals is a rectangle, so $ABCD$ is a rectangle.

Ask yourself Why does a parallelogram with equal diagonals have to be a rectangle?

All four vertices are at the same distance $\sqrt{\alpha^2+\beta^2}$ from O, so they lie on a circle with centre O, and AC and BD are diameters of it. By the angle in a semicircle, each angle of $ABCD$ is a right angle. To check directly instead, $\overrightarrow{AB} = (\beta-\alpha,\ \alpha-\beta)$ and $\overrightarrow{BC} = (-\alpha-\beta,\ -\alpha-\beta)$, with scalar product

$$-(\beta-\alpha)(\alpha+\beta) - (\alpha-\beta)(\alpha+\beta) = (\alpha+\beta)\big[(\alpha-\beta)-(\alpha-\beta)\big] = 0.$$

The area

The side lengths are

$$AB = \sqrt{2(\alpha-\beta)^2} = \sqrt{2}\,(\alpha-\beta), \qquad BC = \sqrt{2(\alpha+\beta)^2} = \sqrt{2}\,(\alpha+\beta),$$

using $\alpha > \beta > 0$ to take the positive square roots. So

$$\text{area} = AB \cdot BC = 2(\alpha-\beta)(\alpha+\beta) = 2(\alpha^2-\beta^2).$$

Now use the fact that A lies on both curves, so $\alpha^4+\beta^4 = u$ and $\alpha\beta = v$. Then

$$(\alpha^2-\beta^2)^2 = \alpha^4+\beta^4-2\alpha^2\beta^2 = u-2v^2.$$

Since $\alpha^2-\beta^2 > 0$, we take the positive root, and

$$\boxed{\text{area of } ABCD = 2\sqrt{u-2v^2}}$$

This also shows that the question's assumption of four distinct points forces $u > 2v^2$. If $u = 2v^2$ then $\alpha = \beta$ and the curves only touch.

For $u = 81$ and $v = 4$: $u - 2v^2 = 81 - 32 = 49$, so the area is $2 \times 7 = 14$, as required.

Checks

For $u = 81$, $v = 4$, solving $x^8 - 81x^4 + 256 = 0$ numerically gives $\alpha \approx 2.9690$ and $\beta = 4/\alpha \approx 1.3472$. Then $2(\alpha^2-\beta^2) \approx 14.000$, and the shoelace formula on the four vertices gives the same area. As u falls towards $2v^2$ the two points in each quadrant merge on $y = x$ and the rectangle flattens to a segment, matching area $2\sqrt{u-2v^2} \to 0$.

Common slips

- Giving the coordinate axes as lines of symmetry of $xy = v$. Reflecting in an axis sends the branch in the first quadrant to the second quadrant, where $xy < 0$.
- Labelling the points in the wrong order, for example taking $B = (-\alpha, -\beta)$. Then $ABCD$ is traversed as a crossed quadrilateral.
- Trying to solve for α and β explicitly. The area needs only $\alpha^2-\beta^2$, which comes straight from u and v.

Beyond STEP

The four reflections and rotations that preserve $x^4 + y^4 = u$ form the symmetry group of a square, the dihedral group of order 8. The hyperbola keeps only a subgroup of order 4 (two reflections, the half-turn and the identity). Intersection points are permuted by the symmetries the two curves share, and finding objects by the action of a group like this is a theme of first-year group theory. Allenby's *Rings, Fields and Groups* is a good place to meet these ideas.

WORKED SOLUTION

STEP II 2009, Question 2

Pure • Curve Sketching and Approximation

STEP II 2009, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q2

The key idea

Because $a > 1$, the function a^u is strictly increasing in u, so y is largest exactly where $\sin(\pi e^x)$ is largest and smallest where it is smallest. Everything about the curve therefore comes from the inner function πe^x, which increases steadily and runs through the peaks and troughs of the sine at points that crowd closer and closer together as x grows. For the area, the trapezia have equal heights at their ends, which makes the sum collapse to a single difference of logarithms.

Solution

Part (i)

Taking logarithms, $\ln y = \ln a \, \sin(\pi e^x)$. Differentiating implicitly,

$$\frac{1}{y}\frac{dy}{dx} = \pi \ln a \, e^x \cos(\pi e^x), \qquad \text{so} \qquad \frac{dy}{dx} = \pi \ln a \, e^x \cos(\pi e^x)\, a^{\sin(\pi e^x)}.$$

Here $\ln a > 0$ (since $a > 1$), $e^x > 0$ and $a^{\sin(\pi e^x)} > 0$, so the derivative vanishes exactly when $\cos(\pi e^x) = 0$, that is, when πe^x is an odd multiple of $\frac{\pi}{2}$. Since $\pi e^x > 0$, only the positive odd multiples are possible:

$$e^x = n + \tfrac{1}{2}, \qquad x = \ln\left(n + \tfrac{1}{2}\right), \qquad n = 0, 1, 2, \ldots$$

At these points $\sin(\pi e^x) = \sin\left((n + \frac{1}{2})\pi\right) = (-1)^n$, so $y = a^{(-1)^n}$. The stationary points are

$$\left(\ln\left(n + \tfrac{1}{2}\right),\ a\right) \ (n \text{ even}), \qquad \left(\ln\left(n + \tfrac{1}{2}\right),\ \tfrac{1}{a}\right) \ (n \text{ odd}).$$

The first set are maxima and the second minima. As a^u is increasing, $y \le a^1 = a$ and $y \ge a^{-1}$ for every x, so points where $y = a$ are global maxima and points where $y = 1/a$ are global minima. Explicitly, the maxima are at $x = \ln\frac{1}{2},\ \ln\frac{5}{2},\ \ln\frac{9}{2}, \ldots$ and the minima at $x = \ln\frac{3}{2},\ \ln\frac{7}{2},\ \ln\frac{11}{2}, \ldots$

Part (ii)

For small x, $e^x \approx 1 + x$, so

$$\sin(\pi e^x) \approx \sin(\pi + \pi x) = -\sin(\pi x) \approx -\pi x,$$

using $\sin(\pi + \theta) = -\sin\theta$ and then $\sin t \approx t$ with $t = \pi x$ small. Hence

$$y \approx a^{-\pi x} = e^{-\pi x \ln a} \approx 1 - \pi x \ln a,$$

where the last step uses $e^t \approx 1 + t$ again, now with the small quantity $t = -\pi x \ln a$. This is the required approximation: the curve crosses the line $y = 1$ at $x = 0$ with gradient $-\pi \ln a$.

Ask yourself Why can't we put $\sin t \approx t$ straight into $\sin(\pi e^x)$?

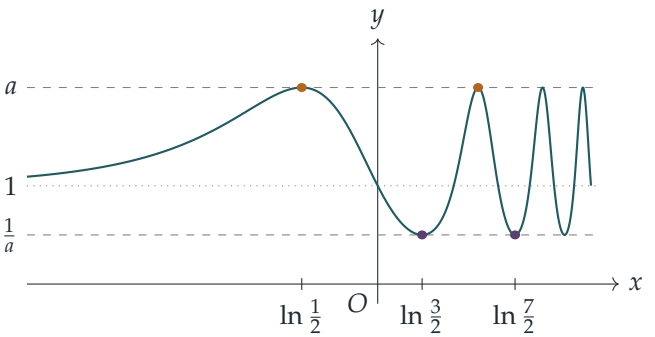
Because πe^x is close to π, not to 0, when x is small. We first have to use $\sin(\pi + \theta) = -\sin\theta$ to move the small quantity πx into the argument of the sine.

Part (iii)

Some features to put on the sketch:

- As $x \to -\infty$, $e^x \to 0$, so $\sin(\pi e^x) \to 0$ and $y \to 1$ from above (for $x < \ln\frac{1}{2}$ we have $0 < \pi e^x < \frac{\pi}{2}$, so the sine is positive and $y > 1$). The line $y = 1$ is an asymptote on the left.
- The curve rises to its first maximum a at $x = \ln\frac{1}{2}$, then oscillates for ever between a and $1/a$, always staying above the x-axis.
- It crosses $y = 1$ wherever e^x is a positive integer, at $x = 0,\ \ln 2,\ \ln 3, \ldots$, and at $x = 0$ the gradient is $-\pi \ln a$, by part (ii).
- The gap between consecutive stationary points is $\ln\frac{n+3/2}{n+1/2} \to 0$, so the oscillations get faster and faster as x increases.

The sketch shows the case $a = 2$.



Part (iv)

Count the maxima from the left: the kth maximum corresponds to $n = 2(k-1)$, so it is at $\mathrm{e}^x = 2k - \frac{3}{2}$. The $(k+1)$th maximum is at $\mathrm{e}^x = 2k + \frac{1}{2}$, and the single minimum between them is at $\mathrm{e}^x = 2k - \frac{1}{2}$. Write

$$x_1 = \ln\left(2k - \tfrac{3}{2}\right), \qquad x_2 = \ln\left(2k - \tfrac{1}{2}\right), \qquad x_3 = \ln\left(2k + \tfrac{1}{2}\right).$$

Joining (x_1, a) to $(x_2, \frac{1}{a})$ and $(x_2, \frac{1}{a})$ to (x_3, a) by straight lines, the region down to the x-axis splits into two trapezia, each with parallel sides a and $\frac{1}{a}$. So the approximate area is

$$\begin{aligned}\tfrac{1}{2}\left(a + \tfrac{1}{a}\right)(x_2 - x_1) + \tfrac{1}{2}\left(a + \tfrac{1}{a}\right)(x_3 - x_2) &= \frac{a^2+1}{2a}\,(x_3 - x_1) \\ &= \frac{a^2+1}{2a}\,\ln\frac{2k+\frac{1}{2}}{2k-\frac{3}{2}} = \frac{a^2+1}{2a}\,\ln\frac{k+\frac{1}{4}}{k-\frac{3}{4}}.\end{aligned}$$

Finally $\dfrac{k+\frac{1}{4}}{k-\frac{3}{4}} = 1 + \dfrac{1}{k-\frac{3}{4}}$, so the area is approximately

$$\left(\frac{a^2+1}{2a}\right)\ln\left(1 + \left(k - \tfrac{3}{4}\right)^{-1}\right),$$

as required. Notice that the minimum's position x_2 cancels completely: only the two maxima matter.

Checks

- Differentiating numerically confirms that $\mathrm{d}y/\mathrm{d}x = 0$ at $x = \ln(n + \frac{1}{2})$ for $n = 0, \ldots, 7$ and nowhere else on $[-6, \ln 8]$, and that the gradient at $x = 0$ is $-\pi \ln a$ (for $a = 2$, both are -2.1776).
- With $a = 2$, $k = 1$ the formula gives $\frac{5}{4}\ln 5 \approx 2.012$, while Simpson's rule gives the true area as 1.954: close, as an approximation should be. The trapezia overestimate slightly: the curve dips below the chords around the minimum by more than it bulges above them near the maxima.
- As $k \to \infty$ the area tends to 0, matching the ever-narrower oscillations.

Common slips

- Including negative odd multiples of $\frac{\pi}{2}$ in part (i): $\pi\mathrm{e}^x$ is always positive.
- Numbering the maxima from $k = 0$ rather than $k = 1$, which shifts the answer to $\ln\left(1 + (k + \frac{1}{4})^{-1}\right)$.
- In part (ii), stopping at $a^{-\pi x}$ without the final linearisation $\mathrm{e}^t \approx 1 + t$.

Beyond STEP

Part (ii) is a first-order Taylor expansion done by composing simpler expansions, which is exactly how undergraduates handle expansions of composite functions. The true area over one oscillation, for large k, is about $(x_3 - x_1)$ times the average of $a^{\sin\theta}$ over a period, an integral that leads to the modified Bessel function $I_0(\ln a)$; comparing it with $\frac{1}{2}(a + \frac{1}{a})$ is a nice exercise in why the trapezium rule overestimates for curves like this. For a careful treatment of approximation and Taylor's theorem, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2009, Question 3

Pure • Trigonometric Identities and Surds

STEP II 2009, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q3

The key idea

The identity $(*)$ turns a tangent you know at an angle x into a tangent at the angle $\frac{1}{4}\pi - \frac{1}{2}x$, which is roughly half as big. So it acts as a half-angle machine. The work is in choosing a starting angle x that lands exactly on the angle you want. After that, the job is careful surd arithmetic: rationalising, expanding and squaring.

Solution

Proving $(*)$

Write $\theta = \frac{1}{4}\pi - \frac{1}{2}x$, so that $x = \frac{1}{2}\pi - 2\theta$. Then

$$\sin x = \cos 2\theta = 1 - 2\sin^2\theta, \qquad \cos x = \sin 2\theta = 2\sin\theta\cos\theta.$$

Hence, wherever both sides are defined (that is, $\cos x \neq 0$),

$$\sec x - \tan x = \frac{1-\sin x}{\cos x} = \frac{2\sin^2\theta}{2\sin\theta\cos\theta} = \frac{\sin\theta}{\cos\theta} = \tan\left(\tfrac{1}{4}\pi - \tfrac{1}{2}x\right).$$

Cancelling $\sin\theta$ is allowed because $\cos x = 2\sin\theta\cos\theta \neq 0$ forces $\sin\theta \neq 0$. This proves $(*)$.

Part (i)

We want $\frac{1}{4}\pi - \frac{1}{2}x = \frac{1}{8}\pi$, which gives $x = \frac{1}{4}\pi$. Since $\sec\frac{1}{4}\pi = \sqrt{2}$ and $\tan\frac{1}{4}\pi = 1$,

$$\tan\tfrac{1}{8}\pi = \sqrt{2} - 1.$$

For the second result, notice that $\frac{11}{24}\pi = \frac{3}{24}\pi + \frac{8}{24}\pi = \frac{1}{8}\pi + \frac{1}{3}\pi$, and we know $\tan\frac{1}{3}\pi = \sqrt{3}$. The addition formula gives

$$\tan\tfrac{11}{24}\pi = \frac{\tan\frac{1}{8}\pi + \tan\frac{1}{3}\pi}{1 - \tan\frac{1}{8}\pi\tan\frac{1}{3}\pi} = \frac{(\sqrt{2}-1)+\sqrt{3}}{1-(\sqrt{2}-1)\sqrt{3}} = \frac{\sqrt{3}+\sqrt{2}-1}{\sqrt{3}-\sqrt{6}+1},$$

which is the required result.

Part (ii)

Rather than rationalise a denominator with three terms, it is quicker to show that

$$(2+\sqrt{2}+\sqrt{3}+\sqrt{6})(\sqrt{3}-\sqrt{6}+1) = \sqrt{3}+\sqrt{2}-1.$$

Multiplying $\sqrt{3}-\sqrt{6}+1$ by each term of the first bracket in turn, using $\sqrt{12}=2\sqrt{3}$ and $\sqrt{18}=3\sqrt{2}$:

$$\begin{aligned}
2(\sqrt{3}-\sqrt{6}+1) &= 2\sqrt{3}-2\sqrt{6}+2,\\
\sqrt{2}(\sqrt{3}-\sqrt{6}+1) &= \sqrt{6}-2\sqrt{3}+\sqrt{2},\\
\sqrt{3}(\sqrt{3}-\sqrt{6}+1) &= 3-3\sqrt{2}+\sqrt{3},\\
\sqrt{6}(\sqrt{3}-\sqrt{6}+1) &= 3\sqrt{2}-6+\sqrt{6}.
\end{aligned}$$

Now we collect terms. The constants give $2+3-6=-1$. The $\sqrt{2}$ terms give $1-3+3=1$. The $\sqrt{3}$ terms give $2-2+1=1$. The $\sqrt{6}$ terms give $-2+1+1=0$. The total is $\sqrt{3}+\sqrt{2}-1$. The denominator $\sqrt{3}-\sqrt{6}+1 \approx 0.28$ is non-zero, so we may divide by it, and

$$\frac{\sqrt{3}+\sqrt{2}-1}{\sqrt{3}-\sqrt{6}+1} = 2+\sqrt{2}+\sqrt{3}+\sqrt{6}.$$

? **Ask yourself** Which value of x should go into $(*)$ for the final part?

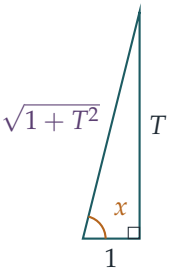
We need $\frac{1}{4}\pi - \frac{1}{2}x = \frac{1}{48}\pi$, so $\frac{1}{2}x = \frac{11}{48}\pi$ and $x = \frac{11}{24}\pi$. This is exactly the angle whose tangent we found in (i) and (ii).

Part (iii)

Let $T = \tan\frac{11}{24}\pi = 2+\sqrt{2}+\sqrt{3}+\sqrt{6}$. By $(*)$ with $x=\frac{11}{24}\pi$,

$$\tan\tfrac{1}{48}\pi = \sec\tfrac{11}{24}\pi - T.$$

For the secant, $\sec^2 x = 1+\tan^2 x$. Since $0 < \frac{11}{24}\pi < \frac{1}{2}\pi$, the cosine is positive, and so we take the positive root, $\sec\frac{11}{24}\pi = \sqrt{1+T^2}$. The triangle below shows the same thing.



Squaring T, the squares of the four terms give $4+2+3+6=15$. The cross terms give

$$2\big(2\sqrt{2}+2\sqrt{3}+2\sqrt{6}+\sqrt{6}+\sqrt{12}+\sqrt{18}\big) = 2\big(5\sqrt{2}+4\sqrt{3}+3\sqrt{6}\big) = 10\sqrt{2}+8\sqrt{3}+6\sqrt{6}.$$

So $1 + T^2 = 16 + 10\sqrt{2} + 8\sqrt{3} + 6\sqrt{6}$, and therefore

$$\tan \tfrac{1}{48}\pi = \sqrt{16 + 10\sqrt{2} + 8\sqrt{3} + 6\sqrt{6}} - 2 - \sqrt{2} - \sqrt{3} - \sqrt{6},$$

as required.

Checks

- $\sqrt{2} - 1 \approx 0.41421$, which matches $\tan 22.5°$.
- $2 + \sqrt{2} + \sqrt{3} + \sqrt{6} \approx 7.59575$, which matches $\tan 82.5°$. The fraction in (i) gives the same value.
- The final expression is $\sqrt{57.45\ldots} - 7.59575 \approx 0.0655435$, which matches $\tan 3.75° \approx 0.0655435$.
- A quick test of $(*)$: at $x = 0$ both sides equal 1, and at $x = \frac{1}{3}\pi$ both sides equal $2 - \sqrt{3} = \tan \frac{1}{12}\pi$.

Common slips

- Solving $\frac{1}{4}\pi - \frac{1}{2}x = \frac{1}{48}\pi$ wrongly and using $x = \frac{11}{48}\pi$. The $\frac{1}{2}$ must be undone.
- Missing cross terms when squaring a four-term surd. There are six of them, each doubled.
- Taking $\sec x = \sqrt{1 + \tan^2 x}$ without saying why the positive root is the right one.

Beyond STEP

Repeatedly halving angles in this way is how exact values such as $\tan \frac{\pi}{48}$, nested square roots and all, are built up. It is the algebraic side of the classical question of which regular polygons can be constructed with ruler and compass. Gauss showed that the answer depends on $\cos \frac{2\pi}{n}$ being expressible using only square roots, and Galois theory explains why. For a first look at fields and field extensions, which is the language of this story, try Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP II 2009, Question 4

Pure • Polynomial Divisibility

STEP II 2009, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q4

The key idea

Divisibility by $(x-1)^5$ is best handled by writing it out: $\mathrm{p}(x) = 1 + (x-1)^5\mathrm{q}(x)$ for some polynomial q. Differentiating shows that a repeated factor of $\mathrm{p}(x) - 1$ survives, one power lower, in $\mathrm{p}'(x)$. In the final part this pins down p' almost completely, so we find p by integrating p' rather than by wrestling with ten unknown coefficients.

Solution

Since $\mathrm{p}(x) - 1$ is divisible by $(x-1)^5$ and has degree 9, we can write

$$\mathrm{p}(x) = 1 + (x-1)^5\,\mathrm{q}(x),$$

where q is a polynomial of degree 4.

Part (i)

Putting $x = 1$ kills the second term, so

$$\mathrm{p}(1) = 1.$$

Part (ii)

Differentiating with the product rule,

$$\begin{aligned}\mathrm{p}'(x) &= 5(x-1)^4\,\mathrm{q}(x) + (x-1)^5\,\mathrm{q}'(x)\\ &= (x-1)^4\big(5\mathrm{q}(x) + (x-1)\mathrm{q}'(x)\big).\end{aligned}$$

The bracket is a polynomial, so $\mathrm{p}'(x)$ is exactly divisible by $(x-1)^4$, as required.

Part (iii)

Exactly the same argument with $x+1$ in place of $x-1$ applies: writing $\mathrm{p}(x) = -1 + (x+1)^5\,\mathrm{r}(x)$ gives $\mathrm{p}(-1) = -1$ and shows that $\mathrm{p}'(x)$ is divisible by $(x+1)^4$.

So $\mathrm{p}'(x)$ is divisible by both $(x-1)^4$ and $(x+1)^4$.

Ask yourself Why does that make $\mathrm{p}'(x)$ divisible by the product $(x-1)^4(x+1)^4$?

Write $\mathrm{p}'(x) = (x-1)^4\mathrm{s}(x)$. Every root of $(x+1)^4$ is -1, which is not a root of $(x-1)^4$, so the factor $(x+1)^4$ must divide $\mathrm{s}(x)$. (Equivalently, the linear factor $x+1$ appears four times in p', and none of those copies can be used up by $(x-1)^4$.) Hence $(x-1)^4(x+1)^4 = (x^2-1)^4$ divides $\mathrm{p}'(x)$.

Now p' has degree 8, the same as $(x^2-1)^4$, so the quotient is a non-zero constant:

$$\mathrm{p}'(x) = k(x^2-1)^4 = k\left(x^8 - 4x^6 + 6x^4 - 4x^2 + 1\right).$$

Integrating,

$$\mathrm{p}(x) = k\,\mathrm{F}(x) + c, \qquad \mathrm{F}(x) = \frac{x^9}{9} - \frac{4x^7}{7} + \frac{6x^5}{5} - \frac{4x^3}{3} + x.$$

Notice that F is odd, so $\mathrm{F}(-1) = -\mathrm{F}(1)$. The two values we know give

$$k\mathrm{F}(1) + c = \mathrm{p}(1) = 1, \qquad -k\mathrm{F}(1) + c = \mathrm{p}(-1) = -1.$$

Adding, $c = 0$; subtracting, $k\mathrm{F}(1) = 1$. With a common denominator of 315,

$$\mathrm{F}(1) = \frac{35 - 180 + 378 - 420 + 315}{315} = \frac{128}{315}, \qquad \text{so} \qquad k = \frac{315}{128}.$$

Therefore

$$\mathrm{p}(x) = \frac{1}{128}\left(35x^9 - 180x^7 + 378x^5 - 420x^3 + 315x\right).$$

We should also check that this polynomial really works, since so far we have only shown that any valid p must have this form. It has degree 9. Also $\mathrm{p}(1) - 1 = 0$ and $\mathrm{p}'(x) = \frac{315}{128}(x-1)^4(x+1)^4$, so the first four derivatives of $\mathrm{p}(x) - 1$ all vanish at $x = 1$; thus $x = 1$ is a root of $\mathrm{p}(x) - 1$ of multiplicity at least 5, i.e. $(x-1)^5$ divides $\mathrm{p}(x) - 1$. Since p is odd, $\mathrm{p}(x) + 1 = -\big(\mathrm{p}(-x) - 1\big)$, which is divisible by $(-x-1)^5 = -(x+1)^5$. So both conditions hold, and this p is the unique answer.

Checks

- The coefficients $35, -180, 378, -420, 315$ sum to 128, confirming $\mathrm{p}(1) = 1$.
- Dividing $\mathrm{p}(x) - 1$ repeatedly by $x - 1$ with exact fractions gives remainder zero exactly five times, and likewise $\mathrm{p}(x) + 1$ by $x + 1$.
- p is odd, as it must be: if p works then so does $-\mathrm{p}(-x)$, and the answer is unique.

Common slips

- Trying to find all ten coefficients of p directly from ten simultaneous equations. It works in principle, but integrating p' is far quicker.
- Forgetting the constant of integration, or not using both $\mathrm{p}(1) = 1$ and $\mathrm{p}(-1) = -1$ to fix it and k.

Beyond STEP

The fact behind part (ii) is that a is a root of multiplicity m of f exactly when $\mathrm{f}(a) = \mathrm{f}'(a) = \cdots = \mathrm{f}^{(m-1)}(a) = 0$. Prescribing values and derivatives at points like this is *Hermite interpolation*, and the polynomial found here is a "smoothstep" curve, rising from -1 to 1 as flatly as possible at both ends, which is used in computer graphics. The argument that coprime factors divide a product is a piece of ring theory: polynomials over a field have unique factorisation, just like the integers. For more, see Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP II 2009, Question 5

Pure • Nested Square Roots

STEP II 2009, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q5

The key idea

The opening expansion is the whole question in disguise: it tells us that $x+2\sqrt{x-1}$ is a perfect square, and the same trick shows $x-2\sqrt{x-1}$ is one too. Once the nested roots are unpicked, every integrand collapses to something elementary. The one thing to watch is that $\sqrt{a^2}=|a|$, not a, so we must keep track of signs, and that is exactly what makes part (ii) interesting.

Solution

The expansion

For $x \geq 1$,

$$(\sqrt{x-1}+1)^2 = (x-1)+2\sqrt{x-1}+1 = x+2\sqrt{x-1}.$$

In the same way $(\sqrt{x-1}-1)^2 = x-2\sqrt{x-1}$. Taking non-negative square roots, and remembering that $\sqrt{a^2}=|a|$,

$$\sqrt{x+2\sqrt{x-1}} = \sqrt{x-1}+1, \qquad \sqrt{x-2\sqrt{x-1}} = \left|\sqrt{x-1}-1\right|.$$

The first needs no modulus because $\sqrt{x-1}+1>0$. The second does: $\sqrt{x-1}-1$ is negative for $1 \leq x < 2$ and non-negative for $x \geq 2$.

Part (i)

On $[5,10]$ we have $\sqrt{x-1} \geq 2 > 1$, so $\left|\sqrt{x-1}-1\right| = \sqrt{x-1}-1$. The numerator is therefore

$$(\sqrt{x-1}+1)+(\sqrt{x-1}-1) = 2\sqrt{x-1},$$

and the integrand is simply 2. Hence

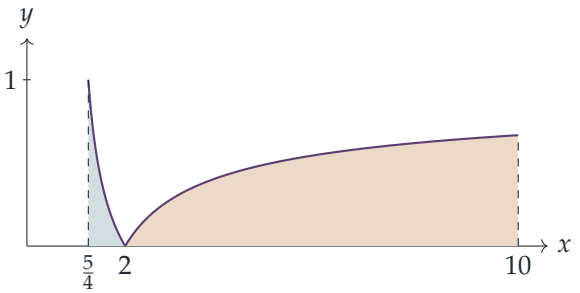
$$\int_5^{10} \frac{\sqrt{x+2\sqrt{x-1}}+\sqrt{x-2\sqrt{x-1}}}{\sqrt{x-1}}\,\mathrm{d}x = \int_5^{10} 2\,\mathrm{d}x = 10.$$

Part (ii)

By the expansion,

$$y = \frac{\left|\sqrt{x-1}-1\right|}{\sqrt{x-1}} = \left|1 - \frac{1}{\sqrt{x-1}}\right|.$$

So $y \geq 0$ throughout, and the curve touches the x-axis at $x = 2$, where $\sqrt{x-1} = 1$.



Ask yourself Why can't we just drop the square root and integrate $1 - \dfrac{1}{\sqrt{x-1}}$ over the whole range?

Because $\sqrt{x - 2\sqrt{x-1}}$ is never negative, but $\sqrt{x-1} - 1$ is negative for $x < 2$. Ignoring the modulus would count the piece on $[\frac{5}{4}, 2]$ as negative area. So we split at $x = 2$.

On $[\frac{5}{4}, 2]$, $\sqrt{x-1} \leq 1$, so $y = \dfrac{1}{\sqrt{x-1}} - 1$, and

$$\int_{5/4}^{2} \left(\frac{1}{\sqrt{x-1}} - 1\right) \mathrm{d}x = \left[2\sqrt{x-1} - x\right]_{5/4}^{2} = (2-2) - \left(1 - \tfrac{5}{4}\right) = \tfrac{1}{4}.$$

On $[2, 10]$, $\sqrt{x-1} \geq 1$, so $y = 1 - \dfrac{1}{\sqrt{x-1}}$, and

$$\int_{2}^{10} \left(1 - \frac{1}{\sqrt{x-1}}\right) \mathrm{d}x = \left[x - 2\sqrt{x-1}\right]_{2}^{10} = (10-6) - (2-2) = 4.$$

The total area is therefore $\frac{1}{4} + 4 = \frac{17}{4}$.

Part (iii)

The second root in the numerator has a different shape, but the same trick works with $x + 1$ in place of $x - 1$:

$$\left(\sqrt{x+1} - 1\right)^2 = (x+1) - 2\sqrt{x+1} + 1 = x - 2\sqrt{x+1} + 2.$$

For $x \geq \frac{5}{4}$ we have $\sqrt{x+1} > 1$, so $\sqrt{x - 2\sqrt{x+1} + 2} = \sqrt{x+1} - 1$, with no modulus needed. Writing $\sqrt{x^2-1} = \sqrt{x-1}\,\sqrt{x+1}$ (valid since both factors are positive for

$x > 1$), the integrand becomes

$$\frac{(\sqrt{x-1}+1)+(\sqrt{x+1}-1)}{\sqrt{x-1}\sqrt{x+1}} = \frac{\sqrt{x-1}+\sqrt{x+1}}{\sqrt{x-1}\sqrt{x+1}} = \frac{1}{\sqrt{x+1}} + \frac{1}{\sqrt{x-1}}.$$

The two ± 1 terms cancel, which is the point of the slightly odd-looking numerator. Hence

$$\begin{aligned}\int_{5/4}^{10}\Big(\frac{1}{\sqrt{x+1}} + \frac{1}{\sqrt{x-1}}\Big)\mathrm{d}x &= \Big[2\sqrt{x+1}+2\sqrt{x-1}\Big]_{5/4}^{10} \\ &= \big(2\sqrt{11}+6\big) - \big(3+1\big) \\ &= 2\sqrt{11}+2,\end{aligned}$$

using $\sqrt{9/4} = \frac{3}{2}$ and $\sqrt{1/4} = \frac{1}{2}$ at the lower limit.

Checks

- At $x = 5$: $\sqrt{5+4} = 3$ and $\sqrt{5-4} = 1$, so the integrand in (i) is $(3+1)/2 = 2$, as claimed.
- In (ii), at $x = \frac{5}{4}$ the curve has height $|1-2| = 1$ and at $x = 10$ height $1 - \frac{1}{3} = \frac{2}{3}$, matching the sketch; the first piece is a small sliver of area $\frac{1}{4}$, which is plausible.
- Simpson's rule on the original integrands gives 10.000, 4.250 and 8.6332, agreeing with 10, $\frac{17}{4}$ and $2\sqrt{11}+2 \approx 8.6332$.

Common slips

- Writing $\sqrt{x-2\sqrt{x-1}} = \sqrt{x-1}-1$ for all x. In (ii) this gives $4 - \frac{1}{4} = \frac{15}{4}$ instead of $\frac{17}{4}$.
- In (iii), not spotting that $x - 2\sqrt{x+1} + 2$ is also a perfect square, and trying a substitution instead.

Beyond STEP

The identity $\sqrt{a^2} = |a|$ is a small example of a big theme in analysis: functions defined by formulas can hide corners and changes of sign, and you must look at where an expression changes sign before integrating it. Denesting radicals such as $\sqrt{x+2\sqrt{x-1}}$ also has a surprisingly deep algebraic theory, connected to field extensions. For the analysis side, Abbott's *Understanding Analysis* is a good next step.

WORKED SOLUTION

STEP II 2009, Question 6

Pure • Bounding Infinite Series

STEP II 2009, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q6

The key idea

We can't sum $\sum 1/F_i$ exactly, but we can trap it. The recurrence gives two easy comparisons: each Fibonacci number is less than twice the one before it, and more than twice the one two places before it. The first comparison bounds the tail of the series below by a geometric series. The second relates the tail to a shifted copy of itself, which bounds it above. Adding a few terms exactly before using either bound tightens the estimates as much as we like.

Solution

Adding successive pairs gives

$$F_3 = 2, \quad F_4 = 3, \quad F_5 = 5, \quad F_6 = 8, \quad F_7 = 13, \quad F_8 = 21, \quad F_9 = 34, \quad F_{10} = 55.$$

All the F_n are positive, and $F_{n+1} = F_n + F_{n-1} > F_n$ for $n \geq 2$, so the sequence is strictly increasing from F_2 onwards.

The series converges. For $i \geq 4$ we have $F_i = F_{i-1} + F_{i-2} > 2F_{i-2}$, since $F_{i-1} > F_{i-2}$ when $i - 2 \geq 2$. As $F_1 = 1 \geq 2^{-1/2}$ and $F_2 = 1 \geq 2^0$, while $F_3 = 2 \geq 2^{1/2}$, induction in steps of two gives $F_i \geq 2^{(i-2)/2}$ for every i. So $0 < 1/F_i \leq (1/\sqrt{2})^{i-2}$, and S converges by comparison with a geometric series of ratio $1/\sqrt{2} < 1$. This lets us manipulate the sums below freely. We'll also use one fact repeatedly. If $a_i \leq b_i$ for every i, and $a_i < b_i$ for at least one i, then $\sum a_i < \sum b_i$ for convergent series.

Part (i)

The lower bound. For $i \geq 4$, $F_{i-2} < F_{i-1}$, so

$$F_i = F_{i-1} + F_{i-2} < 2F_{i-1} \qquad \Longrightarrow \qquad \frac{1}{F_i} > \frac{1}{2F_{i-1}} \qquad (i \geq 4),$$

which is the first result. (At $i = 3$ we get equality, $F_3 = 2F_2$, which is why the inequality starts at $i = 4$.)

Applying this repeatedly, for $i \geq 4$,

$$\frac{1}{F_i} > \frac{1}{2F_{i-1}} > \frac{1}{2^2F_{i-2}} > \cdots > \frac{1}{2^{i-3}F_3} = \frac{1}{2^{i-2}}.$$

Summing over $i \geq 4$,

$$\sum_{i=4}^{\infty} \frac{1}{F_i} > \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots = \frac{1}{2},$$

so

$$S = 1 + 1 + \tfrac{1}{2} + \sum_{i=4}^{\infty} \frac{1}{F_i} > \tfrac{5}{2} + \tfrac{1}{2} = 3.$$

The upper bound. Now use the other comparison: $1/F_i < 1/(2F_{i-2})$ for $i \geq 4$, from $F_i > 2F_{i-2}$ above. We apply it to every term from $i = 5$ onwards and set $j = i - 2$:

$$\sum_{i=5}^{\infty} \frac{1}{F_i} < \frac{1}{2} \sum_{i=5}^{\infty} \frac{1}{F_{i-2}} = \frac{1}{2} \sum_{j=3}^{\infty} \frac{1}{F_j} = \frac{1}{2}(S - 2).$$

The left-hand side is S minus its first four terms, $1 + 1 + \frac{1}{2} + \frac{1}{3} = \frac{17}{6}$. Hence

$$S - \tfrac{17}{6} < \tfrac{1}{2}S - 1 \qquad \Longrightarrow \qquad \tfrac{1}{2}S < \tfrac{11}{6} \qquad \Longrightarrow \qquad S < \tfrac{11}{3} = 3\tfrac{2}{3}.$$

Ask yourself Why start the upper-bound comparison at $i = 5$ rather than at $i = 4$?

Starting at $i = 4$ works but gives $S - \frac{5}{2} < \frac{1}{2}(S - 1)$, that is $S < 4$, which isn't good enough. The more terms we add exactly before estimating, the less the factor $\frac{1}{2}$ costs us. That idea drives part (ii).

Part (ii)

Write $S_k = \sum_{i=1}^{k} 1/F_i$ for the partial sums and $T_k = \sum_{i=k}^{\infty} 1/F_i$ for the tails, so $S = S_{k-1} + T_k$. Part (i) used two general bounds, both valid for $k \geq 3$.

Lower: exactly as before, $1/F_i > 1/(2^{i-k}F_k)$ for $i > k$, so

$$T_{k+1} > \frac{1}{F_k}\left(\tfrac{1}{2} + \tfrac{1}{4} + \cdots\right) = \frac{1}{F_k}, \qquad \text{so} \qquad S > S_k + \frac{1}{F_k}.$$

Upper: $1/F_i < 1/(2F_{i-2})$ for $i \geq k + 2 \geq 5$ gives $T_{k+2} < \frac{1}{2}T_k$, and $T_{k+2} = T_k - \frac{1}{F_k} - \frac{1}{F_{k+1}}$, so

$$T_k < 2\left(\frac{1}{F_k} + \frac{1}{F_{k+1}}\right), \qquad \text{so} \qquad S < S_{k-1} + 2\left(\frac{1}{F_k} + \frac{1}{F_{k+1}}\right).$$

With $k = 3$ these are the bounds 3 and $3\frac{2}{3}$ of part (i). Now take $k = 5$ in both.

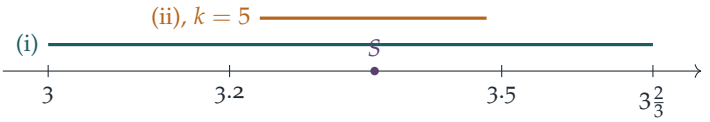
For the lower bound, $S_5 = 1 + 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{5} = \frac{91}{30}$, so

$$S > \frac{91}{30} + \frac{1}{5} = \frac{97}{30} > \frac{96}{30} = 3.2.$$

For the upper bound, $S_4 = \frac{17}{6}$ and $F_5 = 5$, $F_6 = 8$, so

$$S < \frac{17}{6} + 2\left(\frac{1}{5} + \frac{1}{8}\right) = \frac{17}{6} + \frac{13}{20} = \frac{170 + 39}{60} = \frac{209}{60} < \frac{210}{60} = 3.5.$$

Together these give $3.2 < S < 3.5$, as required. ($k = 4$ isn't enough: it gives only $\frac{19}{6} < S < \frac{107}{30}$, about $3.17 < S < 3.57$.)



Checks

- Adding up the first 200 terms numerically gives $S \approx 3.35989$. This lies inside both intervals, and the bounds for $k = 3, \ldots, 6$ all bracket it: $3 < \frac{19}{6} < \frac{97}{30} < \frac{197}{60} < S < \frac{2681}{780} < \frac{209}{60} < \frac{107}{30} < \frac{11}{3}$.
- Test the comparisons on actual values: $F_7 = 13 < 2 \times 8$ and $13 > 2 \times 5$.

Common slips

- Using $1/F_i > 1/(2F_{i-1})$ at $i = 3$, where it's an equality. That breaks the strictness, or produces the wrong geometric sum.
- Manipulating S as a number ("$S - \frac{17}{6} < \frac{1}{2}S - 1$") without first knowing that the series converges.
- Giving up on the upper bound because the comparison started at $i = 4$ only gives $S < 4$. Add more terms exactly before estimating.

Beyond STEP

The value is $S \approx 3.3598856\ldots$, the *reciprocal Fibonacci constant*. It is known to be irrational (a result of André-Jeannin, 1989), but no closed form is known. The comparisons here are ratio tests in disguise. F_{i+1}/F_i tends to the golden ratio $\varphi = \frac{1}{2}(1 + \sqrt{5})$, so the tail behaves like a geometric series with ratio $1/\varphi \approx 0.618$, between our crude ratios $\frac{1}{2}$ and $1/\sqrt{2}$. Comparison tests and tail estimates like these are core to a first analysis course. Abbott's *Understanding Analysis* is an excellent place to meet them properly.

WORKED SOLUTION

STEP II 2009, Question 7

Pure • Integration by Recognition

STEP II 2009, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q7

The key idea

Integrals like these look hopeless until you notice that they are *exactly* the derivative of something. The first result tells us what derivatives of $(x-a)^n e^{bx}\sqrt{1+x^2}$ look like: a power of $(x-a)$, the exponential, a cubic, all over $\sqrt{1+x^2}$. So each integral is really a matching exercise: read off n, a and b, and check the cubic agrees. When the polynomial in the numerator is a quartic rather than a cubic, we have to work out where the extra degree comes from.

Solution

The derivative

Differentiate the product of three factors, writing $\frac{\mathrm{d}}{\mathrm{d}x}\sqrt{1+x^2} = \frac{x}{\sqrt{1+x^2}}$:

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= n(x-a)^{n-1}e^{bx}\sqrt{1+x^2} + b(x-a)^n e^{bx}\sqrt{1+x^2} + (x-a)^n e^{bx}\frac{x}{\sqrt{1+x^2}} \\ &= \frac{(x-a)^{n-1}e^{bx}}{\sqrt{1+x^2}}\Big[\, n(1+x^2) + b(x-a)(1+x^2) + x(x-a)\,\Big],\end{aligned}$$

where we took out the common factor $(x-a)^{n-1}e^{bx}/\sqrt{1+x^2}$ (multiplying the first two terms through by $\sqrt{1+x^2}$ to put them over the common denominator). Expanding the bracket,

$$q(x) = bx^3 + (n+1-ab)x^2 + (b-a)x + (n-ab).$$

Since $b \neq 0$, the coefficient of x^3 is non-zero, so q really is a cubic. This is the required result. We will use it in the form

$$\int \frac{(x-a)^{n-1}e^{bx}\,q(x)}{\sqrt{1+x^2}}\,\mathrm{d}x = (x-a)^n e^{bx}\sqrt{1+x^2} + C. \qquad (*)$$

Part (i)

Matching the factors, we need $a = 4$, $b = 4$ and $n - 1 = 14$, so $n = 15$. With these values

$$q(x) = 4x^3 + (16-16)x^2 + (4-4)x + (15-16) = 4x^3 - 1,$$

which is exactly the cubic in the integrand. So by $(*)$

$$\int \frac{(x-4)^{14}e^{4x}(4x^3-1)}{\sqrt{1+x^2}}\,dx = (x-4)^{15}e^{4x}\sqrt{1+x^2} + C.$$

Part (ii)

Now the polynomial is a quartic, so it can't be q directly.

Ask yourself How can a quartic fit a formula that only produces cubics?

The power of $(x-a)$ in $(*)$ is ours to choose. If the quartic has a factor $(x-1)$, we can move it across to join $(x-1)^{21}$, leaving a cubic. Test $x = 1$: $12 - 1 - 11 = 0$, so $(x-1)$ is indeed a factor, and dividing out,

$$12x^4 - x^2 - 11 = (x-1)(12x^3 + 12x^2 + 11x + 11).$$

(You can also spot this as $(x^2-1)(12x^2+11)$.) The integrand is therefore

$$\frac{(x-1)^{22}e^{12x}(12x^3+12x^2+11x+11)}{\sqrt{1+x^2}}.$$

Take $a = 1$, $b = 12$, $n = 23$:

$$q(x) = 12x^3 + (24-12)x^2 + (12-1)x + (23-12) = 12x^3 + 12x^2 + 11x + 11,$$

which matches. Hence

$$\int \frac{(x-1)^{21}e^{12x}(12x^4-x^2-11)}{\sqrt{1+x^2}}\,dx = (x-1)^{23}e^{12x}\sqrt{1+x^2} + C.$$

Part (iii)

Again the polynomial is a quartic, but this time $x = 2$ gives $64 + 8 - 2 = 70 \neq 0$, so $(x-2)$ is not a factor and the trick from (ii) fails.

Ask yourself If $(x-2)$ isn't a factor, where does the extra degree come from?

Count degrees. The integrand's numerator has degree $6 + 4 = 10$, but $(*)$ with $(x-2)^6$ only gives degree $6 + 3 = 9$. An extra factor of x in the answer would raise the degree by one, so try differentiating xy, where y is the function with $a = 2$, $b = 4$, $n = 7$:

$$y = (x-2)^7e^{4x}\sqrt{1+x^2}, \qquad q(x) = 4x^3 + (8-8)x^2 + (4-2)x + (7-8) = 4x^3 + 2x - 1.$$

By the product rule $(xy)' = y + xy'$. Putting y over the denominator $\sqrt{1+x^2}$ as well,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(xy) &= \frac{(x-2)^7\mathrm{e}^{4x}(1+x^2)}{\sqrt{1+x^2}} + \frac{x(x-2)^6\mathrm{e}^{4x}(4x^3+2x-1)}{\sqrt{1+x^2}} \\ &= \frac{(x-2)^6\mathrm{e}^{4x}}{\sqrt{1+x^2}}\Big[(x-2)(1+x^2) + 4x^4 + 2x^2 - x\Big].\end{aligned}$$

Since $(x-2)(1+x^2) = x^3 - 2x^2 + x - 2$, the bracket is

$$x^3 - 2x^2 + x - 2 + 4x^4 + 2x^2 - x = 4x^4 + x^3 - 2,$$

exactly the quartic we want. Therefore

$$\int \frac{(x-2)^6\mathrm{e}^{4x}(4x^4+x^3-2)}{\sqrt{1+x^2}}\,\mathrm{d}x = x(x-2)^7\mathrm{e}^{4x}\sqrt{1+x^2} + C.$$

Checks

- Differentiate each answer and compare: this is the definitive check, and in each case it is just $(*)$ again (plus the product rule in (iii)).
- Quick degree count: differentiating an answer of polynomial degree d (ignoring the square root and exponential) gives a numerator of degree $d+2$. In (i), $15+2 = 14+3$; in (ii), $23+2 = 21+4$; in (iii), $8+2 = 6+4$.
- Numerically, Simpson's rule for the integral in (iii) over $[-1, 1.2]$ gives $-104.41424\ldots$, agreeing with $F(1.2) - F(-1)$ for $F(x) = x(x-2)^7\mathrm{e}^{4x}\sqrt{1+x^2}$ to about 13 significant figures; (i) and (ii) agree just as well.

Common slips

- Forgetting that n is one more than the power in the integrand: in (i) the answer has $(x-4)^{15}$, not $(x-4)^{14}$.
- In (ii), trying to match the quartic directly and concluding there is no solution, instead of first absorbing the factor $(x-1)$.
- Leaving out the constant of integration.

Beyond STEP

The method here is "integration by recognition": guess the shape of an antiderivative, differentiate it, and match coefficients. Turned into an algorithm, it becomes the Risch algorithm, which computer algebra systems use to decide whether an elementary function has an elementary antiderivative, and to find it if so. A related idea, guessing a solution of a given form with unknown coefficients, is the method of undetermined coefficients for linear differential equations, covered in Boyce and DiPrima, *Elementary Differential Equations*. For more integrals in the STEP style, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 2009, Question 8

Pure • Vector Geometry

STEP II 2009, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q8

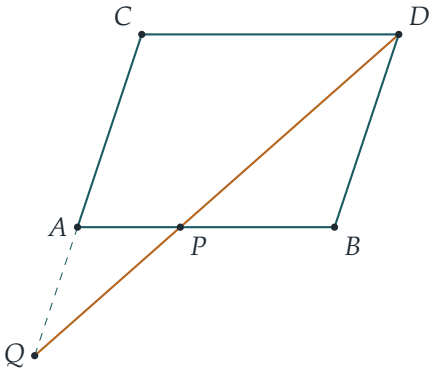
The key idea

Rewrite each definition as "start at one point and move a multiple of a side". Then $\mathbf{p} - \mathbf{b} = \lambda(\mathbf{a} - \mathbf{b})$ and $\mathbf{q} - \mathbf{c} = \mu(\mathbf{a} - \mathbf{c})$, so the lengths BP and CQ are just $\lambda\, AB$ and $\mu\, AC$, and the length condition turns into a very simple equation. To show three points are collinear, show that two of the vectors joining them are parallel.

Solution

The diagram

Since $\mathbf{p} = \mathbf{b} + \lambda(\mathbf{a} - \mathbf{b})$ with $0 < \lambda < 1$, the point P lies on the segment AB, strictly between A and B. Since $\mathbf{q} = \mathbf{c} + \mu(\mathbf{a} - \mathbf{c})$ with $\mu > 1$, the point Q lies on the line CA, beyond A: start at C, go to A, and keep going. The figure (drawn with $\lambda = 0.6$) also shows the point D that appears below.



Finding μ

From the rearrangements above,

$$BP = |\mathbf{p} - \mathbf{b}| = \lambda|\mathbf{a} - \mathbf{b}| = \lambda\, AB, \qquad CQ = |\mathbf{q} - \mathbf{c}| = \mu|\mathbf{a} - \mathbf{c}| = \mu\, AC,$$

using $\lambda > 0$ and $\mu > 0$ to take them out of the modulus. The condition $CQ \times BP = AB \times AC$ becomes $\lambda\mu\, AB \cdot AC = AB \cdot AC$. Since A, B, C are distinct, $AB \cdot AC \neq 0$, so

$$\mu = \frac{1}{\lambda}.$$

This does satisfy $\mu > 1$, because $0 < \lambda < 1$.

The fixed point D

Put $\mathbf{d} = -\mathbf{a} + \mathbf{b} + \mathbf{c}$. We compare $\overrightarrow{PQ}$ and $\overrightarrow{PD}$. First,

$$\begin{aligned}\mathbf{d} - \mathbf{p} &= -\mathbf{a} + \mathbf{b} + \mathbf{c} - \lambda\mathbf{a} - (1-\lambda)\mathbf{b} \\ &= -(1+\lambda)\mathbf{a} + \lambda\mathbf{b} + \mathbf{c}.\end{aligned}$$

Next, with $\mu = 1/\lambda$,

$$\begin{aligned}\mathbf{q} - \mathbf{p} &= \Big(\frac{1}{\lambda} - \lambda\Big)\mathbf{a} + \Big(1 - \frac{1}{\lambda}\Big)\mathbf{c} - (1-\lambda)\mathbf{b} \\ &= \frac{1-\lambda}{\lambda}\Big[(1+\lambda)\mathbf{a} - \lambda\mathbf{b} - \mathbf{c}\Big],\end{aligned}$$

using $\frac{1}{\lambda} - \lambda = \frac{(1-\lambda)(1+\lambda)}{\lambda}$ and $1 - \frac{1}{\lambda} = -\frac{1-\lambda}{\lambda}$.

Ask yourself How do we see that these two vectors are parallel?

The bracket is exactly $-(\mathbf{d} - \mathbf{p})$. So

$$\mathbf{q} - \mathbf{p} = -\frac{1-\lambda}{\lambda}\,(\mathbf{d} - \mathbf{p}), \qquad \text{that is} \qquad \mathbf{d} = \mathbf{p} - \frac{\lambda}{1-\lambda}\,(\mathbf{q} - \mathbf{p}).$$

The coefficient is finite because $\lambda \neq 1$, and $\mathbf{q} \neq \mathbf{p}$ because Q is not on the line AB. Hence D lies on the line PQ. The vector $\mathbf{d}$ involves only $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, so this is the same point for every λ, as required. Because the coefficient $-\lambda/(1-\lambda)$ is negative, P lies between Q and D, as in the figure.

The quadrilateral $ABDC$

We have

$$\mathbf{d} - \mathbf{c} = \mathbf{b} - \mathbf{a} \quad \text{and} \quad \mathbf{d} - \mathbf{b} = \mathbf{c} - \mathbf{a},$$

so $\overrightarrow{CD} = \overrightarrow{AB}$ and $\overrightarrow{BD} = \overrightarrow{AC}$. Opposite sides are equal and parallel, so $ABDC$ is a **parallelogram**, and D is the vertex opposite A. (It is non-degenerate because A, B, C are not collinear.)

Checks

- A geometric cross-check: with $BD \parallel AC$, triangles PAQ and PBD are similar, so $AQ/BD = AP/BP$. Here $AQ = (\mu - 1)AC$, $BD = AC$, $AP = (1 - \lambda)AB$ and $BP = \lambda AB$, giving $\mu - 1 = (1 - \lambda)/\lambda$, which is $\mu = 1/\lambda$ again.
- With $A = (0, 0)$, $B = (4, 0)$, $C = (1, 3)$ and $\lambda = 0.6$ (the figure) we get $P = (1.6, 0)$, $Q = (-\frac{2}{3}, -2)$ and $D = (5, 3)$. The gradients of QP and PD are both $\frac{15}{17}$.
- We also tested 20,000 random triangles in three dimensions. We solved the length condition for μ numerically, and in every case it agreed with $1/\lambda$ to 10^{-13}, with P, Q, D collinear.

Common slips

- Writing $BP = (1 - \lambda)AB$. The coefficient of $\mathbf{b}$ in $\mathbf{p}$ is $1 - \lambda$, but it is the weight at A that measures the distance from B.
- Calling $ABDC$ "a parallelogram" without checking the order of the vertices. Here it is $A \to B \to D \to C$, so AD and BC are the diagonals.

Beyond STEP

Expressions like $\lambda\mathbf{a} + (1 - \lambda)\mathbf{b}$, whose coefficients add up to 1, are *affine combinations*. They are the natural language of points (as opposed to vectors), and collinearity is an affine property: it does not depend on the choice of origin. This leads on to barycentric coordinates and to classical results such as Menelaus' theorem, which describes exactly when a line cuts the three sides of a triangle. Strang's *Introduction to Linear Algebra* develops the vector side, and Siklos's *Advanced Problems in Mathematics* has more STEP-style vector geometry.

WORKED SOLUTION

STEP II 2009, Question 9

Mechanics • Centre of Mass

STEP II 2009, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q9

The key idea

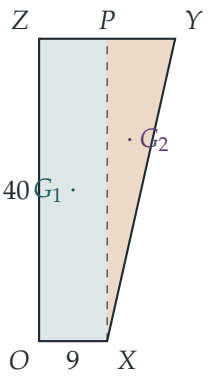
Every distance we need is measured perpendicular to one fixed plane: the edge OZ in part (i), and the back of the tank in part (ii), which contains OZ. So we only ever need one coordinate. Split each shape into pieces whose centres of mass we know (rectangles and a triangle), take moments about that line or plane, and divide by the total mass. The tank is the same calculation one level up: its pieces are two trapezia and three rectangles of sheet metal, and then a block of liquid.

Solution

Part (i)

Take O as the origin, with the x-axis along OX and the y-axis along OZ. Then $X = (9, 0)$, $Z = (0, 40)$ and, since YZ is parallel to OX with length 18, $Y = (18, 40)$. As a check, $XY^2 = 9^2 + 40^2 = 1681 = 41^2$, as it should be. The distance from OZ is just the x-coordinate.

Draw the line from X parallel to OZ, meeting ZY at $P = (9, 40)$. This splits the lamina into a rectangle $OXPZ$ and a right-angled triangle XYP.



Let the lamina have mass 1 per cm^2, so mass equals area.

- Rectangle $OXPZ$: area $9 \times 40 = 360$, centre G_1 at $x = 4.5$.
- Triangle XYP: area $\frac{1}{2} \times 9 \times 40 = 180$. The centre of mass of a uniform triangle is at the mean of its vertices, so G_2 has $x = \frac{1}{3}(9 + 18 + 9) = 12$.

The whole lamina has area 540. Taking moments about OZ,

$$540\,\bar{x} = 360 \times 4.5 + 180 \times 12 = 1620 + 2160 = 3780,$$

so $\bar{x} = 7$. The centre of mass is 7 cm from OZ, as required.

Part (ii)

Now let the sheet metal have mass 1 per cm^2. The back $OTWZ$ lies in the plane $x = 0$, and every point of the tank keeps the x-coordinate it had in part (i), since the width d runs perpendicular to the end faces. So we take moments about the back. Each rectangular face is uniform, so its centre of mass is at its centre, which has the same x-coordinate as the midpoint of the corresponding edge of the trapezium.

Piece	Area (cm^2)	x of centre (cm)	Moment
Two ends	$2 \times 540 = 1080$	7	7560
Base $OXUT$	$9d$	4.5	$40.5d$
Back $OTWZ$	$40d$	0	0
Front $XUVY$	$41d$	$\frac{1}{2}(9 + 18) = 13.5$	$553.5d$
Total	$1080 + 90d$		$7560 + 594d$

Ask yourself Where is the centre of the sloping front face?

The front is a rectangle with one pair of edges XU and YV, which lie at $x = 9$ and $x = 18$. Its centre is halfway between them, at $x = 13.5$. Its area uses the slant length $XY = 41$, not the horizontal width 9.

So the empty tank has

$$\bar{x} = \frac{7560 + 594d}{1080 + 90d}.$$

Every coefficient is divisible by 18: $7560 = 18 \times 420$, $594 = 18 \times 33$, $1080 = 18 \times 60$ and $90 = 18 \times 5$. So

$$\bar{x} = \frac{420 + 33d}{60 + 5d} = \frac{3(140 + 11d)}{5(12 + d)} \text{ cm},$$

which is the required distance from the back.

The filled tank

The liquid fills a prism with cross-section the trapezium of part (i) and length d. Its volume is $540d$, so its mass is $540dk$ in our units (mass per unit volume is k times the

metal's mass per unit area, which we took to be 1). Every cross-section is the same trapezium, so the liquid's centre of mass is at $x = 7$, as in part (i).

With $d = 20$, the tank has mass $1080 + 1800 = 2880$ and moment $7560 + 594 \times 20 = 19440$ about the back. The liquid has mass $10800k$ and moment $75600k$. So the filled tank has

$$\bar{x} = \frac{19440 + 75600k}{2880 + 10800k} = \frac{27 + 105k}{4 + 15k} \text{ cm},$$

after dividing top and bottom by 720. This is the distance of the centre of mass of the filled tank from the back.

Checks

- With $k = 0$ (no liquid, or liquid of negligible density) the answer is $\frac{27}{4} = 6.75$, and the empty-tank formula at $d = 20$ gives $\frac{3\times360}{5\times32} = 6.75$ too.
- As $k \to \infty$ the liquid dominates and $\bar{x} \to \frac{105}{15} = 7$, the liquid's own centre of mass.
- The answer always lies between 6.75 and 7, as a weighted average of the two must. For example, $k = 1$ gives $\frac{132}{19} \approx 6.947$.
- A numerical check (shoelace centroid of the trapezium, a Monte Carlo estimate giving $\bar{x} \approx 6.9994$, and exact fractions for both tank formulae) agrees with all the results.

Common slips

- Using $9d$ instead of $41d$ for the area of the front: the face is sloping, so its width is the slant length XY.
- Leaving out the back because its moment is zero. It still contributes $40d$ to the total mass.
- Counting only one end, or giving the tank a lid.

Beyond STEP

Here we found centres of mass by splitting into pieces with known centres. At university, the centre of mass of a lamina or solid is defined by integrals such as $\bar{x} = \frac{1}{M}\iint x\,\rho\,\mathrm{d}A$, and the "split into pieces" method is just the fact that integrals add over a region cut into parts. The same integrals give moments of inertia, which control how bodies rotate. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both ideas carefully.

WORKED SOLUTION

STEP II 2009, Question 10

Mechanics • Collisions and Restitution

STEP II 2009, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q10

The key idea

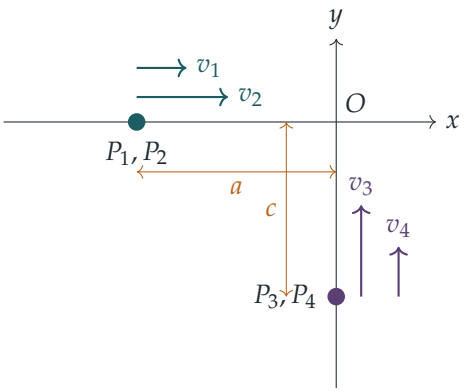
There are four collisions, but only the first two need any mechanics: they fix the four speeds just afterwards. Everything else is timing. Two particles that "collide at O" while travelling along different axes simply reach O at the same instant, so each later collision gives one equation of the form distance/speed = distance/speed. Divide one timing equation by the other and the unknown distances drop out, leaving a condition on the masses.

Solution

Setting up

We model the four objects as particles moving on smooth axes, so between collisions each moves with constant velocity. Let P_2 start at distance a from O on the negative x-axis and P_3 at distance c from O on the negative y-axis. We must show $a = c$.

Take $t = 0$ to be the moment of the first two (simultaneous) collisions. These happen where P_2 and P_3 are sitting, so at $t = 0$ the particles P_1 and P_2 are both at distance a from O, and P_3 and P_4 are both at distance c from O.



The first two collisions

Suppose a particle of mass M moving with speed u hits a stationary particle of mass m directly, with coefficient of restitution e, and afterwards they have velocities V and w in the original direction of motion. Conservation of momentum and Newton's experimental law give

$$Mu = MV + mw, \qquad w - V = eu,$$

so

$$w = \frac{M(1+e)u}{M+m}, \qquad V = \frac{(M-em)u}{M+m}.$$

Applying this to P_1 hitting P_2, and to P_4 hitting P_3, the velocities towards O just after $t = 0$ are

$$v_1 = \frac{(m_1 - em_2)u}{m_1+m_2}, \qquad v_2 = \frac{m_1(1+e)u}{m_1+m_2},$$
$$v_4 = \frac{(m_4 - em_3)u}{m_3+m_4}, \qquad v_3 = \frac{m_4(1+e)u}{m_3+m_4}.$$

Clearly $v_2, v_3 > 0$. Since P_1 and P_4 do later reach O, they must still be moving towards it, so $v_1 > 0$ and $v_4 > 0$ too.

Using the later collisions

Ask yourself How can two particles on perpendicular axes "collide"?

The only point the two axes share is O. So "P_2 and P_3 collide at O" just means they arrive at O at the same time, and likewise for P_1 and P_4. Between $t = 0$ and their arrival at O each particle moves at constant speed (the question tells us the next collision any of them is involved in happens at O), so the arrival times are distance over speed:

$$\frac{a}{v_2} = \frac{c}{v_3} \qquad (P_2 \text{ and } P_3 \text{ meet at } O), \tag{1}$$
$$\frac{a}{v_1} = \frac{c}{v_4} \qquad (P_1 \text{ and } P_4 \text{ meet at } O). \tag{2}$$

Dividing (1) by (2), the distances cancel:

$$\frac{v_1}{v_2} = \frac{v_4}{v_3}.$$

From the formulae above, the factors u and $m_1 + m_2$ (or $m_3 + m_4$) cancel in each ratio, so this says

$$\frac{m_1 - em_2}{m_1(1+e)} = \frac{m_4 - em_3}{m_4(1+e)},$$

and multiplying through by $m_1 m_4(1+e)$,

$$m_4(m_1 - em_2) = m_1(m_4 - em_3), \qquad \text{so} \qquad e\,m_2 m_4 = e\,m_1 m_3.$$

Since $e > 0$ we may divide by e, so $m_1 m_3 = m_2 m_4$, that is,

$$\frac{m_1}{m_2} = \frac{m_4}{m_3} = k, \text{ say.}$$

Now go back to v_2 and v_3. Dividing top and bottom by m_2 and m_3 respectively,

$$v_2 = \frac{k(1+e)u}{k+1} = v_3.$$

So P_2 and P_3 leave the first collisions with equal speeds, and then (1) gives $a = c$. Hence P_2 and P_3 were initially equidistant from O, as required.

Checks

- Take $m_1 = 2$, $m_2 = 1$, $m_3 = 3$, $m_4 = 6$ (so $k = 2$), $e = \frac{1}{2}$, $u = 1$ and $a = c = 1.7$. Then $v_2 = v_3 = 1$ and $v_1 = v_4 = \frac{1}{2}$: P_2 and P_3 reach O together at $t = 1.7$, and P_1 and P_4 together at $t = 3.4$, later as the question says. A step-by-step simulation agrees.
- Note $v_2 - v_1 = eu > 0$, so P_1 really does arrive after P_2, consistent with "some time later".
- If $e = 0$ the argument breaks, as it should: then $v_1 = v_2$ and $v_3 = v_4$, so (2) is the same as (1), and any a and c with $a/c = v_2/v_3$ work. That is exactly why the question insists $e > 0$.

Common slips

- Measuring P_1's journey from its starting point. After $t = 0$ it starts from where P_2 was, at distance a from O, which is why the same a appears in both (1) and (2).
- Dividing by e without comment. The case $e = 0$ really is different.
- Trying to analyse the collision at O between P_2 and P_3. Nothing about its outcome is needed; only the fact that they get there together.

Beyond STEP

The mass condition $m_1 m_3 = m_2 m_4$ says the two collisions are scaled copies of each other: velocities after a direct impact depend only on the mass ratio and e, a first glimpse of dimensional analysis and similarity. At university, collisions are cleanest in the centre-of-mass frame, where the coefficient of restitution simply scales the relative velocity. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops momentum and collisions in this way.

WORKED SOLUTION

STEP II 2009, Question 11

Mechanics • Power and Resistance

STEP II 2009, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q11

The key idea

Treat the whole train as one body of mass $(n+1)M$ with total resistance $(n+1)R$. At power P the driving force at speed v is P/v, so Newton's second law gives the acceleration straight away. To find a *time* to reach a speed, write the acceleration as $\mathrm{d}v/\mathrm{d}t$, separate the variables and integrate. The last part is an energy balance: the engine's work goes into kinetic energy and into work against resistance.

Solution

Model the engine and trucks as particles moving together (the couplings stay taut and rigid), so the train is a single body. Write

$$m = (n+1)M, \qquad k = (n+1)R$$

for its total mass and total resistance.

The acceleration

Power is force times speed, so at maximum power and speed v the engine's tractive force is P/v. Resolving along the track, Newton's second law for the whole train gives

$$ma = \frac{P}{v} - k, \qquad \text{so} \qquad a = \frac{P-(n+1)Rv}{(n+1)Mv}.$$

The time to reach speed V

Put $a = \dfrac{\mathrm{d}v}{\mathrm{d}t}$ and separate the variables:

$$\frac{\mathrm{d}t}{\mathrm{d}v} = \frac{mv}{P-kv}, \qquad T = m\int_0^V \frac{v}{P-kv}\,\mathrm{d}v.$$

To integrate, split the fraction by writing the numerator in terms of the denominator, $v = \frac{1}{k}\big(P - (P - kv)\big)$:

$$\frac{v}{P - kv} = -\frac{1}{k} + \frac{P}{k} \cdot \frac{1}{P - kv}.$$

Hence

$$T = m\Big[-\frac{v}{k} - \frac{P}{k^2}\ln(P - kv)\Big]_0^V = -\frac{mV}{k} - \frac{mP}{k^2}\ln\Big(1 - \frac{kV}{P}\Big).$$

Since $m/k = M/R$ and $m/k^2 = M/\big((n+1)R^2\big)$,

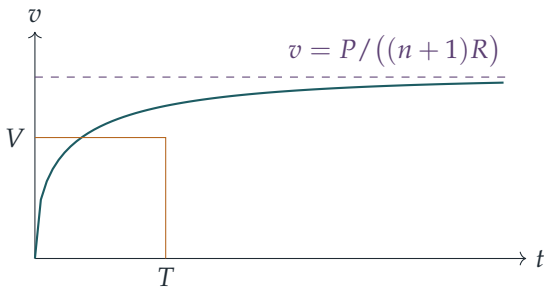
$$T = -\frac{MV}{R} - \frac{MP}{(n+1)R^2}\ln\Big(1 - \frac{(n+1)RV}{P}\Big).$$

Ask yourself Why must P exceed $(n+1)RV$, rather than this just being a condition for the logarithm to make sense?

Physically, the acceleration $(P - kv)/(mv)$ is positive while $v < P/k$ and would be zero at $v = P/k$. The train starts at $v = 0 < P/k$ and speeds up, but the acceleration dies away as v approaches P/k. The formula shows that it never gets there: as $V \to P/k$ from below, $\ln(1 - kV/P) \to -\infty$, so $T \to \infty$. So P/k is a terminal speed that the train approaches but never reaches in finite time, and speeds at or above it are never reached. Therefore speed V can be reached only if $V < P/k$, that is,

$$P > (n+1)RV,$$

as required. (Equivalently, $P \le kV$ would mean the engine at full power can't even balance the resistance at speed V.)



(The sketch is qualitative: the speed rises steeply at first, where the driving force P/v is large, and levels off towards the terminal speed.)

Part (i)

Let $x = \dfrac{(n+1)RV}{P} = \dfrac{kV}{P}$, which is small. Then $\ln(1-x) \approx -x - \frac{1}{2}x^2$, and using $V = Px/k$,

$$T = -\frac{mV}{k} - \frac{mP}{k^2}\ln(1-x) \approx -\frac{mP}{k^2}x + \frac{mP}{k^2}\Big(x + \tfrac{1}{2}x^2\Big) = \frac{mP}{2k^2}x^2 = \frac{mP}{2k^2} \cdot \frac{k^2V^2}{P^2} = \frac{mV^2}{2P}.$$

The first-order terms cancel exactly, which is why we needed the x^2 term. Hence

$$PT \approx \tfrac{1}{2}(n+1)MV^2.$$

Interpretation. PT is the work done by the engine in time T, and $\tfrac{1}{2}(n+1)MV^2$ is the kinetic energy gained by the train. So when $(n+1)RV$ is small compared with P (the resistance force times the speed is small next to the engine's power), almost all of the engine's work goes into kinetic energy and very little is lost to resistance: the train behaves as if there were no resistance.

Part (ii)

In time T the engine, working at constant power P, does work PT. Over the distance X the constant total resistance $(n+1)R$ does work $-(n+1)RX$; in other words the train does work $(n+1)RX$ against it. The train starts from rest and ends with speed V, and the track is horizontal, so there is no change in potential energy. By the work–energy principle,

$$PT = \tfrac{1}{2}(n+1)MV^2 + (n+1)RX.$$

Rearranging,

$$X = \frac{2PT - (n+1)MV^2}{2(n+1)R},$$

as required. This matches part (i): if resistance is negligible, the $(n+1)RX$ term drops out and we are left with $PT \approx \tfrac{1}{2}(n+1)MV^2$.

Checks

- Take $n = 3$, $M = 2$, $R = 1$, $P = 40$, $V = 6$ (so $(n+1)RV = 24 < 40$). The formula gives $T \approx 6.3258$, and then $X \approx 27.258$. Stepping the equation of motion forward in time numerically gives the same values to four significant figures.
- Dimensions: MV/R is (mass)(speed)/(force), which is a time, and so is MP/R^2; the logarithm is dimensionless.
- As $V \to P/\big((n+1)R\big)$, T grows without bound (for the numbers above, $T \approx 28$, 72, 118 at 90%, 99%, 99.9% of the terminal speed), which confirms the condition.
- For small x, the ratio $PT/\tfrac{1}{2}(n+1)MV^2$ is about $1 + \tfrac{2}{3}x$, tending to 1, as part (i) predicts.

Common slips

- Using the mass and resistance of the engine alone instead of the whole train: the engine pulls all $n+1$ vehicles.
- Writing $a = \mathrm{d}v/\mathrm{d}x$, or using $v\,\mathrm{d}v/\mathrm{d}x$, when a *time* is wanted. That route gives distance, not time.

- Expanding the logarithm only to first order in part (i): the x terms cancel, so you get $T \approx 0$ and lose the answer.

Beyond STEP

The equation $m\,\mathrm{d}v/\mathrm{d}t = P/v - k$ is a first-order nonlinear ODE that happens to be separable, and the terminal speed P/k is a stable equilibrium that solutions approach but never reach in finite time. This behaviour is typical near equilibria of smooth ODEs, as uniqueness of solutions shows. Part (ii) is the work–energy theorem, which becomes a central organising principle in university mechanics. For more, see Kleppner and Kolenkow, *An Introduction to Mechanics*, and for the ODE side Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP II 2009, Question 12

Probability and Statistics • Half-Normal Distribution

STEP II 2009, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q12

The key idea

The exponent $-2x^2$ is a disguised normal density. Writing $-2x^2 = -\dfrac{x^2}{2\sigma^2}$ gives $\sigma = \frac{1}{2}$, so f is exactly twice the density of $Y \sim \mathrm{N}(0, \frac{1}{4})$, cut off at 0. In other words, X has the same distribution as $|Y|$. Once you spot that, the normalising constant, $\mathrm{E}(X^2)$ and the median all come straight from facts about the normal distribution. Only $\mathrm{E}(X)$ needs a (very easy) integral.

Solution

Throughout, let $Y \sim \mathrm{N}(0, \frac{1}{4})$ and $Z \sim \mathrm{N}(0,1)$, so that Y has the same distribution as $\frac{1}{2}Z$ and its density is

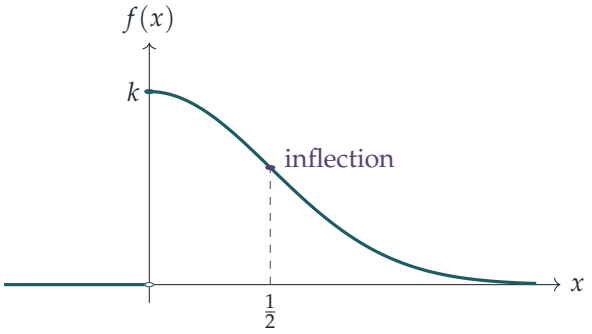
$$g(y) = \frac{1}{\frac{1}{2}\sqrt{2\pi}}\,\mathrm{e}^{-y^2/(2\cdot\frac{1}{4})} = \frac{2}{\sqrt{2\pi}}\,\mathrm{e}^{-2y^2}.$$

Part (i)

For $x \geq 0$, $f'(x) = -4kx\,\mathrm{e}^{-2x^2} \leq 0$, so f decreases from its maximum value k at $x = 0$, with zero gradient there. Also

$$f''(x) = k(16x^2 - 4)\,\mathrm{e}^{-2x^2},$$

which changes sign at $x = \frac{1}{2}$, so there is a point of inflection at $x = \frac{1}{2}$. As $x \to \infty$, $f(x) \to 0$ very rapidly, and $f(x) = 0$ for $x < 0$. The graph is the right-hand half of a bell curve (with $k \approx 1.60$ from part (ii)), and there is a jump at $x = 0$ from 0 up to k.



Part (ii)

The total probability must be 1. Since g is symmetric about 0 and integrates to 1 over the whole real line, its integral over $[0,\infty)$ is $\frac{1}{2}$:

$$\int_0^\infty \frac{2}{\sqrt{2\pi}}\,\mathrm{e}^{-2x^2}\,\mathrm{d}x = \frac{1}{2} \quad\Longrightarrow\quad \int_0^\infty \mathrm{e}^{-2x^2}\,\mathrm{d}x = \frac{\sqrt{2\pi}}{4}.$$

So $k\cdot\dfrac{\sqrt{2\pi}}{4} = 1$, giving

$$k = \frac{4}{\sqrt{2\pi}} = 2\sqrt{\frac{2}{\pi}} \approx 1.596.$$

Notice that $f(x) = 2g(x)$ for $x \geq 0$, which confirms that X is distributed like $|Y|$.

Part (iii)

For the mean, the factor x makes the integral elementary, because $x\,\mathrm{e}^{-2x^2}$ is a multiple of the derivative of e^{-2x^2}:

$$\mathrm{E}(X) = k\int_0^\infty x\,\mathrm{e}^{-2x^2}\,\mathrm{d}x = k\Big[-\tfrac{1}{4}\mathrm{e}^{-2x^2}\Big]_0^\infty = \frac{k}{4} = \frac{1}{\sqrt{2\pi}} \approx 0.399.$$

Ask yourself How do we find $\displaystyle\int_0^\infty x^2\,\mathrm{e}^{-2x^2}\,\mathrm{d}x$ without a long integration by parts?

Use the normal distribution again. The function $x^2 g(x)$ is even, so

$$\mathrm{E}(X^2) = \int_0^\infty x^2\cdot 2g(x)\,\mathrm{d}x = \int_{-\infty}^\infty x^2 g(x)\,\mathrm{d}x = \mathrm{E}(Y^2) = \mathrm{Var}(Y) = \frac{1}{4},$$

since $\mathrm{E}(Y) = 0$. (Integration by parts with $u = x$, $\mathrm{d}v = x\mathrm{e}^{-2x^2}\,\mathrm{d}x$ gives the same: $\int_0^\infty x^2\mathrm{e}^{-2x^2}\,\mathrm{d}x = \frac{1}{4}\int_0^\infty \mathrm{e}^{-2x^2}\,\mathrm{d}x = \frac{1}{4k}$.) Hence

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - \big(\mathrm{E}(X)\big)^2 = \frac{1}{4} - \frac{1}{2\pi} = \frac{\pi-2}{4\pi} \approx 0.0908.$$

Part (iv)

The median m satisfies $\mathrm{P}(X \le m) = \frac{1}{2}$. Since X is distributed like $|Y|$ and Y like $\frac{1}{2}Z$,

$$\mathrm{P}(X \le m) = \mathrm{P}(|Z| \le 2m) = 2\Phi(2m) - 1,$$

using the symmetry of the standard normal distribution. (Directly: $\mathrm{P}(X \le m) = \int_0^m 2g(x)\,\mathrm{d}x = 2\big(\mathrm{P}(Y \le m) - \frac{1}{2}\big)$.) Setting this equal to $\frac{1}{2}$ gives

$$\Phi(2m) = \frac{3}{4}.$$

From tables, $\Phi^{-1}(0.75) = 0.6745$, so $2m = 0.6745$ and

$$m \approx 0.337 \quad \text{(to 3 s.f.)}.$$

Checks

A numerical (Simpson's rule) integration of f over $[0, 10]$ with $k = 2\sqrt{2/\pi}$ gives total probability 1.0000, $\mathrm{E}(X) = 0.39894$ and $\mathrm{Var}(X) = 0.090845$, matching $1/\sqrt{2\pi}$ and $(\pi - 2)/(4\pi)$. Solving $\int_0^m f(x)\,\mathrm{d}x = \frac{1}{2}$ numerically gives $m = 0.33724$. A simulation of $|Y|$ for 200 000 normal values with standard deviation $\frac{1}{2}$ gave mean 0.399 and variance 0.091, with 50.0% of values below 0.337. Sense check: the median 0.337 is below the mean 0.399, as it should be for a distribution with a long right tail.

Common slips

- Taking $\sigma^2 = 2$ or $\sigma = \frac{1}{4}$ from the exponent: match $2x^2$ with $x^2/(2\sigma^2)$, which gives $\sigma^2 = \frac{1}{4}$.
- Forgetting the factor 2: over half the line, the normal density only integrates to $\frac{1}{2}$.
- Solving $\Phi(2m) = \frac{1}{2}$, which gives $m = 0$. The half of X's probability below m corresponds to the middle half of Y's, so the condition is $\Phi(2m) = \frac{3}{4}$.

Beyond STEP

X is a *half-normal* random variable, and $|Y|$ for Y normal is the simplest example of a *transformation of a random variable*. In general, if $X = h(Y)$ then the density of X is obtained by summing the density of Y over all preimages, which is the factor 2 here. The same idea shows that Z^2 has a chi-squared distribution with one degree of freedom, and that is the start of a lot of statistics. Grimmett and Welsh, *Probability: An Introduction*, treats functions of random variables carefully.

WORKED SOLUTION

STEP II 2009, Question 13

Probability and Statistics • Binomial Expected Cost

STEP II 2009, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q13

The key idea

The number of engines that fail is binomial, so the repair cost is a simple function of a binomial random variable, and its expectation is a short sum. The trick for reaching the neat form $(*)$ is to avoid summing the "replace everything" cases one by one: use the complement, and then factor $1-q^n = p(1+q+\cdots+q^{n-1})$ to pull out the factor of p.

Solution

We assume, as the question intends, that engines fail independently of one another, each with probability p, and write $q = 1-p$. All costs are measured in units of K until the end.

The Andover

Let X be the number of Andover engines that fail, so $X \sim \mathrm{B}(4,p)$, and let C_A be the repair cost. Then

$$C_A = \begin{cases} 0 & X = 0, \\ K & X = 1, \\ 4K & X \geq 2. \end{cases}$$

Now $\mathrm{P}(X=0) = q^4$ and $\mathrm{P}(X=1) = 4pq^3$, so $\mathrm{P}(X \geq 2) = 1 - q^4 - 4pq^3$. Hence

$$\begin{aligned} \mathrm{E}(C_A) &= K \cdot 4pq^3 + 4K\left(1 - q^4 - 4pq^3\right) \\ &= 4K(1-q^4) - 12Kpq^3. \end{aligned}$$

Since $1-q^4 = (1-q)(1+q+q^2+q^3) = p(1+q+q^2+q^3)$,

$$\begin{aligned} \mathrm{E}(C_A) &= 4Kp\left(1+q+q^2+q^3\right) - 4Kp \cdot 3q^3 \\ &= 4Kp\left(1+q+q^2-2q^3\right), \end{aligned}$$

which is $(*)$, as required.

The Basingstoke

Let $Y \sim \mathrm{B}(6, p)$ be the number of Basingstoke engines that fail, and C_B the repair cost. Then $C_B = YK$ when $Y \le 2$ and $C_B = 6K$ when $Y \ge 3$. With

$$\mathrm{P}(Y=0) = q^6, \qquad \mathrm{P}(Y=1) = 6pq^5, \qquad \mathrm{P}(Y=2) = 15p^2q^4,$$

we get

$$\begin{aligned}\mathrm{E}(C_B) &= K\cdot 6pq^5 + 2K\cdot 15p^2q^4 + 6K\bigl(1-q^6-6pq^5-15p^2q^4\bigr)\\ &= 6K(1-q^6) - 30Kpq^5 - 60Kp^2q^4.\end{aligned}$$

Using $1-q^6 = p(1+q+q^2+q^3+q^4+q^5)$ and taking out $6Kp$,

$$\mathrm{E}(C_B) = 6Kp\bigl(1+q+q^2+q^3+q^4+q^5-5q^5-10pq^4\bigr).$$

> **Ask yourself** The bracket still contains a p. How do we make it "a form similar to $(*)$"?

Form $(*)$ is a factor of p times a polynomial in q alone, so we replace the remaining p by $1-q$: $10pq^4 = 10q^4 - 10q^5$. Collecting terms,

$$\boxed{\mathrm{E}(C_B) = 6Kp\bigl(1+q+q^2+q^3-9q^4+6q^5\bigr).}$$

When is the Andover cost $\frac{2}{3}$ of the Basingstoke cost?

We need $\mathrm{E}(C_A) = \frac{2}{3}\mathrm{E}(C_B)$, that is

$$4Kp\bigl(1+q+q^2-2q^3\bigr) = 4Kp\bigl(1+q+q^2+q^3-9q^4+6q^5\bigr).$$

Both sides have the factor $4Kp$, and $K>0$, so this holds exactly when $p=0$ or

$$-2q^3 = q^3-9q^4+6q^5 \iff 3q^3-9q^4+6q^5=0 \iff 3q^3(1-q)(1-2q)=0.$$

So $q=0$, $q=1$ or $q=\frac{1}{2}$, giving $p=1$, $p=0$ or $p=\frac{1}{2}$. All three lie in $[0,1]$, so the required values are

$$p=0, \qquad p=\tfrac{1}{2}, \qquad p=1.$$

The value $p=0$ is the dull one: no engine ever fails, and both expected costs are zero. At $p=1$ every engine fails, costing $4K$ and $6K$. The interesting value is $p=\frac{1}{2}$.

Checks

At $p=\frac{1}{2}$, $(*)$ gives $\mathrm{E}(C_A) = 2K\bigl(1+\frac{1}{2}+\frac{1}{4}-\frac{1}{4}\bigr) = 3K$, and the Basingstoke formula gives $\mathrm{E}(C_B) = 3K\bigl(1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}-\frac{9}{16}+\frac{6}{32}\bigr) = \frac{9}{2}K$, and indeed $3 = \frac{2}{3}\cdot\frac{9}{2}$. We also checked both formulae exactly against direct enumeration over the binomial distribution for 200 rational values of p, and by simulation: at $p=0.3$, 2×10^5 simulated launches gave average costs $1.811K$ and $2.483K$, against the formulae's $1.805K$ and $2.485K$.

Common slips

- Charging XK rather than $4K$ when two or more Andover engines fail: when the threshold is passed, *all* engines are replaced, including the good ones.
- Leaving a p inside the Basingstoke bracket, so the answer isn't in the form asked for.
- Dividing by p in the final equation and losing $p = 0$, or dividing by q^3 and losing $p = 1$.

Beyond STEP

This is a small example of expected cost under a threshold policy, the bread and butter of reliability theory and operational research: a manager chooses the threshold to balance the cost of replacing good parts against the cost of more frequent repairs. The tail-sum manipulation we used, writing $\mathrm{P}(X \geq k)$ as one minus the first few terms, is the same idea that gives $\mathrm{E}(X) = \sum_{k \geq 1} \mathrm{P}(X \geq k)$ for non-negative integer variables. Grimmett and Welsh, *Probability: An Introduction*, develops these ideas carefully.

WORKED SOLUTION

STEP III 2009, Question 1

Pure • Coordinate Geometry of Circles

STEP III 2009, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

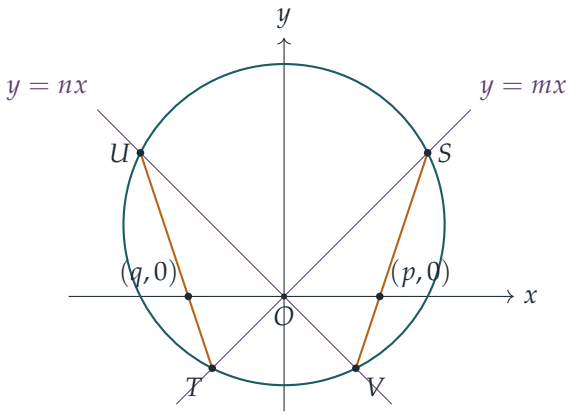
step.maths.org/questions/09-s3-q1

The key idea

All four points lie on lines through the origin, so each is described by a single number: its x-coordinate. Intersecting a circle with the line $y = mx$ gives a quadratic whose two roots are s and t, and we never need the roots themselves, only their sum and product. The final identity then comes from grouping the numerator of $p + q$ so that only st, $s + t$, uv and $u + v$ appear.

Solution

Throughout, S and T lie on the line $y = mx$ and U and V lie on $y = nx$, both lines through the origin O. We assume $m \neq n$ (otherwise all four points are on one line), and that SV and UT are not horizontal, so that they do meet $y = 0$; that is, $ms \neq nv$ and $mt \neq nu$.



The intercepts p and q

The line through $S(s, ms)$ and $V(v, nv)$ has gradient $\dfrac{nv - ms}{v - s}$, so its points are

$$y - ms = \frac{nv - ms}{v - s}(x - s).$$

Putting $y = 0$ and $x = p$ gives $p - s = -\dfrac{ms(v - s)}{nv - ms}$, so

$$p = \frac{s(nv - ms) - ms(v - s)}{nv - ms} = \frac{nsv - msv}{nv - ms} = \frac{(m - n)sv}{ms - nv},$$

as required. (If $v = s$ the line SV is vertical, so $p = s$; the formula still gives $\frac{(m-n)s^2}{(m-n)s} = s$.)

The line UT joins $T(t, mt)$ on $y = mx$ to $U(u, nu)$ on $y = nx$, exactly as SV joins S to V. Replacing s by t and v by u,

$$q = \frac{(m - n)tu}{mt - nu}.$$

The quadratic for s and t

A point (x, mx) lies on $x^2 + (y - c)^2 = r^2$ exactly when $x^2 + (mx - c)^2 = r^2$, that is,

$$(1 + m^2)x^2 - 2mc\,x + c^2 - r^2 = 0.$$

Both s and t satisfy this. Taking S and T to be the two (distinct) intersections of the line with the circle, s and t are the two roots, so by the relations between roots and coefficients,

$$st = \frac{c^2 - r^2}{1 + m^2}, \qquad s + t = \frac{2mc}{1 + m^2}.$$

Showing $p + q = 0$

The same argument on the line $y = nx$ shows that u and v are the roots of $(1 + n^2)x^2 - 2nc\,x + c^2 - r^2 = 0$, so

$$uv = \frac{c^2 - r^2}{1 + n^2}, \qquad u + v = \frac{2nc}{1 + n^2}.$$

Now add the two intercepts over a common denominator:

$$p + q = (m - n)\,\frac{sv(mt - nu) + tu(ms - nv)}{(ms - nv)(mt - nu)}.$$

Ask yourself How do we tame the numerator, which involves all four of s, t, u, v?

Expand it and pair terms by their coefficient m or n:

$$\begin{aligned} sv(mt-nu)+tu(ms-nv) &= m\,stv - n\,suv + m\,stu - n\,tuv \\ &= m\,st(u+v) - n\,uv(s+t). \end{aligned}$$

Every term is now a symmetric function we know. Substituting,

$$\begin{aligned} m\,st(u+v) - n\,uv(s+t) &= m\cdot\frac{c^2-r^2}{1+m^2}\cdot\frac{2nc}{1+n^2} - n\cdot\frac{c^2-r^2}{1+n^2}\cdot\frac{2mc}{1+m^2} \\ &= \frac{2mnc(c^2-r^2)}{(1+m^2)(1+n^2)} - \frac{2mnc(c^2-r^2)}{(1+m^2)(1+n^2)} = 0. \end{aligned}$$

The denominator $(ms-nv)(mt-nu)$ is non-zero by our assumption, so $p+q=0$, as required.

Checks

Take $c=1$, $r^2=5$, $m=1$, $n=-1$, as in the figure. The quadratics are $x^2-x-2=0$ and $x^2+x-2=0$, so $s=2$, $t=-1$ and $u=-2$, $v=1$. Then $st=-2=\frac{1-5}{2}$ and $s+t=1=\frac{2}{2}$, as predicted. The line from $S(2,2)$ to $V(1,-1)$ has gradient 3 and crosses $y=0$ at $p=\frac{4}{3}$, matching $\frac{2\cdot2\cdot1}{2+1}$; the line from $U(-2,2)$ to $T(-1,-1)$ crosses at $q=-\frac{4}{3}$. A computer check with 16 000 random circles and lines gave $|p+q|$ below 10^{-12} every time.

Common slips

- Pairing the wrong points: q comes from U and T, so the swap is $s\to t$, $v\to u$; swapping $s\leftrightarrow v$ instead gives p again.
- Trying to solve the quadratics for s,t,u,v explicitly. The square roots make the algebra hopeless; the grouping $m\,st(u+v)-n\,uv(s+t)$ is what makes it collapse.

Beyond STEP

The line $y=0$ meets the circle in a chord whose midpoint is O, since the centre $(0,c)$ is on the y-axis. So this question proves the *butterfly theorem*: if two chords ST and UV pass through the midpoint O of a chord, then SV and UT cut that chord at points equidistant from O. Our proof never used $c^2<r^2$, so it even works when O is outside the circle, which is a hint that the natural setting is projective geometry, where the result is a case of Desargues' involution theorem. Coxeter and Greitzer's *Geometry Revisited* is a lovely place to read more.

WORKED SOLUTION

STEP III 2009, Question 2

Pure • Power Series Solutions

STEP III 2009, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q2

The key idea

If y is a power series, then so are y', y'' and x^3y, and the differential equation says that one combined power series is identically zero. A power series that vanishes for every x has every coefficient zero, so the differential equation becomes a list of simple equations linking the a_n. The real work is bookkeeping: shift indices so that every series is written in the same power of x before comparing.

Solution

Part (i)

Differentiating term by term (we are told all the series converge, so we take this as allowed),

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(a_n x^n\right) = n a_n x^{n-1}.$$

The $n = 0$ term a_0 is a constant, and its derivative is 0, so the sum starts at $n = 1$:

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

That is the required result. Doing the same again, the $n = 1$ term a_1 of y' is constant, so

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}.$$

Written out as far as a_3:

$$\begin{aligned} y &= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots, \\ y' &= a_1 + 2a_2 x + 3a_3 x^2 + \cdots, \\ y'' &= 2a_2 + 6a_3 x + \cdots. \end{aligned}$$

Part (ii)

We write every term of $xy'' - y' + 4x^3y$ as a series in x^{n-1}. First,

$$xy'' = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-1}.$$

The $n = 1$ term of this sum would be $1 \cdot 0 \cdot a_1 = 0$, so we may start it at $n = 1$ too. Then

$$xy'' - y' = \sum_{n=1}^{\infty} \big[n(n-1) - n\big] a_n x^{n-1} = \sum_{n=1}^{\infty} n(n-2)a_n x^{n-1}.$$

For the last term, $4x^3y = \sum_{m=0}^{\infty} 4a_m x^{m+3}$, and putting $m = n - 4$ makes the power x^{n-1}:

$$4x^3y = \sum_{n=4}^{\infty} 4a_{n-4}\, x^{n-1}.$$

? **Ask yourself** Why can we compare coefficients at all?

The equation says $xy'' - y' + 4x^3y = 0$ for every x in the interval where the series converge, so the combined power series is the zero function there. A convergent power series is zero on an interval only if all its coefficients are zero (put $x = 0$ to get the constant term, divide by x, and repeat). So the coefficient of each power of x must vanish.

The coefficients are:

- x^0 ($n = 1$): $1 \cdot (-1)\, a_1 = 0$, so $a_1 = 0$, as required.
- x^1 ($n = 2$): $2 \cdot 0 \cdot a_2 = 0$, which holds automatically, so a_2 is free.
- x^2 ($n = 3$): $3 \cdot 1 \cdot a_3 = 0$, so $a_3 = 0$.
- x^{n-1} for $n \geq 4$: $n(n-2)a_n + 4a_{n-4} = 0$.

Since $n(n-2) \neq 0$ for $n \geq 4$, we may divide by it, which gives

$$a_n = -\frac{4}{n(n-2)}\, a_{n-4} \qquad (n \geq 4),$$

as required. (a_0 is also free, because no equation involves it on its own.)

The case $a_0 = 1$, $a_2 = 0$. The recurrence links a_n only to a_{n-4}, so the coefficients split into four chains according to n modulo 4. The chains starting at $a_1 = 0$, $a_2 = 0$ and $a_3 = 0$ are zero throughout, so only a_{4k} can be non-zero. Putting $n = 4k$ with $k \geq 1$,

$$a_{4k} = -\frac{4}{4k(4k-2)}\, a_{4k-4} = -\frac{1}{2k(2k-1)}\, a_{4k-4}.$$

Starting from $a_0 = 1$, each step multiplies by -1 and divides by the next two factors of a factorial, so

$$a_{4k} = \frac{(-1)^k}{(2k)!}.$$

(Check: $a_4 = -\frac{1}{2}$, $a_8 = -\frac{1}{12}\cdot(-\frac{1}{2}) = \frac{1}{24}$.) Hence

$$y = \sum_{k=0}^{\infty} \frac{(-1)^k x^{4k}}{(2k)!} = \sum_{k=0}^{\infty} \frac{(-1)^k (x^2)^{2k}}{(2k)!} = \cos(x^2),$$

by the Maclaurin series $\cos u = \sum_k (-1)^k u^{2k}/(2k)!$ with $u = x^2$. This is the required result.

The case $a_0 = 0$, $a_2 = 1$. Now the chains starting at a_0, a_1 and a_3 are all zero, and only a_{4k+2} survive. Putting $n = 4k+2$ with $k \geq 1$,

$$a_{4k+2} = -\frac{4}{(4k+2)\,4k}\, a_{4k-2} = -\frac{1}{(2k+1)\,2k}\, a_{4k-2},$$

and from $a_2 = 1$ we get $a_{4k+2} = (-1)^k/(2k+1)!$. Therefore

$$y = \sum_{k=0}^{\infty} \frac{(-1)^k x^{4k+2}}{(2k+1)!} = \sum_{k=0}^{\infty} \frac{(-1)^k (x^2)^{2k+1}}{(2k+1)!} = \sin(x^2).$$

Checks

Substitute directly. For $y = \sin(x^2)$, $y' = 2x\cos(x^2)$ and $y'' = 2\cos(x^2) - 4x^2\sin(x^2)$, so

$$xy'' - y' + 4x^3 y = 2x\cos(x^2) - 4x^3\sin(x^2) - 2x\cos(x^2) + 4x^3\sin(x^2) = 0.$$

The same works for $\cos(x^2)$. Numerically, the first 40 terms of the cosine series at $x = 1.2$ give $0.1304237\ldots$, matching $\cos(1.44)$. Also, both answers make sense at $x = 0$: $\cos 0 = 1 = a_0$, and $\sin(x^2) \approx x^2$ near 0, so $a_2 = 1$.

Common slips

- Comparing coefficients before shifting indices, so that a_n in one series is matched with a_n in another even though they multiply different powers of x.
- Forgetting that $a_3 = 0$, which is needed to kill the chain $a_3, a_7, a_{11}, \ldots$.
- Dividing by $n(n-2)$ without noting that it is non-zero for $n \geq 4$. At $n = 2$ it is zero, which is exactly why a_2 is free.

Beyond STEP

This is the method of Frobenius in its simplest form. The point $x = 0$ is a *regular singular point* of this equation, because the coefficient of y'' vanishes there, and the general theory explains why one ordinary power series succeeds here when it sometimes fails (the indicial roots, 0 and 2, differ by an integer). The same method gives Bessel and Legendre

functions. Justifying term-by-term differentiation needs uniform convergence, which is first-year analysis. For both ideas, see Boyce and DiPrima, *Elementary Differential Equations*, and Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2009, Question 3

Pure • Limits and Curve Sketching

STEP III 2009, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q3

The key idea

The function is undefined at $t = 0$ only because of a $0/0$, so power series tell us what it "should" be there. Part (ii) is a matter of putting everything over a common denominator; the even function that appears is $\frac{t}{2}\coth\frac{t}{2}$ in disguise. For part (iii), the sign of $f'(t)$ is decided entirely by its numerator, and that numerator is exactly $\mathrm{e}^t(1-t)-1$, which is why the inequality is there.

Solution

Part (i)

Using $\mathrm{e}^t = 1 + t + \frac{1}{2}t^2 + \frac{1}{6}t^3 + \cdots$, for $t \neq 0$ we may cancel a factor of t:

$$f(t) = \frac{t}{t + \frac{1}{2}t^2 + \frac{1}{6}t^3 + \cdots} = \frac{1}{1 + \frac{1}{2}t + \frac{1}{6}t^2 + \cdots} \longrightarrow 1 \quad \text{as } t \to 0,$$

since the denominator tends to 1. So $\lim_{t\to 0} f(t) = 1$.

By the quotient rule,

$$f'(t) = \frac{(\mathrm{e}^t - 1) - t\,\mathrm{e}^t}{(\mathrm{e}^t - 1)^2}.$$

Both numerator and denominator tend to 0, so expand each. The numerator is

$$\begin{aligned}\mathrm{e}^t - 1 - t\,\mathrm{e}^t &= \left(t + \tfrac{1}{2}t^2 + \tfrac{1}{6}t^3 + \cdots\right) - \left(t + t^2 + \tfrac{1}{2}t^3 + \cdots\right) \\ &= -\tfrac{1}{2}t^2 - \tfrac{1}{3}t^3 + \cdots,\end{aligned}$$

and the denominator is $\left(t + \frac{1}{2}t^2 + \cdots\right)^2 = t^2\left(1 + \frac{1}{2}t + \cdots\right)^2$. Dividing top and bottom by t^2,

$$f'(t) = \frac{-\frac{1}{2} - \frac{1}{3}t + \cdots}{\left(1 + \frac{1}{2}t + \cdots\right)^2} \longrightarrow -\frac{1}{2} \quad \text{as } t \to 0.$$

Part (ii)

Let $g(t) = f(t) + \frac{1}{2}t$. Over a common denominator,

$$g(t) = \frac{t}{e^t - 1} + \frac{t}{2} = \frac{2t + t(e^t - 1)}{2(e^t - 1)} = \frac{t\,(e^t + 1)}{2(e^t - 1)}.$$

Now replace t by $-t$ and multiply top and bottom by e^t:

$$g(-t) = \frac{-t\,(e^{-t} + 1)}{2(e^{-t} - 1)} = \frac{-t\,(1 + e^t)}{2(1 - e^t)} = \frac{t\,(e^t + 1)}{2(e^t - 1)} = g(t).$$

So $f(t) + \frac{1}{2}t$ is even, as required. (Dividing top and bottom by $e^{t/2}$ shows $g(t) = \frac{t}{2}\coth\frac{t}{2}$, a product of two odd functions, which is another way to see it.)

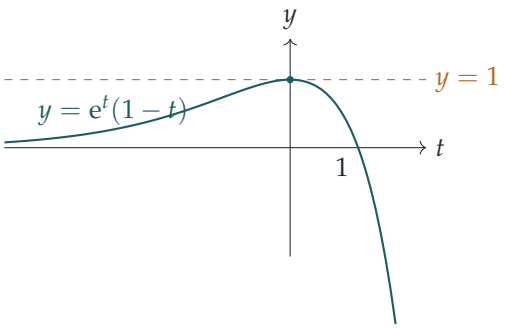
A useful consequence for the sketch: $f(t) - f(-t) = -t$, so the graph of f is the graph of an even function tilted by the line $y = -\frac{1}{2}t$.

Part (iii)

Let $h(t) = e^t(1 - t)$. Then

$$h'(t) = e^t(1 - t) - e^t = -t\,e^t,$$

which is positive for $t < 0$ and negative for $t > 0$. So h increases up to $t = 0$ and decreases afterwards: its only turning point is a maximum, $h(0) = 1$. Also $h(t) \to 0$ as $t \to -\infty$, and $h(t) < 0$ for $t > 1$.



The curve touches $y = 1$ at $t = 0$ and lies strictly below it everywhere else, so

$$e^t(1 - t) \le 1, \quad \text{with equality only at } t = 0.$$

Ask yourself How does this inequality tell us anything about $f'(t)$?

Look again at the numerator of $f'(t)$ from part (i):

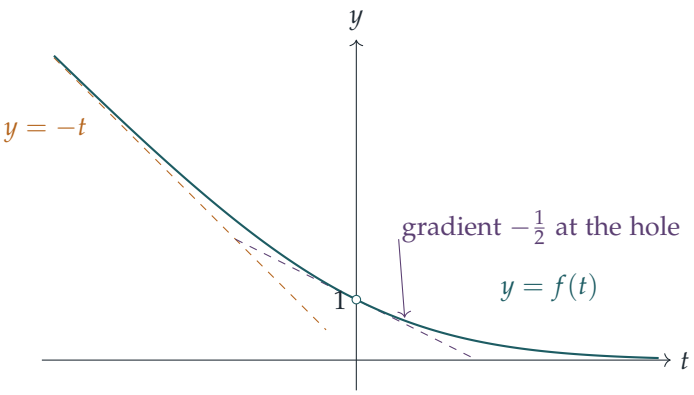
$$e^t - 1 - t\,e^t = e^t(1 - t) - 1.$$

For $t \neq 0$ this is strictly negative, by the strict inequality above. The denominator $(e^t - 1)^2$ is strictly positive for $t \neq 0$. Hence $f'(t) < 0$ for every $t \neq 0$; in particular $f'(t) \neq 0$, so f has no stationary points and is decreasing on each side of 0.

The sketch of f

Collect the facts.

- $f(t) > 0$ for all $t \neq 0$, since t and $\mathrm{e}^t - 1$ have the same sign.
- Near 0 the graph approaches $(0, 1)$ with gradient $-\frac{1}{2}$ (part (i)); the point itself is missing, as $f(0)$ is undefined.
- f is decreasing throughout (part (iii)).
- As $t \to \infty$, $f(t) = t/(\mathrm{e}^t - 1) \to 0^+$, because the exponential beats t.
- As $t \to -\infty$, $f(t) + t = \dfrac{t\,\mathrm{e}^t}{\mathrm{e}^t - 1} \to 0$, so $y = -t$ is an oblique asymptote. This also follows from part (ii): $f(t) = f(-t) - t$ and $f(-t) \to 0$.



Checks

- Part (ii) predicts $f(-1) - f(1) = 1$. Numerically $f(1) = 1/(\mathrm{e} - 1) \approx 0.58198$ and $f(-1) = \mathrm{e}/(\mathrm{e} - 1) \approx 1.58198$, which differ by exactly 1.
- The derivative formula evaluated at $t = \pm 10^{-4}$ gives -0.49998 and -0.50002, consistent with the limit $-\frac{1}{2}$.
- The coth form gives $g(t) \approx 1 + \frac{1}{12}t^2$ for small t, so $f(t) \approx 1 - \frac{1}{2}t + \frac{1}{12}t^2$, matching both limits in part (i).

Common slips

- Using L'Hôpital once on $f'(t)$ and stopping at another $0/0$. Expanding both numerator and denominator to order t^2 avoids this.
- Proving only $\mathrm{e}^t(1 - t) \leq 1$ and then claiming $f'(t) \neq 0$. You need the inequality to be *strict* for $t \neq 0$, which comes from $t = 0$ being the only stationary point of h.
- Drawing the sketch through $(0, 1)$ as a solid point: f is not defined there.

Beyond STEP

The expansion $\dfrac{t}{\mathrm{e}^t - 1} = \sum_{n \geq 0} B_n \dfrac{t^n}{n!}$ defines the Bernoulli numbers B_n. Part (i) shows $B_0 = 1$ and $B_1 = -\frac{1}{2}$, and part (ii) shows that $B_n = 0$ for every odd $n \geq 3$. These numbers turn up in the Euler–Maclaurin summation formula, in sums of powers $1^k + 2^k + \cdots + n^k$, and in Euler's formula for $\sum 1/n^{2k}$. Removing a "hole" like the one at $t = 0$ is the idea of a removable singularity; for a careful treatment of limits and continuity, read Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2009, Question 4

Pure • Laplace Transforms

STEP III 2009, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q4

The key idea

Parts (i)–(iii) are each one line of calculus applied to the defining integral: (i) just merges two exponentials, (ii) is a substitution and (iii) is integration by parts. Part (iv) can be done by integrating by parts twice, but it is neater to use (iii) twice and let the transforms of sin and cos pin each other down. For the final part, think of the target $\mathrm{e}^{-pt}\cos qt$ as built in three steps, $\sin t \to \cos t \to \cos qt \to \mathrm{e}^{-pt}\cos qt$, each step being one of the four results.

Solution

Write $\mathcal{L}[g](s)$ for the Laplace transform of a function g, so that $F = \mathcal{L}[f]$. "Suitable" means the integrals converge; in particular we assume $\mathrm{e}^{-st}f(t) \to 0$ as $t \to \infty$ for $s > 0$.

Part (i)

Combining the exponentials,

$$\mathcal{L}\left[\mathrm{e}^{-bt}f(t)\right](s) = \int_0^\infty \mathrm{e}^{-st}\mathrm{e}^{-bt}f(t)\,\mathrm{d}t = \int_0^\infty \mathrm{e}^{-(s+b)t}f(t)\,\mathrm{d}t = F(s+b),$$

which is the definition of F evaluated at $s+b$. This is legitimate because $s+b > 0$.

Part (ii)

Substitute $u = at$, so $t = u/a$ and $\mathrm{d}t = \mathrm{d}u/a$. Since $a > 0$ the limits 0 and ∞ stay as they are, so

$$\mathcal{L}\left[f(at)\right](s) = \int_0^\infty \mathrm{e}^{-st}f(at)\,\mathrm{d}t = \frac{1}{a}\int_0^\infty \mathrm{e}^{-(s/a)u}f(u)\,\mathrm{d}u = \frac{1}{a}F\left(\frac{s}{a}\right),$$

again using $s/a > 0$.

Part (iii)

Integrate by parts, differentiating e^{-st} and integrating $f'(t)$:

$$\begin{aligned}\mathcal{L}[f'](s) &= \Big[e^{-st}f(t)\Big]_0^\infty - \int_0^\infty (-s)e^{-st}f(t)\,dt \\ &= \big(0 - f(0)\big) + s\int_0^\infty e^{-st}f(t)\,dt \\ &= sF(s) - f(0),\end{aligned}$$

where the upper limit vanishes because $e^{-st}f(t) \to 0$ as $t \to \infty$ (part of f being suitable).

Part (iv)

Let $S(s) = \mathcal{L}[\sin t](s)$ and $C(s) = \mathcal{L}[\cos t](s)$. Both integrals converge for $s > 0$, because $|e^{-st}\sin t| \le e^{-st}$, and similarly for cos; and $e^{-st}\sin t$ and $e^{-st}\cos t$ both tend to 0, so (iii) applies to each.

Applying (iii) with $f = \sin$, whose derivative is cos and with $\sin 0 = 0$:

$$C(s) = sS(s).$$

Applying (iii) with $f = \cos$, whose derivative is $-\sin$ and with $\cos 0 = 1$:

$$-S(s) = sC(s) - 1.$$

(The transform of $-\sin t$ is $-S(s)$, since a constant factor comes straight out of the integral.) Substituting the first equation into the second,

$$-S(s) = s^2S(s) - 1 \quad\Longrightarrow\quad (s^2+1)S(s) = 1 \quad\Longrightarrow\quad S(s) = \frac{1}{s^2+1},$$

as required. As a by-product,

$$C(s) = \frac{s}{s^2+1}.$$

The final part

? **Ask yourself** How do we turn $\sin t$ into $\cos qt$ without any new integration?

Differentiation turns sin into cos (that's (iii)), rescaling t turns $\cos t$ into $\cos qt$ (that's (ii)), and multiplying by an exponential is (i). So we go in three steps.

Step 1: $\cos t$. By (iii) with $f(t) = \sin t$, using (iv) for F,

$$\mathcal{L}[\cos t](s) = s\cdot\frac{1}{s^2+1} - \sin 0 = \frac{s}{s^2+1}.$$

(This is the by-product of part (iv).)

Step 2: $\cos qt$. By (ii) with $f(t) = \cos t$ and $a = q > 0$,

$$\mathcal{L}[\cos qt](s) = \frac{1}{q}\cdot\frac{s/q}{(s/q)^2+1} = \frac{1}{q}\cdot\frac{sq}{s^2+q^2} = \frac{s}{s^2+q^2}.$$

Step 3: $\mathrm{e}^{-pt}\cos qt$. By (i) with $f(t)=\cos qt$ and $b=p>0$, we replace s by $s+p$:

$$\boxed{\mathcal{L}\left[\mathrm{e}^{-pt}\cos qt\right](s)=\frac{s+p}{(s+p)^2+q^2}} \qquad (s>0).$$

Checks

- Putting $p\to 0$ and $q=1$ gives $s/(s^2+1)$, the transform of $\cos t$ from Step 1. Putting $q\to 0$ gives $1/(s+p)$, and indeed $\int_0^\infty \mathrm{e}^{-(s+p)t}\,\mathrm{d}t = 1/(s+p)$.
- As $s\to\infty$, $sF(s)\to 1=\cos 0$, as it must: for large s the weight $s\mathrm{e}^{-st}$ concentrates near $t=0$, so $sF(s)$ tends to the value of the function at 0.
- Numerically, with $s=1$, $p=\frac{1}{2}$, $q=2$ the formula gives $1.5/6.25=0.24$, and Simpson's rule on the integral agrees. The computer checked (i)–(iv) and the final answer at random parameter values to within 10^{-12}.

Common slips

- In (ii), forgetting the factor $1/a$ from $\mathrm{d}t=\mathrm{d}u/a$, or writing $F(as)$ instead of $F(s/a)$.
- In Step 3, shifting only some of the s's: *every* s becomes $s+p$, so the numerator is $s+p$, not s.
- Getting the sign of $f(0)$ wrong in (iii): the boundary term is $0-f(0)$.

Beyond STEP

These rules are the engine of the Laplace transform method for differential equations: (iii) turns derivatives into multiplication by s, so a linear ODE with constant coefficients becomes an algebraic equation for $F(s)$, which you solve and then "look up" in a table built from results like the one here. The shift rule (i) is exactly why damped oscillations $\mathrm{e}^{-pt}\cos qt$ show up as poles at $s=-p\pm \mathrm{i}q$. Boyce and DiPrima, *Elementary Differential Equations*, has a full chapter on the method.

WORKED SOLUTION

STEP III 2009, Question 5

Pure • Symmetric Functions of Roots

STEP III 2009, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q5

The key idea

Everything here is about the three *elementary symmetric functions*

$$e_1 = x + y + z, \qquad e_2 = yz + zx + xy, \qquad e_3 = xyz,$$

because once we know them, x, y and z are the roots of a known cubic $t^3 - e_1 t^2 + e_2 t - e_3 = 0$. Each new piece of information comes from multiplying two of the given power sums together and seeing which cross terms appear. The recurrence then falls out of the cubic: each of x, y, z satisfies it, so each of their powers does too.

Solution

Write $S_n = x^n + y^n + z^n$, so we are given $S_1 = 1$, $S_2 = 2$, $S_3 = 3$.

The value of $yz + zx + xy$

Squaring S_1,

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2(yz + zx + xy),$$

so $1 = 2 + 2(yz + zx + xy)$, giving

$$yz + zx + xy = -\tfrac{1}{2},$$

as required.

The value of $\sum x^2 y$, and then xyz

Write $T = x^2y + x^2z + y^2z + y^2x + z^2x + z^2y$ for the sum of all six terms of the form (one variable squared)$\times$(a different variable). Multiplying S_2 by S_1 gives nine terms: the three cubes and the six terms of T:

$$(x^2 + y^2 + z^2)(x + y + z) = (x^3 + y^3 + z^3) + T.$$

Hence $2 \times 1 = 3 + T$, and so

$$T = -1,$$

which is the second result.

Ask yourself Which product of known quantities contains both T and xyz?

Try $(x+y+z)(yz+zx+xy)$. Multiplying out gives nine terms. Six of them have the shape $x \cdot xy = x^2y$, and these are exactly the six terms of T. The other three are $x \cdot yz$, $y \cdot zx$ and $z \cdot xy$, each equal to xyz. So

$$(x+y+z)(yz+zx+xy) = T + 3xyz.$$

Substituting the values we know, $1 \times \left(-\frac{1}{2}\right) = -1 + 3xyz$, so $3xyz = \frac{1}{2}$ and

$$xyz = \tfrac{1}{6},$$

as required.

The recurrence

We now know $e_1 = 1$, $e_2 = -\frac{1}{2}$ and $e_3 = \frac{1}{6}$. Since

$$(t-x)(t-y)(t-z) = t^3 - e_1t^2 + e_2t - e_3 = t^3 - t^2 - \tfrac{1}{2}t - \tfrac{1}{6},$$

each of x, y, z is a root of this cubic. In particular

$$x^3 = x^2 + \tfrac{1}{2}x + \tfrac{1}{6},$$

and similarly for y and z. Multiplying by x^{n-2} gives

$$x^{n+1} = x^n + \tfrac{1}{2}x^{n-1} + \tfrac{1}{6}x^{n-2},$$

and the same with y or z in place of x. Adding the three equations,

$$S_{n+1} = S_n + \tfrac{1}{2}S_{n-1} + \tfrac{1}{6}S_{n-2},$$

so

$$a = 1, \qquad b = \tfrac{1}{2}, \qquad c = \tfrac{1}{6}.$$

Ask yourself Does "for all n" cause any trouble with negative powers?

Multiplying by x^{n-2} is harmless for $n \geq 2$. For smaller n we need negative powers of x, y and z, and these exist because none of them is zero: their product is $\frac{1}{6} \neq 0$. So the relation holds for every integer n. (It holds even if x, y, z are not all real, and in fact they aren't: the cubic has one real root, about 1.431, and two complex conjugate roots.)

Checks

- With $n = 2$ the recurrence should reproduce the given data, using $S_0 = 3$: indeed $S_2 + \frac{1}{2}S_1 + \frac{1}{6}S_0 = 2 + \frac{1}{2} + \frac{1}{2} = 3 = S_3$.
- The recurrence predicts $S_4 = 3 + 1 + \frac{1}{6} = \frac{25}{6}$ and $S_5 = \frac{25}{6} + \frac{3}{2} + \frac{1}{3} = 6$. Solving the cubic numerically (roots 1.43085 and $-0.21542 \pm 0.26471\,\mathrm{i}$) and adding fourth and fifth powers gives 4.16667 and 6.00000, in agreement. The same roots satisfy the recurrence for negative n too.

Common slips

- Getting the signs in the cubic wrong: it is $t^3 - e_1 t^2 + e_2 t - e_3$, so here $b = -e_2 = +\frac{1}{2}$ and $c = e_3 = \frac{1}{6}$.
- Forgetting that $S_0 = 3$, not 1, when testing the recurrence at $n = 2$.

Beyond STEP

These are Newton's identities, which link power sums to elementary symmetric functions for any number of variables. Behind them is the *fundamental theorem of symmetric polynomials*: every symmetric polynomial in $x_1, \ldots, x_k$ is a polynomial in $e_1, \ldots, e_k$. The recurrence for S_n is a linear recurrence whose characteristic equation is the cubic itself, the same idea that solves linear difference equations and gives the Fibonacci formula. Allenby's *Rings, Fields and Groups* develops polynomials and their roots in this spirit.

WORKED SOLUTION

STEP III 2009, Question 6

Pure • Ptolemy's Theorem

STEP III 2009, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q6

The key idea

The distance between two points on the unit circle depends only on the angle between them, and the trick for finding it is to pull out the "average" angle: $e^{i\beta} - e^{i\alpha}$ is $e^{i(\alpha+\beta)/2}$ times something purely imaginary. Once every chord is a sine of a half-angle difference, the big identity becomes a trigonometric identity, which product-to-sum formulae dispatch in a couple of lines. The four points then form a quadrilateral inscribed in a circle, and the identity is a famous theorem about it.

Solution

The chord length

Factor out $e^{i(\alpha+\beta)/2}$:

$$\begin{aligned} e^{i\beta} - e^{i\alpha} &= e^{i(\alpha+\beta)/2}\left(e^{i(\beta-\alpha)/2} - e^{-i(\beta-\alpha)/2}\right) \\ &= e^{i(\alpha+\beta)/2} \cdot 2i \sin \tfrac{1}{2}(\beta-\alpha). \end{aligned}$$

Since $\left|e^{i(\alpha+\beta)/2}\right| = 1$ and $|2i| = 2$,

$$\left|e^{i\beta} - e^{i\alpha}\right| = 2\left|\sin \tfrac{1}{2}(\beta-\alpha)\right|.$$

Finally, $0 < \alpha < \beta < 2\pi$ gives $0 < \frac{1}{2}(\beta-\alpha) < \pi$, so the sine is positive and the modulus signs can go: $\left|e^{i\beta} - e^{i\alpha}\right| = 2\sin \frac{1}{2}(\beta-\alpha)$, as required.

Ask yourself Why is the condition $\alpha < \beta$ there at all?

Without it, $\frac{1}{2}(\beta-\alpha)$ could lie in $(-\pi, 0)$ and the sine would be negative, so the formula would give a negative length. Ordering the angles is exactly what lets us drop the modulus. Note also that $|e^{i\alpha} - e^{i\beta}| = |e^{i\beta} - e^{i\alpha}|$, so the order inside the modulus doesn't matter.

The identity

Because $0 < \alpha < \beta < \gamma < \delta < 2\pi$, every pair appearing in the identity is in increasing order, so the first result applies to all six moduli. Dividing through by the common factor $2 \cdot 2 = 4$, we need to show

$$\sin\tfrac{1}{2}(\beta-\alpha)\sin\tfrac{1}{2}(\delta-\gamma) + \sin\tfrac{1}{2}(\gamma-\beta)\sin\tfrac{1}{2}(\delta-\alpha) = \sin\tfrac{1}{2}(\gamma-\alpha)\sin\tfrac{1}{2}(\delta-\beta).$$

The natural variables are the three gaps between consecutive points. Put

$$x = \tfrac{1}{2}(\beta-\alpha), \qquad y = \tfrac{1}{2}(\gamma-\beta), \qquad z = \tfrac{1}{2}(\delta-\gamma),$$

so that $\frac{1}{2}(\gamma-\alpha) = x+y$, $\frac{1}{2}(\delta-\beta) = y+z$ and $\frac{1}{2}(\delta-\alpha) = x+y+z$. The claim is

$$\sin x \sin z + \sin y \sin(x+y+z) = \sin(x+y)\sin(y+z).$$

Using $2\sin P \sin Q = \cos(P-Q) - \cos(P+Q)$ on each product:

$$\begin{aligned} 2\sin x \sin z &= \cos(x-z) - \cos(x+z), \\ 2\sin y \sin(x+y+z) &= \cos(x+z) - \cos(x+2y+z), \\ 2\sin(x+y)\sin(y+z) &= \cos(x-z) - \cos(x+2y+z). \end{aligned}$$

Adding the first two lines, the $\cos(x+z)$ terms cancel and we get exactly the third line. So the trigonometric identity holds (for all x, y, z, in fact), and multiplying back by 4 and using the chord formula gives

$$\left|e^{i\alpha} - e^{i\beta}\right|\left|e^{i\gamma} - e^{i\delta}\right| + \left|e^{i\beta} - e^{i\gamma}\right|\left|e^{i\alpha} - e^{i\delta}\right| = \left|e^{i\alpha} - e^{i\gamma}\right|\left|e^{i\beta} - e^{i\delta}\right|,$$

as required.

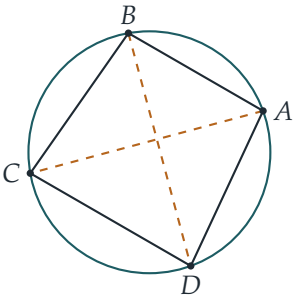
Ask yourself The trig identity holds for all x, y, z. So where did we use the ordering $\alpha < \beta < \gamma < \delta$?

In replacing each modulus by $2\sin(\text{half-difference})$ *without* modulus signs. If the points were out of order, some of those sines would be negative and the identity between the lengths would generally fail. For example, with angles $0.3, 2, 1, 4$ (so $\gamma < \beta$) the left side is about 4.84 and the right side about 1.15.

Interpretation

Let A, B, C, D be the points $e^{i\alpha}, e^{i\beta}, e^{i\gamma}, e^{i\delta}$. They lie on the unit circle, and the ordering of the angles means we meet them in the order A, B, C, D going anticlockwise. So $ABCD$ is a (convex) cyclic quadrilateral with sides AB, BC, CD, DA and diagonals AC, BD, and the identity says

$$AB \cdot CD + BC \cdot AD = AC \cdot BD.$$



Any circle is obtained from the unit circle by a scaling and a translation, and both sides of this equation scale by the same factor (the square of the scale factor), so the result holds for every cyclic quadrilateral: *in a cyclic quadrilateral, the sum of the products of the two pairs of opposite sides equals the product of the diagonals.* This is Ptolemy's theorem.

Checks

- A square inscribed in the unit circle ($\alpha, \beta, \gamma, \delta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$) has sides $\sqrt{2}$ and diagonals 2: $\sqrt{2}\cdot\sqrt{2} + \sqrt{2}\cdot\sqrt{2} = 4 = 2\cdot 2$.
- A rectangle with sides a, b gives $a^2 + b^2 = (\text{diagonal})^2$: Ptolemy contains Pythagoras.
- Numerically, with 10^5 random ordered quadruples of angles the two sides agreed to within 2×10^{-15}.

Common slips

- Writing $|e^{i\beta} - e^{i\alpha}| = 2\sin\frac{1}{2}(\beta - \alpha)$ without checking the sine is positive.
- Expanding $\sin\frac{1}{2}(\delta - \alpha)$ and friends with compound-angle formulae in terms of the original four angles: it works, but is far messier than using the three gaps x, y, z.
- Forgetting that the identity needs the vertices in cyclic order: it is the diagonals, not any two chords, on the right-hand side.

Beyond STEP

For four points in general position the same quantities satisfy only Ptolemy's *inequality* $AB\cdot CD + BC\cdot AD \geq AC\cdot BD$, with equality exactly when the points are concyclic in that order; a slick proof uses the complex identity $(a-b)(c-d) + (a-d)(b-c) = (a-c)(b-d)$ and the triangle inequality. Behind this lies the cross-ratio, an invariant of Möbius transformations, which sends circles to circles and is central to complex geometry. Needham's *Visual Complex Analysis* develops these ideas beautifully.

WORKED SOLUTION

STEP III 2009, Question 7

Pure • Repeated Differentiation and Induction

STEP III 2009, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q7

The key idea

Everything flows from one innocent identity: $(1+x^2)f_0(x) = 1$. Differentiating it repeatedly is what generates relations between consecutive derivatives, and a relation that holds for one n can be differentiated once more to give the next one, which is exactly the shape of an induction step. In part (ii), multiplying by the right power of $1+x^2$ clears every denominator, and a second induction tracks the leading term to pin down the degree.

Solution

Part (i)

Write f_n for $f_n(x)$ and $P(n)$ for the statement

$$(1+x^2)f_{n+1} + 2(n+1)xf_n + n(n+1)f_{n-1} = 0.$$

Base case. Since $(1+x^2)f_0 = 1$ for all x, differentiating (with $f_0' = f_1$) gives

$$(1+x^2)f_1 + 2xf_0 = 0.$$

Differentiating again, using the product rule on each term,

$$(1+x^2)f_2 + 2xf_1 + 2xf_1 + 2f_0 = 0, \qquad \text{that is,} \qquad (1+x^2)f_2 + 4xf_1 + 2f_0 = 0,$$

which is $P(1)$.

Inductive step. Suppose $P(n)$ holds for some $n \geq 1$, for all x. Since it is an identity in x, we may differentiate it. Using $f_k' = f_{k+1}$,

$$\begin{aligned} 0 &= (1+x^2)f_{n+2} + 2xf_{n+1} + 2(n+1)f_n + 2(n+1)xf_{n+1} + n(n+1)f_n \\ &= (1+x^2)f_{n+2} + 2(n+2)xf_{n+1} + (n+1)(n+2)f_n, \end{aligned}$$

since $2+2(n+1)=2(n+2)$ and $2(n+1)+n(n+1)=(n+1)(n+2)$. This is $P(n+1)$. So by induction $P(n)$ holds for all $n \geq 1$, as required.

(Equivalently, apply Leibniz's rule to differentiate $(1+x^2)f_0 = 1$ exactly $n+1$ times: only three terms survive, because the third derivative of $1+x^2$ is zero, and they are the three terms of $P(n)$.)

Part (ii)

The first three. We have $P_0 = (1+x^2)f_0 = 1$. Next,

$$f_1 = \frac{-2x}{(1+x^2)^2}, \qquad \text{so} \qquad P_1 = (1+x^2)^2 f_1 = -2x.$$

Differentiating f_1 by the quotient rule,

$$f_2 = \frac{-2(1+x^2)^2 + 2x \cdot 2(1+x^2)\cdot 2x}{(1+x^2)^4} = \frac{-2(1+x^2)+8x^2}{(1+x^2)^3} = \frac{6x^2-2}{(1+x^2)^3},$$

so $P_2 = (1+x^2)^3 f_2 = 6x^2 - 2$. (Part (i) with $n=1$ gives the same: $(1+x^2)f_2 = -4xf_1 - 2f_0$.)

The recurrence. Differentiate $P_n = (1+x^2)^{n+1} f_n$ by the product rule:

$$P_n' = 2(n+1)x(1+x^2)^n f_n + (1+x^2)^{n+1} f_{n+1}.$$

Multiplying through by $1+x^2$ turns each term into a P:

$$(1+x^2)P_n' = 2(n+1)x\,(1+x^2)^{n+1} f_n + (1+x^2)^{n+2} f_{n+1} = 2(n+1)xP_n + P_{n+1}.$$

Rearranging,

$$P_{n+1} - (1+x^2)P_n' + 2(n+1)xP_n = 0$$

for every $n \geq 0$, as required. No induction is needed here; it is a direct calculation valid for each n.

> **Ask yourself** How do we know the degree is exactly n, and not smaller because the top terms cancel?

We prove a slightly stronger statement, which carries the information we need through the induction: *P_n is a polynomial of degree n with leading coefficient $a_n \neq 0$.* The case $n=0$ holds, since $P_0 = 1$.

Suppose it holds for some $n \geq 0$, so $P_n = a_n x^n + (\text{lower powers})$ with $a_n \neq 0$. By the recurrence,

$$P_{n+1} = (1+x^2)P_n' - 2(n+1)xP_n.$$

The right-hand side is a polynomial, because P_n and so P_n' are. Now $P_n' = na_n x^{n-1} + \cdots$ (this is 0 when $n=0$, which matches $na_n = 0$), so $(1+x^2)P_n'$ has degree at most $n+1$ with x^{n+1} coefficient na_n. Also $2(n+1)xP_n$ has degree $n+1$ with leading coefficient $2(n+1)a_n$. Hence P_{n+1} has degree at most $n+1$, and its x^{n+1} coefficient is

$$a_{n+1} = na_n - 2(n+1)a_n = -(n+2)a_n.$$

This is nonzero, because $a_n \neq 0$ and $n+2 > 0$. So the top terms cannot cancel, and P_{n+1} has degree exactly $n+1$. By induction, P_n is a polynomial of degree n for every $n \geq 0$.

As a bonus, $a_{n+1} = -(n+2)a_n$ with $a_0 = 1$ gives $a_n = (-1)^n(n+1)!$, matching $a_1 = -2$ and $a_2 = 6$ above.

Checks

- Part (i) at $x = 0$, $n = 1$: from P_n we get $f_0(0) = 1$, $f_1(0) = 0$, $f_2(0) = -2$, and $f_2 + 0 + 2f_0 = -2 + 2 = 0$.
- The recurrence with $n = 1$: $(1+x^2)P_1' - 4xP_1 = -2 - 2x^2 + 8x^2 = 6x^2 - 2 = P_2$.
- Numerically at $x = 0.7$, a central difference gives $f_0''(0.7) \approx 0.284164$, and $(6x^2 - 2)/(1+x^2)^3 = 0.284164$. We also checked part (i) and the recurrence exactly in rational arithmetic up to $n = 12$ at 200 random rational points.

Common slips

- Not proving a base case that matches the stated range: the identity in part (i) starts at $n = 1$, so the base case must be $n = 1$, which needs f_2.
- Saying "the degree goes up by one each time" without checking that the x^{n+1} terms of $(1+x^2)P_n'$ and $2(n+1)xP_n$ don't cancel. They don't, because $n \neq 2(n+1)$, but it has to be said.

Beyond STEP

Splitting into partial fractions with complex numbers, $\frac{1}{1+x^2} = \frac{1}{2\mathrm{i}}\left(\frac{1}{x-\mathrm{i}} - \frac{1}{x+\mathrm{i}}\right)$, makes every derivative easy and gives the closed form $P_n(x) = (-1)^n\, n!\, \operatorname{Im}(x+\mathrm{i})^{n+1}$ (try it for $n = 1, 2$). Writing $x = \cot\theta$ then shows that the zeros of P_n are $\cot\frac{k\pi}{n+1}$ for $k = 1, \ldots, n$: all real and distinct, like the zeros of the classical orthogonal polynomials, which are also generated by repeatedly differentiating a simple function. Needham's *Visual Complex Analysis* is a lovely place to see why complex numbers tame calculations like this.

WORKED SOLUTION

STEP III 2009, Question 8

Pure • Limits and Reduction Formulae

STEP III 2009, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q8

This question has a full guided treatment in *Beyond STEP*, Section 4.5, *The Sophomore's Dream*: https://beyondstep.keithmcnulty.org/ch04/sophomores-dream.html

The key idea

Every part is about the logarithm being "weak": near 0, $\ln x$ blows up so slowly that any positive power of x beats any power of $\ln x$. The substitution $x = \mathrm{e}^{-t}$ turns this into the given fact about exponentials beating powers. That limit then kills the boundary terms when we integrate by parts. In the last part, the series for the exponential, applied to $x^x = \mathrm{e}^{x \ln x}$, produces exactly the integrals of part (ii).

Solution

Part (i)

Since $\ln x$ is defined only for $x > 0$, the limit means $x \to 0^+$. Put $x = \mathrm{e}^{-t}$, so that $\ln x = -t$, and $x \to 0^+$ exactly when $t \to \infty$. Then

$$x^m (\ln x)^n = \mathrm{e}^{-mt}(-t)^n = (-1)^n \mathrm{e}^{-mt} t^n.$$

The factor $(-1)^n$ is just a fixed sign, and $\mathrm{e}^{-mt} t^n \to 0$ as $t \to \infty$ by the given result. Hence

$$\lim_{x \to 0} x^m (\ln x)^n = 0.$$

For the second result, take $m = n = 1$: this gives $x \ln x \to 0$ as $x \to 0$. Since the exponential function is continuous,

$$x^x = \mathrm{e}^{x \ln x} \to \mathrm{e}^0 = 1 \qquad \text{as } x \to 0,$$

as required.

Part (ii)

First, the integral makes sense. The integrand $x^m(\ln x)^n$ is continuous on $(0,1]$, and by part (i) it tends to 0 as $x \to 0$. So it is bounded on $(0,1]$, and I_n is an ordinary (convergent) integral.

Integrate I_{n+1} by parts with $u = (\ln x)^{n+1}$ and $\dfrac{\mathrm{d}v}{\mathrm{d}x} = x^m$, so that $\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{(n+1)(\ln x)^n}{x}$ and $v = \dfrac{x^{m+1}}{m+1}$:

$$\begin{aligned} I_{n+1} &= \left[\frac{x^{m+1}(\ln x)^{n+1}}{m+1}\right]_0^1 - \int_0^1 \frac{x^{m+1}}{m+1} \cdot \frac{(n+1)(\ln x)^n}{x}\,\mathrm{d}x \\ &= \left[\frac{x^{m+1}(\ln x)^{n+1}}{m+1}\right]_0^1 - \frac{n+1}{m+1}\int_0^1 x^m(\ln x)^n\,\mathrm{d}x. \end{aligned}$$

? **Ask yourself** What is the boundary term at $x = 0$, where $\ln x$ isn't even defined?

The lower limit really means the limit as $x \to 0^+$. By part (i), with $m+1$ and $n+1$ in place of m and n (both allowed, since $m+1 \geq 1$ and $n+1 \geq 0$), $x^{m+1}(\ln x)^{n+1} \to 0$. At the upper limit, $\ln 1 = 0$ and $n+1 \geq 1$, so that term is 0 too. The bracket therefore vanishes, and

$$I_{n+1} = -\frac{n+1}{m+1}\,I_n,$$

as required.

To evaluate I_n, start from

$$I_0 = \int_0^1 x^m\,\mathrm{d}x = \frac{1}{m+1}.$$

Each step of the recurrence multiplies by $-\dfrac{k}{m+1}$ for $k = 1, 2, \ldots, n$, so

$$I_n = \left(-\frac{1}{m+1}\right)\left(-\frac{2}{m+1}\right)\cdots\left(-\frac{n}{m+1}\right)\frac{1}{m+1} = \frac{(-1)^n\,n!}{(m+1)^{n+1}}.$$

(A one-line induction confirms this: if it holds for n, the recurrence gives $-\frac{n+1}{m+1} \cdot \frac{(-1)^n n!}{(m+1)^{n+1}} = \frac{(-1)^{n+1}(n+1)!}{(m+1)^{n+2}}$.)

Part (iii)

For $0 < x \leq 1$, write $x^x = \mathrm{e}^{x\ln x}$ and use the exponential series:

$$x^x = \sum_{n=0}^{\infty} \frac{(x\ln x)^n}{n!} = \sum_{n=0}^{\infty} \frac{x^n(\ln x)^n}{n!}.$$

(The value at $x = 0$ does not affect the integral; by part (i) the function x^x tends to 1 there, so it is bounded and the integral is an ordinary one.)

Now integrate term by term. The $n = 0$ term is $\int_0^1 1\,\mathrm{d}x = 1$. For $n \geq 1$, the nth term is I_n from part (ii) with $m = n$ (allowed, since then m is a positive integer):

$$\frac{1}{n!}\int_0^1 x^n (\ln x)^n \,\mathrm{d}x = \frac{1}{n!}\cdot\frac{(-1)^n\, n!}{(n+1)^{n+1}} = \frac{(-1)^n}{(n+1)^{n+1}}.$$

This formula gives 1 when $n = 0$ as well. So

$$\int_0^1 x^x \,\mathrm{d}x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+1)^{n+1}} = 1 - \left(\tfrac{1}{2}\right)^2 + \left(\tfrac{1}{3}\right)^3 - \left(\tfrac{1}{4}\right)^4 + \cdots,$$

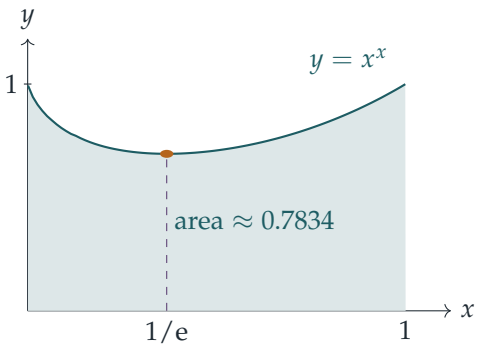
which is the required result.

Ask yourself Are we allowed to integrate an infinite sum term by term?

Not automatically, but here it is safe because $x \ln x$ stays small. Its derivative is $\ln x + 1$, so on $(0, 1]$ it has its minimum at $x = 1/\mathrm{e}$, and $|x \ln x| \leq 1/\mathrm{e} < 1$. If we stop the series after the u^N term, with $u = x \ln x$, the remainder is

$$\left|\sum_{n>N} \frac{u^n}{n!}\right| \leq \sum_{n>N} \frac{|u|^n}{n!} \leq \sum_{n>N} \frac{1}{n!} \leq \frac{2}{(N+1)!}$$

for every x in $(0, 1]$. Integrating over an interval of length 1, the error in the truncated term-by-term integral is at most $2/(N+1)!$, which tends to 0. So the integral equals the sum of the series.



Checks

- Take $m = 1$, $n = 1$ in part (ii): $\int_0^1 x \ln x \,\mathrm{d}x = -\frac{1}{4}$, which matches direct integration by parts ($[\frac{1}{2}x^2 \ln x - \frac{1}{4}x^2]_0^1 = -\frac{1}{4}$).
- Numerically, $\int_0^1 x^x \,\mathrm{d}x \approx 0.7834305$ by the midpoint rule, and the partial sums $1 - \frac{1}{4} + \frac{1}{27} - \frac{1}{256} + \frac{1}{3125} - \cdots$ settle to 0.7834305 within a handful of terms, since the terms shrink extremely fast.
- The graph sits between its minimum $\mathrm{e}^{-1/\mathrm{e}} \approx 0.69$ and 1, so the area must be between 0.69 and 1, as it is.

Common slips

- Dropping the boundary term at 0 without comment. It is the whole point of part (i), and it needs $m+1 \geq 1$.
- Getting the sign wrong in I_n: $\ln x < 0$ on $(0,1)$, so I_n has the sign of $(-1)^n$.
- Forgetting the $n = 0$ term in part (iii), which is where the leading 1 comes from.

Beyond STEP

The final identity is known as the "sophomore's dream", found by Johann Bernoulli in 1697. The step we had to justify, swapping an integral and an infinite sum, is a central theme of first-year analysis: it is safe when the series converges *uniformly*, which is what our remainder bound shows (it is an instance of the Weierstrass M-test). Abbott's *Understanding Analysis* gives a very readable account of uniform convergence and when term-by-term integration works.

WORKED SOLUTION

STEP III 2009, Question 9

Mechanics • Projectile Trajectory Angles

STEP III 2009, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

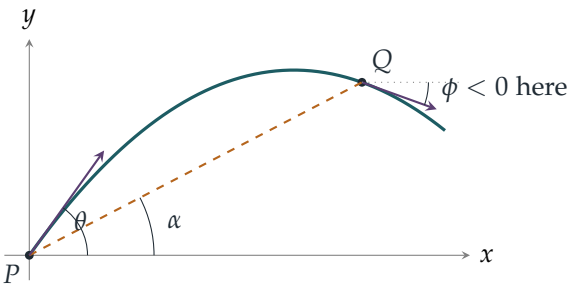
step.maths.org/questions/09-s3-q9

The key idea

In projectile motion the horizontal velocity never changes and the vertical velocity changes at a constant rate. So the average vertical velocity between P and Q is just the mean of its values at the two ends, and the gradient of the chord PQ is the mean of the gradients at the ends. That gives part (i) in one line. For part (ii), the trajectory equation is a quadratic in $\tan\theta$ (once we write $\sec^2\theta = 1 + \tan^2\theta$), and sums and products of roots do the rest.

Solution

We model the particle as moving freely under constant gravity g, with no air resistance. Take P as the origin, with the x-axis along the positive horizontal direction and the y-axis vertically up. Let u be the speed of projection and let $Q = (X, Y)$. We assume $X > 0$: Q lies on the positive side of P, so the horizontal velocity $u\cos\theta$ is positive and all the angles $\theta, \phi, \theta', \phi'$ lie in $(-\frac{\pi}{2}, \frac{\pi}{2})$. (If Q were vertically above or below P, the angles at P and Q would both be $\pm\frac{\pi}{2}$ and the question would say nothing.) Then $\tan\alpha = Y/X$.



Part (i)

The horizontal velocity is constant, $u\cos\theta$. The vertical velocity is $u\sin\theta$ at P, say v_Q at Q, and it decreases at the constant rate g, so it is a linear function of time. Hence, if the flight from P to Q takes time T,

$$X = (u\cos\theta)\,T, \qquad Y = \frac{u\sin\theta + v_Q}{2}\,T,$$

because the average of a linear function over an interval is the mean of its end values. At each point the direction of motion is the direction of the velocity, so $\tan\theta = \dfrac{u\sin\theta}{u\cos\theta}$ and $\tan\phi = \dfrac{v_Q}{u\cos\theta}$. Dividing,

$$\tan\alpha = \frac{Y}{X} = \frac{1}{2}\left(\frac{u\sin\theta}{u\cos\theta} + \frac{v_Q}{u\cos\theta}\right) = \tfrac{1}{2}(\tan\theta + \tan\phi),$$

which is $\tan\theta + \tan\phi = 2\tan\alpha$, as required.

Part (ii)

The quadratic. Eliminating t from $x = ut\cos\theta$, $y = ut\sin\theta - \frac{1}{2}gt^2$ gives the trajectory

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta} = x\tan\theta - \frac{gx^2}{2u^2}\left(1+\tan^2\theta\right).$$

Write $k = \dfrac{gX^2}{2u^2}$, which is positive. Since the path passes through (X, Y), $T = \tan\theta$ satisfies

$$kT^2 - XT + (Y + k) = 0.$$

The second trajectory has the same u, X and Y, so $\tan\theta'$ satisfies the same equation. The two trajectories are different, so $\theta \neq \theta'$, and since tan is one-to-one on $(-\frac{\pi}{2}, \frac{\pi}{2})$, $\tan\theta$ and $\tan\theta'$ are the *two* roots. Therefore

$$\tan\theta + \tan\theta' = \frac{X}{k}, \qquad \tan\theta\tan\theta' = \frac{Y+k}{k} = 1 + \frac{Y}{k}.$$

The first result. By the compound-angle formula,

$$\tan(\theta+\theta') = \frac{\tan\theta + \tan\theta'}{1 - \tan\theta\tan\theta'} = \frac{X/k}{-Y/k} = -\frac{X}{Y} = -\cot\alpha.$$

This needs $Y \neq 0$. If $Y = 0$ (so $\alpha = 0$ and $\cot\alpha$ is undefined), the product of the roots is 1, which says $\theta + \theta' = \frac{\pi}{2}$, the familiar pair of complementary angles for the same range on level ground. In that sense the result still holds.

The second result. Write $t = \tan\alpha = Y/X$ and $s = \tan\theta + \tan\theta' = X/k$. Then $Y/k = ts$, so $\tan\theta\tan\theta' = 1 + ts$. By part (i), applied to each trajectory,

$$\tan\phi = 2t - \tan\theta, \qquad \tan\phi' = 2t - \tan\theta'.$$

So

$$\tan\phi + \tan\phi' = 4t - s,$$
$$1 - \tan\phi\tan\phi' = 1 - \left(4t^2 - 2ts + \tan\theta\tan\theta'\right) = 1 - 4t^2 + 2ts - 1 - ts = t(s - 4t).$$

These can't both be zero: if $\tan\phi' = -\tan\phi$ then $\tan\phi\tan\phi' = -\tan^2\phi \le 0 \ne 1$. When $Y \ne 0$ (so $t \ne 0$) and $s \ne 4t$,

$$\tan(\phi + \phi') = \frac{4t - s}{t(s - 4t)} = -\frac{1}{t} = -\cot\alpha = \tan(\theta + \theta').$$

In the other cases both $1 - \tan\phi\tan\phi'$ and $1 - \tan\theta\tan\theta'$ are zero ($t = 0$), or both $\tan(\phi + \phi')$ and $\tan(\theta + \theta')$ are infinite, so in every case $\phi + \phi'$ and $\theta + \theta'$ have the same tangent or are both odd multiples of $\frac{\pi}{2}$. Either way,

$$\theta + \theta' - (\phi + \phi') = n\pi \quad \text{for some integer } n.$$

Ask yourself Equal tangents only pin the angle down up to a multiple of π. How do we know the multiple is exactly one π?

We use the sizes of the angles. On each trajectory the gradient is $\tan\theta - \dfrac{gx}{u^2\cos^2\theta}$, which strictly decreases as x increases. Since $X > 0$, we get $\tan\phi < \tan\theta$, so $\phi < \theta$; likewise $\phi' < \theta'$. All four angles lie in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $0 < \theta - \phi < \pi$ and $0 < \theta' - \phi' < \pi$. Adding,

$$0 < n\pi = (\theta + \theta') - (\phi + \phi') < 2\pi,$$

so $n = 1$, and $\theta + \theta' = \pi + \phi + \phi'$, as required.

Checks

- Take $X = 1$, $Y = \frac{1}{2}$ and $u^2 = 4g$, so $k = \frac{1}{8}$. The quadratic becomes $T^2 - 8T + 5 = 0$, with roots $4 \pm \sqrt{11}$. Then $\tan(\theta + \theta') = 8/(1 - 5) = -2 = -\cot\alpha$. Numerically, $\theta + \theta' \approx 2.0344$ and $\pi + \phi + \phi' \approx 2.0344$, where $\tan\phi = 1 - \tan\theta$ and $\tan\phi' = 1 - \tan\theta'$.
- Level ground ($Y = 0$): $\theta + \theta' = \frac{\pi}{2}$, and by symmetry $\phi = -\theta$, $\phi' = -\theta'$. So $\pi + \phi + \phi' = \pi - \frac{\pi}{2} = \frac{\pi}{2}$, as it should be.
- We stepped the motion numerically to confirm part (i), and tested both results of part (ii) on 20,000 random choices of u, X and Y that admit two trajectories.

Common slips

- Concluding $\theta + \theta' = \pi + \phi + \phi'$ from equal tangents alone. Equal tangents give the angles only up to a multiple of π, and you need the inequalities $\phi < \theta$ and $\phi' < \theta'$ to settle which multiple.
- Forgetting to say why $\tan\theta$ and $\tan\theta'$ are the *two different* roots of the same quadratic. They are different because the trajectories are, and tan is one-to-one on $(-\frac{\pi}{2}, \frac{\pi}{2})$.

Beyond STEP

Part (ii) has a pleasant reading in terms of reversibility. Run either path backwards and you get a projectile from Q to P launched at angle $\phi + \pi$. So the four angles are tied together by a symmetry between the two ends, and the relation $\theta + \theta' = \pi + \phi + \phi'$ expresses it. Time-reversal symmetry of Newton's laws (without friction or drag) is a theme that runs right through university mechanics. For a thorough first-year treatment of projectiles and much more, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP III 2009, Question 10

Mechanics • Simple Harmonic Motion

STEP III 2009, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q10

The key idea

The equilibrium compression d is there to tell us the stiffness of the spring without ever naming it: in equilibrium the spring force balances the weight, so $k/m = g/d$. After that, Newton's second law gives simple harmonic motion about the equilibrium position, and the drop height h only enters through the initial speed. The particle leaves the spring when the spring can no longer push on it, which happens when the compression returns to zero.

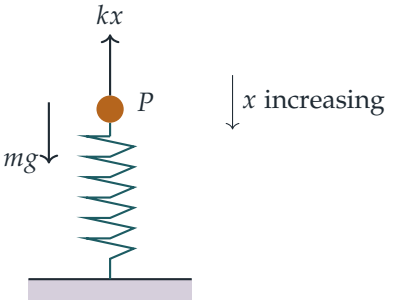
Solution

The differential equation

Model P as a particle of mass m and the spring as light, obeying Hooke's law with stiffness k, and ignore air resistance. When P rests on the spring, the forces on it are its weight mg down and the spring force kd up, so

$$kd = mg, \qquad \text{that is,} \qquad \frac{k}{m} = \frac{g}{d}.$$

Now take the dropped case. Measure x downwards (it is the compression), with $t = 0$ at the moment of impact. While P is in contact, the forces on it are as shown.



Because the spring is light, the force it exerts on P is exactly kx upwards (a light spring transmits the force at its ends). Newton's second law, taking downwards as positive, gives $m\ddot{x} = mg - kx$, so

$$\ddot{x} + \frac{g}{d}x = g, \qquad 0 \le t \le T.$$

Solving it

This is simple harmonic motion about $x = d$ with

$$\omega = \sqrt{g/d}.$$

A particular solution is the constant $x = d$, and the complementary function is $B\cos\omega t + C\sin\omega t$, so the general solution is

$$x = d + B\cos\omega t + C\sin\omega t.$$

The initial conditions come from the fall. At $t = 0$ the spring has just been touched, so $x(0) = 0$; and P has fallen freely through height h, so its speed is $\sqrt{2gh}$, downwards, which is the direction of increasing x. Hence

$$\begin{aligned} x(0) = d + B = 0 \qquad &\Rightarrow \quad B = -d, \\ \dot{x}(0) = C\omega = \sqrt{2gh} \qquad &\Rightarrow \quad C = \sqrt{2gh}\,\sqrt{d/g} = \sqrt{2hd}. \end{aligned}$$

So

$$x = d - d\cos\omega t + \sqrt{2hd}\,\sin\omega t, \qquad A = d, \quad B = -d, \quad C = \sqrt{2hd}, \quad \omega = \sqrt{g/d}.$$

The time of separation

Ask yourself When exactly does P lose contact?

The spring is not attached to P, so it can push but never pull. The contact force on P is the spring force kx, which stays positive while $x > 0$ and falls to zero when x returns to 0, with P moving upwards. If the motion carried on beyond that point, the equation would need a downward pull from the spring, which it can't give. So T is the first positive time with $x = 0$.

To find it, write the oscillating part as a single cosine. Let $R = \sqrt{d^2 + 2hd}$ and define the acute angle φ by

$$\cos\varphi = \frac{d}{R}, \qquad \sin\varphi = \frac{\sqrt{2hd}}{R},$$

so that $\tan\varphi = \sqrt{2h/d}$ and $\varphi = \arctan\sqrt{2h/d}$. Then $d\cos\omega t - \sqrt{2hd}\sin\omega t = R\cos(\omega t + \varphi)$, and

$$x = d - R\cos(\omega t + \varphi).$$

So $x = 0$ exactly when $\cos(\omega t + \varphi) = d/R = \cos\varphi$. As t increases from 0, the angle $\omega t + \varphi$ starts at φ, and on the interval $\varphi < \omega t + \varphi < 2\pi - \varphi$ we have $\cos(\omega t + \varphi) <$

$\cos\varphi$, so $x > 0$ there. The first return to $\cos\varphi$ is at

$$\omega T + \varphi = 2\pi - \varphi, \qquad \text{that is,} \qquad \omega T = 2\pi - 2\varphi.$$

Dividing by $\omega = \sqrt{g/d}$,

$$T = \sqrt{d/g}\left(2\pi - 2\arctan\sqrt{2h/d}\right),$$

as required.

Checks

- If $h = 0$ the particle is placed on the spring at its natural length with zero speed, $\varphi = 0$, and $T = 2\pi\sqrt{d/g}$: a full period, after which P is back at the top with zero speed. That is exactly right.
- As $h \to \infty$, $\varphi \to \frac{\pi}{2}$ and $\omega T \to \pi$: a very fast particle spends only half a period on the spring, because the equilibrium shift d becomes negligible next to the amplitude.
- By energy conservation, P should leave with speed $\sqrt{2gh}$. Indeed $\dot{x}(T) = R\omega\sin(2\pi - \varphi) = -R\omega\sin\varphi = -\sqrt{2gh}$.
- Numerically, stepping the motion with $d = 0.1\,\text{m}$, $h = 0.2\,\text{m}$ and $g = 9.8\,\text{m}\,\text{s}^{-2}$ gives a contact time of $0.411\,\text{s}$, matching the formula. Random values of d and h agree to six decimal places.

Common slips

- Measuring from the equilibrium position and then forgetting that x in the question is the compression, which gives $A = 0$.
- Taking the initial speed as $\sqrt{2gh}$ but with the wrong sign: downwards is the direction of increasing x.
- Assuming separation happens at the equilibrium position $x = d$, or at half a period. The spring keeps pushing until it is back to its natural length.

Beyond STEP

This is the simplest example of a *unilateral constraint*: the spring can push but not pull. The motion obeys different differential equations in different phases, which are stitched together by matching position and velocity, just as the free fall here is matched to the spring phase. The same idea runs through the theory of impacts and bouncing systems. For more on oscillators and on energy methods like the check above, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP III 2009, Question 11

Mechanics • Variable Mass Motion

STEP III 2009, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q11

The key idea

When mass changes, $F = ma$ is the wrong law: we must use force = rate of change of momentum. The dust is stationary, so it brings no momentum of its own, and the comet's momentum $M(1+bx)v$ changes only because of the external force. That gives a first integral straight away. Writing $v = \mathrm{d}x/\mathrm{d}t$ then turns it into a separable equation whose solution is a quadratic in x.

Solution

We model the comet as a particle. The dust it collects was at rest, so it carries no momentum into the comet, and the only force on the system (comet plus collected dust) is the gravitational force Mf along the direction of motion.

Part (i)

With $f = 0$ there is no external force, so momentum is conserved. At entry the momentum is MV; at time t it is $M(1+bx)v$. Hence

$$(1+bx)\,v = V.$$

Now put $v = \dfrac{\mathrm{d}x}{\mathrm{d}t}$ and separate:

$$\int_0^x (1+bu)\,\mathrm{d}u = \int_0^t V\,\mathrm{d}s \quad\Longrightarrow\quad x + \tfrac{1}{2}bx^2 = Vt,$$

using $x = 0$ when $t = 0$. So $bx^2 + 2x - 2Vt = 0$, and since $x \geq 0$ we need the root with the positive square root:

$$x = \frac{\sqrt{1+2bVt} - 1}{b}.$$

Part (ii)

Now the external force is Mf, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[M(1+bx)v\right] = Mf.$$

Since M and f are constants, integrating from 0 to t gives $M(1+bx)v - MV = Mft$ (at $t=0$, $x=0$ and $v=V$). Therefore

$$v = \frac{ft+V}{1+bx},$$

as required.

Ask yourself Why can't we just write $M(1+bx)\dfrac{\mathrm{d}v}{\mathrm{d}t} = Mf$?

That equation would ignore the momentum needed to bring each new speck of dust from rest up to speed v. The product rule shows the difference: the rate of change of momentum is $M(1+bx)\dot{v} + Mb\dot{x}\,v$, and the second term is exactly that "catching-up" cost.

As in part (i), write $v = \mathrm{d}x/\mathrm{d}t$ and separate:

$$\int_0^x (1+bu)\,\mathrm{d}u = \int_0^t (fs+V)\,\mathrm{d}s \quad\Longrightarrow\quad x + \tfrac{1}{2}bx^2 = \tfrac{1}{2}ft^2 + Vt.$$

So $bx^2 + 2x - (ft^2 + 2Vt) = 0$ and, taking the non-negative root again,

$$x = \frac{\sqrt{1+bft^2+2bVt}-1}{b}.$$

(Setting $f=0$ recovers part (i).)

Constant speed. The comet moves with constant speed V exactly when $x = Vt$ for all t, that is, when $\sqrt{1+bft^2+2bVt} = 1+bVt$. Squaring, this needs

$$1 + bft^2 + 2bVt = 1 + 2bVt + b^2V^2t^2 \quad \text{for all } t,$$

which holds precisely when $f = bV^2$. Then $v = \dfrac{bV^2t+V}{1+bVt} = V$, so with this choice the comet does move at constant speed: the gravitational pull exactly supplies the momentum needed to accelerate the dust it sweeps up.

Limiting speed. In general, using $1+bx = \sqrt{1+bft^2+2bVt}$ from the formula for x,

$$v = \frac{ft+V}{\sqrt{1+2bVt+bft^2}} = \frac{f+V/t}{\sqrt{bf+2bV/t+1/t^2}} \longrightarrow \frac{f}{\sqrt{bf}} = \sqrt{\frac{f}{b}} \quad \text{as } t\to\infty,$$

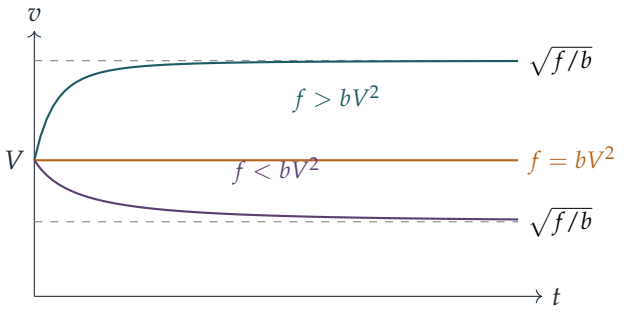
where we divided top and bottom by $t>0$ and used $bf>0$ (we need $f>0$, since the force acts in the direction of motion). So the speed tends to the constant $\sqrt{f/b}$.

Ask yourself Is this consistent with the constant-speed case?

Yes: if $f = bV^2$ then $\sqrt{f/b} = V$, the speed the comet had all along. In fact differentiating v^2 gives

$$\frac{\mathrm{d}(v^2)}{\mathrm{d}t} = \frac{2(ft + V)(f - bV^2)}{(1 + 2bVt + bft^2)^2},$$

so the speed rises steadily towards $\sqrt{f/b}$ if $f > bV^2$, falls steadily towards it if $f < bV^2$, and stays at V if $f = bV^2$.



Checks

- Units: b has units of $1/\text{length}$ and f is an acceleration, so f/b has units of speed squared and $\sqrt{f/b}$ is a speed; bft^2 and bVt are dimensionless, as they must be inside the square root.
- With $b = 1$, $V = 1$, $f = 3$, stepping the equations $\dot{x} = p/(1 + bx)$, $\dot{p} = f$ numerically agrees with the formulas for x and v to eight decimal places, and at $t = 10^6$ the speed is $1.7320508\ldots \approx \sqrt{3}$.
- As $b \to 0$ (no dust), $x = \left(\sqrt{1 + bft^2 + 2bVt} - 1\right)/b \to \frac{1}{2}ft^2 + Vt$, the usual constant-acceleration result.

Common slips

- Using $(\text{mass}) \times \dot{v} = \text{force}$ for a body whose mass is changing.
- Taking the negative root of the quadratic, which gives $x < 0$.
- Forgetting the constant of integration MV when integrating the momentum equation.

Beyond STEP

Variable-mass problems are a classic first-year topic: the same momentum bookkeeping gives the rocket equation, and raindrops growing as they fall through a cloud lead to

equations very like these. The limiting speed $\sqrt{f/b}$ is a terminal velocity, where the force exactly balances the rate at which momentum is spent on new mass. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats systems of variable mass carefully.

WORKED SOLUTION

STEP III 2009, Question 12

Probability and Statistics • Probability Generating Functions

STEP III 2009, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q12

The key idea

In part (i) each round of tossing halves the expected number of coins, but the number of tosses is itself random, so we condition on it: given the previous count, the next count is binomial and its mean is easy. That turns the expectations into a geometric sequence. In part (ii) the total is a sum of independent, identically distributed pieces, and a probability generating function turns "sum of independent variables" into "product of generating functions".

Solution

Part (i)

Each toss is a head with probability $\frac{1}{2}$, independently, so $X_1 \sim \mathrm{B}(k, \frac{1}{2})$ and

$$\mathrm{E}(X_1) = \tfrac{1}{2}k.$$

Given $X_1 = x_1$, Albert makes x_1 fresh, independent tosses, so $X_2 \mid X_1 = x_1 \sim \mathrm{B}(x_1, \frac{1}{2})$ and

$$\mathrm{E}(X_2 \mid X_1 = x_1) = \tfrac{1}{2}x_1.$$

(If $x_1 = 0$ he makes no tosses and $X_2 = 0$, which agrees.) Averaging over the possible values of X_1, by the law of total expectation,

$$\mathrm{E}(X_2) = \sum_{x_1=0}^{k} \mathrm{E}(X_2 \mid X_1 = x_1)\,\mathrm{P}(X_1 = x_1) = \tfrac{1}{2}\sum_{x_1=0}^{k} x_1\,\mathrm{P}(X_1 = x_1) = \tfrac{1}{2}\,\mathrm{E}(X_1) = \tfrac{1}{4}k,$$

as required.

? **Ask yourself** Why can't we just say "X_2 is $\mathrm{B}(\frac{1}{2}k, \frac{1}{2})$"?

Because the number of tosses in the second round is the random variable X_1, not the number $\frac{1}{2}k$; indeed $\frac{1}{2}k$ need not even be an integer. Conditioning on X_1 fixes the number of tosses, and only then is the binomial mean available.

Exactly the same argument works at every stage: given $X_i = x_i$, $X_{i+1} \sim \mathrm{B}(x_i, \frac{1}{2})$, so $\mathrm{E}(X_{i+1}) = \frac{1}{2}\mathrm{E}(X_i)$. By induction,

$$\mathrm{E}(X_i) = \frac{k}{2^i} \qquad (i \geq 1),$$

and summing the geometric series (ratio $\frac{1}{2}$),

$$\sum_{i=1}^{\infty} \mathrm{E}(X_i) = k\left(\tfrac{1}{2} + \tfrac{1}{4} + \tfrac{1}{8} + \cdots\right) = k.$$

Part (ii)

For one coin, $Y_i = n$ means n heads followed by a tail, so

$$\mathrm{P}(Y_i = n) = \left(\tfrac{1}{2}\right)^n \cdot \tfrac{1}{2} = \left(\tfrac{1}{2}\right)^{n+1}, \qquad n = 0, 1, 2, \ldots$$

Its probability generating function is a geometric series, convergent for $|t| < 2$:

$$G_{Y_i}(t) = \sum_{n=0}^{\infty} \left(\tfrac{1}{2}\right)^{n+1} t^n = \frac{\frac{1}{2}}{1 - \frac{1}{2}t} = \frac{1}{2 - t}.$$

The k coins are tossed separately, so $Y_1, \ldots, Y_k$ are independent, and the generating function of a sum of independent variables is the product of their generating functions. Hence

$$G_Y(t) = \mathrm{E}\big(t^{Y_1 + \cdots + Y_k}\big) = \prod_{i=1}^{k} \mathrm{E}\big(t^{Y_i}\big) = (2 - t)^{-k}.$$

Mean and variance. Differentiating,

$$G_Y'(t) = k(2 - t)^{-k-1}, \qquad G_Y''(t) = k(k+1)(2 - t)^{-k-2},$$

so $G_Y'(1) = k$ and $G_Y''(1) = k(k+1)$. Therefore

$$\mathrm{E}(Y) = G_Y'(1) = k,$$

and, since $G_Y''(1) = \mathrm{E}\big(Y(Y-1)\big)$,

$$\mathrm{Var}(Y) = G_Y''(1) + G_Y'(1) - \big(G_Y'(1)\big)^2 = k(k+1) + k - k^2 = 2k.$$

The distribution. $\mathrm{P}(Y = r)$ is the coefficient of t^r in $G_Y(t)$. Write

$$G_Y(t) = 2^{-k}\left(1 - \tfrac{1}{2}t\right)^{-k},$$

and use the binomial series for a negative index (valid for $|t| < 2$):

$$(1-x)^{-k} = \sum_{r=0}^{\infty} \binom{k+r-1}{r} x^r.$$

(The coefficient is $\dfrac{k(k+1)\cdots(k+r-1)}{r!}$, which is $\binom{k+r-1}{r}$.) With $x = \frac{1}{2}t$,

$$\mathrm{P}(Y=r) = \binom{k+r-1}{r}\frac{1}{2^{k+r}}, \qquad r = 0, 1, 2, \ldots$$

Ask yourself Is there a way to see this formula directly?

Yes. Think of all the tosses in one long line: $Y = r$ means that the kth tail appears on toss number $k+r$. That particular sequence of $k+r$ tosses has probability $2^{-(k+r)}$, and the last toss must be a tail, so we only choose which r of the first $k+r-1$ tosses are heads: $\binom{k+r-1}{r}$ ways. This is the negative binomial distribution.

Checks

- With $k = 1$, $\mathrm{P}(Y=r) = 2^{-(r+1)}$, the single-coin geometric distribution, and $\mathrm{E}(Y) = 1$, $\mathrm{Var}(Y) = 2$, the familiar values for a geometric variable with $p = \frac{1}{2}$ counted from 0.
- $G_Y(1) = 1$, as every generating function must satisfy.
- With $k = 2$, $\mathrm{P}(Y=0) = \frac{1}{4}$ (both first tosses tails) and $\mathrm{P}(Y=1) = \binom{2}{1}\frac{1}{8} = \frac{1}{4}$ (head then tail on one coin, tail on the other: two orders, each $\frac{1}{8}$).
- Exact computation with fractions confirmed $\mathrm{E}(X_i) = k/2^i$ for $k \le 7$ and $i < 40$ (so the partial sums approach k), and the formula for $\mathrm{P}(Y=r)$, the mean k and variance $2k$ for $k \le 6$.

Common slips

- Taking $\mathrm{P}(Y_i = n) = (\frac{1}{2})^n$, forgetting the final tail; the probabilities then sum to 2.
- Using $\mathrm{Var}(Y) = G''(1) - \big(G'(1)\big)^2$, which misses the $G'(1)$ term and gives k instead of $2k$.
- Writing the coefficient as $\binom{k+r}{r}$: the last toss is fixed as a tail, so only $k+r-1$ tosses are free.

Beyond STEP

Part (i) is a branching process: each "individual" (a head) has a random number of "offspring" in the next generation, here $\mathrm{B}(1, \frac{1}{2})$, and since the mean number of offspring

is less than 1 the population dies out. The conditioning step $\mathrm{E}(X_{i+1}) = \mathrm{E}\big(\mathrm{E}(X_{i+1} \mid X_i)\big)$ is the tower property, and in a first-year course the generating function of X_i is shown to be an i-fold composition of the offspring generating function. Grimmett and Welsh, *Probability: An Introduction*, treats both generating functions and branching processes.

WORKED SOLUTION

STEP III 2009, Question 13

Probability and Statistics • Central Limit Theorem

STEP III 2009, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q13

The key idea

Everything is a function of the single angle θ: $X = \cos\theta$ and $Y = \sin\theta$. So the distribution of X comes from the cumulative distribution function (work out which arc of the circle gives $\cos\theta \leq x$), and the means, variances and covariances are just averages of trigonometric functions over a full turn. In part (ii), $\overline{X}$ is an average of many independent, identically distributed variables, which is exactly the situation where the central limit theorem applies.

Solution

Part (i)

Since P is on the unit circle, $X = \cos\theta$ and $Y = \sin\theta$, where θ has density $\frac{1}{2\pi}$ on $[0, 2\pi)$.

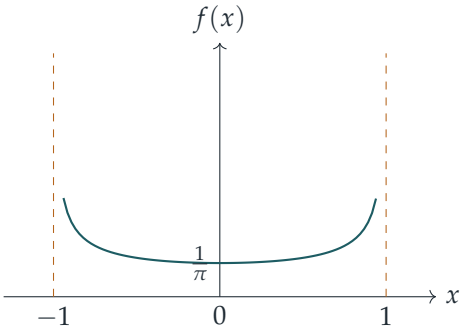
The density of X. Clearly $-1 \leq X \leq 1$. Take $-1 < x < 1$ and let $\alpha = \arccos x \in (0, \pi)$. On $[0, 2\pi)$ we have $\cos\theta > x$ exactly when $\theta < \alpha$ or $\theta > 2\pi - \alpha$, an arc of total length 2α. Hence

$$F(x) = \mathrm{P}(X \leq x) = 1 - \frac{2\alpha}{2\pi} = 1 - \frac{\arccos x}{\pi}.$$

Differentiating, using $\dfrac{\mathrm{d}}{\mathrm{d}x}\arccos x = -\dfrac{1}{\sqrt{1-x^2}}$,

$$f(x) = \frac{1}{\pi\sqrt{1-x^2}} \qquad (-1 < x < 1),$$

and $f(x) = 0$ otherwise. (A quick check: $\int_{-1}^{1} f(x)\,\mathrm{d}x = \frac{1}{\pi}\left[\arcsin x\right]_{-1}^{1} = 1$.)



The density is smallest at $x = 0$ and blows up at $x = \pm 1$: the point P moves horizontally slowest near the ends of the horizontal diameter, so X spends most of its time there.

The variance. By symmetry (f is even), $\mathrm{E}(X) = 0$. Working directly with θ,

$$\mathrm{E}(X^2) = \frac{1}{2\pi}\int_0^{2\pi} \cos^2\theta \,\mathrm{d}\theta = \frac{1}{2\pi}\int_0^{2\pi} \tfrac{1}{2}(1+\cos 2\theta)\,\mathrm{d}\theta = \frac{1}{2},$$

so

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - \big(\mathrm{E}(X)\big)^2 = \tfrac{1}{2}.$$

By the same argument (or since $\theta - \frac{\pi}{2}$ is also uniform on a full turn), $\mathrm{E}(Y) = 0$ and $\mathrm{Var}(Y) = \frac{1}{2}$.

Uncorrelated. We have

$$\mathrm{E}(XY) = \frac{1}{2\pi}\int_0^{2\pi} \cos\theta \sin\theta \,\mathrm{d}\theta = \frac{1}{4\pi}\int_0^{2\pi} \sin 2\theta \,\mathrm{d}\theta = 0,$$

so $\mathrm{Cov}(X, Y) = \mathrm{E}(XY) - \mathrm{E}(X)\mathrm{E}(Y) = 0 - 0 = 0$. Hence X and Y are uncorrelated.

Not independent. They are certainly not independent, since $X^2 + Y^2 = 1$: once X is known, Y is pinned down to one of $\pm\sqrt{1 - X^2}$. For a concrete event, $\mathrm{P}(X > 0.8) > 0$ and $\mathrm{P}(Y > 0.8) > 0$, but $\mathrm{P}(X > 0.8 \text{ and } Y > 0.8) = 0$ because then $X^2 + Y^2 > 1.28$. Independence would require this joint probability to equal the (positive) product. Zero correlation only says there is no *linear* relationship; here the relationship is a circle, which is symmetric enough to have no linear trend at all.

Part (ii)

Each pair (X_i, Y_i) has the distribution in part (i), and pairs with different indices are independent because the points are chosen independently.

$\overline{X}$ and $\overline{Y}$ are uncorrelated. Covariance is bilinear, so

$$\mathrm{Cov}(\overline{X}, \overline{Y}) = \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} \mathrm{Cov}(X_i, Y_j).$$

For $i \neq j$, X_i depends only on P_i and Y_j only on P_j, which are independent, so $\mathrm{Cov}(X_i, Y_j) = 0$. For $i = j$, $\mathrm{Cov}(X_i, Y_i) = 0$ by part (i). Every term vanishes, so $\mathrm{Cov}(\overline{X}, \overline{Y}) = 0$: $\overline{X}$ and $\overline{Y}$ are uncorrelated.

The approximate probability. The X_i are independent and identically distributed with mean 0 and variance $\frac{1}{2}$. So

$$\mathrm{E}(\overline{X}) = 0, \qquad \mathrm{Var}(\overline{X}) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{1}{n^2}\cdot\frac{n}{2} = \frac{1}{2n},$$

where we may add the variances because the X_i are independent.

> **Ask yourself** X is nothing like normal (look at its density). Why can we use a normal approximation?

The central limit theorem needs only independent, identically distributed terms with finite variance, whatever their shape. So for large n, $\overline{X}$ is approximately $\mathrm{N}\left(0, \frac{1}{2n}\right)$, with standard deviation $\sigma = \dfrac{1}{\sqrt{2n}}$. Now

$$\sqrt{\frac{2}{n}} = \frac{2}{\sqrt{2n}} = 2\sigma,$$

so with $Z \sim \mathrm{N}(0,1)$,

$$\begin{aligned}\mathrm{P}\left(|\overline{X}| \leq \sqrt{\tfrac{2}{n}}\right) &= \mathrm{P}\left(\frac{|\overline{X}|}{\sigma} \leq 2\right) \approx \mathrm{P}(|Z| \leq 2)\\ &= 2\Phi(2) - 1 = 2(0.9772) - 1 \approx 0.954 \approx 0.95,\end{aligned}$$

as required.

Checks

- The variance $\frac{1}{2}$ makes sense: $X^2 + Y^2 = 1$ and, by symmetry, $\mathrm{E}(X^2) = \mathrm{E}(Y^2)$, so each must be $\frac{1}{2}$. No integration needed!
- A simulation of 4×10^5 random angles matched $F(x) = 1 - \frac{1}{\pi}\arccos x$ to about three decimal places at several values of x, and gave $\mathrm{Var}(X) \approx 0.5007$ and $\mathrm{Cov}(X, Y) \approx 0.001$.
- With $n = 50$ and 40 000 repetitions, the simulated $\mathrm{Var}(\overline{X})$ was 0.00997 (theory 0.01), and the proportion with $|\overline{X}| \leq \sqrt{2/50} = 0.2$ was 0.954, against $\mathrm{P}(|Z| \leq 2) = 0.9545$.

Common slips

- Differentiating $1 - \frac{1}{\pi}\arccos x$ with the wrong sign and getting a negative "density". A density can never be negative, so check the sign.

- Concluding independence from zero covariance. Independence implies zero covariance, but not the other way round.
- Using $\mathrm{Var}(\overline{X}) = \frac{1}{2}$ or $\frac{1}{2} \cdot \frac{1}{\sqrt{n}}$: the variance of a mean of n independent copies is the single variance divided by n; it is the *standard deviation* that scales like $1/\sqrt{n}$.

Beyond STEP

The density $1/(\pi\sqrt{1-x^2})$ is the *arcsine distribution*, which turns up surprisingly often, for example as the limiting law of the fraction of time a random walk spends positive. The rest of the question is about the gap between uncorrelated and independent, and about the central limit theorem, which you will prove using moment generating or characteristic functions in a first-year probability course. In fact $(\overline{X}, \overline{Y})$ is approximately bivariate normal, and for jointly normal variables zero correlation *does* imply independence. Grimmett and Welsh, *Probability: An Introduction*, covers all of this well.

Beyond STEP

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